Optimal Carbon Tax with Dynamic Skill Heterogeneity

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July 31, 2026†
Work in Progress: Latest version

Abstract

How should a government price carbon when it also cares about redistribution? I extend the dynamic Mirrlees model to two goods, one polluting, over a life cycle of privately observed skill shocks. Under weak separability, the dirty good should be taxed at the social cost of carbon (SCC) while labor and savings distortions keep their standard forms. Only non-separability makes it non-linear, raising it above the SCC when the dirty good is a weaker labor complement. In a US calibration with energy standing in for the dirty good however, uniform taxation captures 98% of the welfare gain of an optimal non-linear carbon tax. When labor and capital distortions are frozen at the levels implied by the US tax system, this sufficiency result is overturned as the best uniform tax only captures 1.5% of the welfare gain of the best non-linear carbon tax. Here, a flat tax at an SCC of \$190/tCO2 would be welfare-reducing and the best flat tax would deflate to an equivalent of \$38/tCO2. This happens to be in the ballpark of where many jurisdictions price carbon, consistent with a government optimizing the carbon margin alone.

Keywords: *Carbon tax, dynamic Mirrlees taxation, second-best environmental policy*
JEL Classification: *D62; D82; H21; H23*

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†I thank my advisor Guillaume Sublet for his invaluable guidance and support, as well as the members of his workshop and the participants of the CIREQ Weekly PhD Student Research Workshop, the McGill University microbreakfast, the University of Montréal macro brownbag, the CIREQ-CIRANO-RRECQ Environmental and Natural Resource Economics Workshop, the 19th CIREQ PhD Students' Conference, the 59th CEA Annual Conference, the 35th CREEA Annual Conference, the 65th SCSE Conference, the 60th CEA Annual Conference, the 25th LAGV conference, and the 7th WCERE conference for their comments and feedback. I am grateful for the support and funding from the Fonds de Recherche du Québec Société et Culture (FRQSC) through the Doctoral Training Scholarships program. DOI: https://doi.org/10.69777/329014.

1 Introduction

Carbon pricing is the economist's favorite climate policy, and the voter's least. France abandoned its carbon tax plans after the *gilets jaunes* protests and Canada repealed its consumer carbon tax in 2025 after it became an electoral wedge issue on the grounds that the tax unfairly burdens low-income households. This disconnect might in part be driven by the fact that most carbon policy is informed by integrated assessment models (Environment and Climate Change Canada (2016), Interagency Working Group on Social Cost of Greenhouse Gases (2021)), which are representative agent models that are not designed to address this concern.

Whether carbon taxes are regressive is a topic of intense study, with one aspect of the debate relating to the choice of measure. When households are ranked by annual income, taxes on gasoline and other energy appear to fall more heavily on the poor. However, when households are ranked by lifetime income, the burden appears much flatter because households smooth consumption and today's low earners include the young and temporarily unfortunate (Poterba 1989; Hassett et al. 2009). The welfare-relevant burden of a carbon tax is thus a lifetime object, not an annual one. It's therefore important to consider the timing of an optimal carbon tax.

However, most research focuses on the incidence of existing carbon taxes rather than asking how they should be designed in the first place. Barrage (2019) and Douenne et al. (2026) are the exceptions, but the first ties the tax system to an *ad hoc* revenue target rather than optimal redistribution and the second restricts attention to linear instruments even though income taxes are in practice non-linear. How then should a government price carbon when it cares about redistribution, and does the presence of pollution change how income should be taxed?

The answer turns on how the income tax is modeled. The Ramsey tradition treats it as a schedule of linear instruments and usually finds that carbon should be priced below the Pigouvian level because the carbon tax compounds the distortions the rest of the tax system already imposes. The Mirrleesian tradition allows the government to levy nonlinear income taxes and usually finds that the carbon tax should be set at the Pigouvian level, barring exceptions in the structure of preferences.

I take the Mirrleesian approach into a dynamic setting with two goods, one clean and one dirty, in which agents draw privately observed skill shocks over their lifecycle and their consumption of the dirty good feeds a stock externality. To my knowledge, this is the first dynamic-Mirrleesian derivation of the optimal consumer carbon tax. Compared to the static irrelevance theorems of Kaplow (2012) and Jacobs and Mooij (2015), dynamics introduce savings and history-dependence.

I find that the static irrelevance result holds in a dynamic environment. Under weak separability, the carbon wedge equals the social cost of carbon, the labor wedge keeps the Golosov et al. (2016) ABCD form, and the savings wedge keeps its inverse-Euler form. Redistributing through the optimal nonlinear income tax restores the split between corrective and redistributive taxation.

That is, the carbon price should not be adjusted to perform redistribution.

I then quantify the separation by solving the optimal mechanism. The optimal wedge on energy is the social cost of carbon scaled up by a history-dependent Corlett-Hague premium: it starts at the Pigouvian level, rises with age as the labor wedge does, and ends the working life about 9% above the Pigouvian price. That premium is however small in welfare terms. A flat carbon tax at the social cost of carbon captures 98% of the welfare gain of the full optimum and the wedge's entire Mirrleesian fine structure is worth only a few thousandths of a percent of consumption. This near-sufficiency is the separation in welfare terms: pricing carbon at the social cost of carbon and redistributing through the income tax is close to optimal even when preferences are nonseparable and energy is a leisure complement.

The deviations from the Pigouvian benchmark that do arise are small, signed by how each good complements labor, and progressive in skill. Late in the working life the optimal energy wedge reaches about 70% for the top skill quintile against 57% for the bottom, both above the Pigouvian level near 50%. The optimal carbon wedge rises with skill and the regressivity of real carbon taxes is therefore a property of flat instruments. This is the normative counterpart the positive incidence debate does not answer to, reaching from the Mirrleesian side the near-Pigouvian result that Douenne et al. (2026) and Douenne et al. (2025) reach from the Ramsey side.

These results assume every margin is set optimally. Real systems are not jointly optimal, and Section 5 prices carbon against an income tax that is given rather than chosen. Holding the labor and savings wedges to exogenous schedules, a single statistic carries the whole departure from the social cost of carbon: the net shadow price of the given income tax, the labor error net of the savings error. The two errors reach the carbon price through the same channel and can offset. The labor error sets the level of the deviation within a date, pulling the carbon price below the social cost of carbon where the schedule overtaxes labor and above it where the schedule undertaxes. The savings error cannot share one sign across the life cycle, so it tilts the price across dates rather than moving its level, and there is generically an age at which the two cancel and pricing is exactly Pigouvian. Run in the other direction, a carbon price legislated below the social cost of carbon raises the optimal labor wedge by the marginal budget share of energy times the uncollected damage: the income tax collects the carbon the carbon price misses. The optimal linear instruments of the Ramsey tradition sit inside this family, at the levels where the constraint multipliers average to zero.

Section 2 reviews the literature. Sections 3 to 4 build the two-good dynamic Mirrlees economy and derive the separation and the modified social cost of carbon. Section 5 takes one side of the tax system as given and re-derives the other: first the labor and savings wedges held to the schedules an actual income tax implies, then the carbon price held to a legislated or anonymous rate, the form real carbon taxes take. Section 6 solves the mechanism numerically on US data. Section 7 concludes.

2 Literature review

This paper sits between two literatures that reach opposite verdicts on how distortionary taxation should bend the carbon price and draws on a third that has the tools to adjudicate between them, yet has never been aimed at the environment. Should a government facing a distortionary tax system price carbon below marginal damages because the environmental tax compounds distortions elsewhere, above marginal damages because the environmental tax can be used to ease distortions elsewhere, or exactly at marginal damages?

The Ramsey tradition typically finds that the optimal carbon tax should lie below marginal damages, but that the structure of the optimal income tax is unchanged (Sandmo 1975; Bovenberg and Mooij 1994; Bovenberg and Smulders 1996). Barrage (2019) extends their static results with a dynamic climate-economy Ramsey model and finds that the optimal carbon tax should be 8-24% below the first-best Pigouvian level in the presence of distortionary income taxation. However, this analysis assumes a representative agent, focusing on efficiency and ignoring any distributional concerns.

Adding heterogeneous agents to this model overturns this result. Douenne et al. (2026) extend Barrage's dynamic Ramsey model to heterogeneous agents, taking inspiration from the model developed in Werning (2007), and find that the optimal carbon tax is nearly Pigouvian. Douenne et al. (2025) show that this is robust to the inclusion of uninsurable income risk. Whether such a flat carbon tax is regressive and whether revenue recycling is progressive is still an open debate (Klenert et al. 2018; Goulder et al. 2019) and a parallel policy and ethics literature presses for differentiated carbon taxation (Chancel 2022; Dietsch 2024).

Without personalized carbon taxes, the exact restrictions on a uniform carbon tax reshape both its level and incidence. Using a heterogeneous agent microsimulation model calibrated to German data and focusing on revenue-neutral carbon tax reforms, Ploeg et al. (2025) find that the carbon tax should exceed marginal damages when agents have high inequality aversion. In a model calibrated to German data where households differ in income and energy efficiency, Hänsel et al. (2022) show that the first-best keeps the carbon tax uniform at the Pigouvian level and carries all redistribution through lump-sum transfers conditioned on the household's income–energy type while leaving the income tax undistorted.

Separation can also fail for a reason orthogonal to instrument constraints. When climate damages fall unequally across households and compensation is incomplete, the social cost of carbon itself becomes distribution-dependent (Dennig et al. 2015; Kornek et al. 2021). I hold damages homogeneous to isolate the redistributive-instrument channel, so the separation I study is the Atkinson–Stiglitz one.

A static Mirrleesian literature establishes that separation and overturns with it the Ramsey results: the below-damages result is not intrinsic to distortionary taxation but rather an artifact of the linear-instrument restriction. Once the income tax is the optimal nonlinear schedule of Mirrlees (1971), the logic of Atkinson and Stiglitz (1976) applies and differential commodity taxes are redundant under weak separability. Cremer et al. (1998), Cremer et al. (2001), and Cremer et al. (2010) develop the environmental version of the problem where a nonlinear income tax alongside a strictly Pigouvian levy is efficient, so the public-funds correction of the Ramsey tradition disappears. Kaplow (2012) shows that any environmental reform can be made distribution-neutral by a type-by-type adjustment of the income tax, so a move to Pigouvian pricing is Pareto-improving whatever the schedule. Jacobs and Mooij (2015) show that, when goods and environmental quality are weakly separable from labor, the optimal pollution tax equals marginal damages exactly and the marginal cost of public funds is one.

When separability fails, the tax follows the logic established in Corlett and Hague (1953). A good is taxed above the Pigouvian level for a leisure complement, the case West and Williams (2007) substantiate for gasoline, and below it for a leisure substitute. Saez (2002) extends the benchmark to preferences that vary with ability. In a linear-tax setting, Jacobs and Ploeg (2019) show that the pollution tax stays Pigouvian when Engel curves are linear and slips below only when the poor spend disproportionately on the dirty good under nonlinear Engel curves. Every result in this literature is static, with no savings and no lifecycle dynamics. Therefore, it cannot speak to the age profile or the time path of a carbon tax.

The tools for this dynamic treatment come from the New Dynamic Public Finance literature. Golosov et al. (2003) derive a rationale for positive capital taxation in a dynamic Mirrlees economy, since it is harder to elicit future work from households that save privately. This overturns the classic zero-capital-tax benchmark common in the Ramsey tradition (Judd 1985; Chamley 1986). Moreover, the authors find that the Atkinson-Stiglitz uniform-commodity result carries over into the dynamic setting. Kocherlakota (2005) shows how to implement the capital wedge into a wealth-tax and Albanesi and Sleet (2006) derive a tractable recursive form of the problem. Kapička (2013) extends the first-order approach to persistent shocks, Pavan et al. (2014) give the general envelope characterization, Farhi and Werning (2013) characterize the labor wedge over the life cycle, and Golosov et al. (2016) extend the ABC form of the labor wedge derived in the static frameworks of Mirrlees (1971) and Saez (2001) into a dynamic ABCD form. The optimal wedges the dynamic Mirrlees model delivers vary with age, connecting to the age-dependent-tax literature (Weinzierl 2011; Bastani et al. 2013; Erosa and Gervais 2002), which prices age-varying labor rather than a commodity. None of this machinery has been aimed at commodity taxation under non-separable preferences or corrective taxation.

I extend the dynamic Mirrleesian apparatus from one consumption good to two, letting the second pollute, and characterize the optimal carbon tax alongside the labor and savings wedges over the life cycle. To my knowledge this is the first dynamic Mirrleesian derivation of the optimal carbon tax. The heterogeneous-agent Ramsey results and this paper share the Jacobs and Mooij (2015) and Kaplow (2012) irrelevance result, which both approaches carry into a climate economy by different routes. Douenne et al. (2026) carry it with an optimally-set linear transfer, showing analytically that the marginal cost of public funds averages to one across a balanced-growth path and equals one period by period when preferences are logarithmic. Douenne et al. (2025) find that this result still exists when agents face uninsurable labor risk. I consider uninsurable labor risk with a fully nonlinear income tax and obtain a carbon wedge equal to the social cost of carbon at every history under weak separability and any preference satisfying mild conditions, alongside a Mirrleesian labor wedge and a non-zero savings wedge following the inverse Euler condition. The two approaches practically converge in their policy conclusions however: pricing carbon at the social cost of carbon and redistributing through the income tax captures almost all the gains of a non-linear carbon wedge.

3 Environment

The economy is a two-good extension of the dynamic Mirrlees setting of Golosov et al. (2003), Farhi and Werning (2013), and Golosov et al. (2016). The two goods are a clean numéraire and a polluting good. To isolate how household income inequality shapes the optimal carbon tax, I abstract from international and intergenerational equity, and consider a single country as well as a single generation living for $T+1$ periods.

Each household consumes a clean good c_t and a polluting good q_t, supplies labor l_t, and is exposed to the aggregate pollution stock S_t. Expected lifetime utility is

$$\mathbb{E}_0\left[\sum_{t=0}^{T}\beta^t U(c_t, q_t, l_t, S_t)\right]$$

$\beta \in (0,1)$ is the discount factor and utility is non-separable between its arguments. The clean good c is the numéraire, with prices normalized to 1. A unit of output transforms one-for-one into either good, so the producer price of the dirty good is also 1. The clean good is emission-free, whereas each unit of the dirty good emits a unit of carbon, so that date-t emissions equal aggregate dirty-good consumption Q_t. Emissions accumulate into a persistent pollution stock S_t, a function of current and past aggregate emissions.

$$S_t = S\left(Q_0, Q_1, \ldots, Q_t\right)$$

where $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta]\, dF_0(\theta)$ is aggregate consumption of the dirty good at date t. I assume that S is differentiable and non-decreasing in each argument, but otherwise leave it unrestricted. The linear special case $S_t = \sum_{j=0}^{t}(1-d_j)\, Q_{t-j}$ in which each past emission flow leaves an additive trace $(1-d_j)$ in the current stock, is the carbon-cycle representation of

Golosov et al. (2014b). The general form above nests it and allows nonlinear accumulation. A convenient leading example keeps a fraction δ of the stock from one period to the next.

$$S_t = \sum_{j=0}^{t} \delta^j Q_{t-j}$$

This is the recursion $S_t = \delta S_{t-1} + Q_t$ from an initial pollution stock. To keep the state finite, I assume emissions stop affecting the pollution stock after s periods.

$$S_t = S\left(Q_{t-s}, ..., Q_t\right)$$

Aside from the polluting good, the model follows the standard dynamic Mirrlees setup. At $t = 0$, agents draw an initial skill (type) θ_0 from a distribution $F_0(\theta)$. Later skills follow a Markov process $F_t(\theta|\theta_{t-1})$. At period t, each agent knows their skill draws up to time t, which I denote by a skill history vector $\theta^t = (\theta_0, \ldots, \theta_t)$. Θ^t is the set of all possible skill histories.

An agent of type θ_t who supplies l_t units of labor produces $y_t = \theta_t l_t$ units of output. Everyone observes total output, but skill θ_t and labor supply l_t are private. Consumption of both goods is also publicly observable[1]. Let $c_t(\theta^t), q_t(\theta^t) : \Theta^t \to \mathbb{R}_+$ and $y_t(\theta^t) : \Theta^t \to \mathbb{R}_+$ be the agent's allocation of the clean good, the polluting good, and output, respectively. Let $\sigma^t(\theta^t) : \Theta^t \to \Theta^t$ describe how an agent with skill history θ^t reports it, and let Σ^t be the set of all possible reporting strategies. Capital transfers resources across periods through a linear technology with exogenous constant gross return $R > 0$. The social welfare function is

$$\int_0^\infty \alpha(\theta)\, \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U(c_t, q_t, l_t, S_t) \right] dF_0(\theta)$$

$\alpha(\theta)$ is a Pareto weight on the welfare of an agent with initial skill θ, normalized so that $\int_0^\infty \alpha(\theta)\, dF_0(\theta) = 1$. A utilitarian planner sets $\alpha(\theta) = 1$ for all θ. Let U_c and U_q denote the partial derivatives with respect to the clean and polluting goods, and U_l with respect to labor. Since $y = \theta l$, I can write $U(c, q, l, S)$ as $U\left(c, q, \frac{y}{\theta}, S\right)$. Let U_y and U_θ be the partial derivatives of utility with respect to y and θ.

Assumption 3.1 (preferences).
U is twice continuously differentiable with $U_c, U_q > 0$, $U_l, U_S < 0$, $U_{cc}, U_{qq}, U_{ll}, U_{SS} \leq 0$, strictly

[1] This is a plausible assumption for some goods such as electricity and air travel. For goods whose consumption cannot be observed, the planner is further constrained and the unrestricted wedges of Section 4 no longer apply. The anonymous price of Section 5.2 survives: an anonymous linear price requires observing total expenditure, not its composition.

quasi-concave in (c, q) with $2U_c U_q U_{cq} - U_c^2 U_{qq} - U_q^2 U_{cc} > 0$ (regular strict quasi-concavity: a strictly convex, non-degenerate indifference map), and satisfies Inada conditions in (c, q). Single crossing holds at every S.

$$\frac{\partial}{\partial \theta} \frac{U_l(c, q, y/\theta)}{U_c(c, q, y/\theta, S)} \geq 0 \qquad \text{for all } (c, q, y, \theta)$$

At the optimal allocation, $1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) > 0$, where $\epsilon_t, \gamma_{c,t}$ are defined in (1) below.

Assumption 3.2 (shock process)**.**
For all $t \geq 1$, f_t is differentiable in both arguments with $f_t' = \partial f_t / \partial \theta$ and $f_{2,t} = \partial f_t / \partial \theta_{t-1}$. For all θ_{t-1}, the ratio

$$\frac{\theta_{t-1} \int_\theta^\infty f_{2,t}(x|\theta_{t-1})\, dx}{\theta f_t(\theta|\theta_{t-1})}$$

is bounded, and $\lim_{\theta \to \infty} \frac{1-F_t(\theta|\theta_{t-1})}{\theta f_t(\theta|\theta_{t-1})}$ is finite.

Any admissible allocation must be incentive compatible: the agent must weakly prefer truthful reporting to any other strategy. The pollution stock S_t is an aggregate, so no single agent's deviation moves it and it enters both sides of the constraint identically. For every initial type θ and every reporting strategy,

$$\begin{aligned} &\mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \\ &\qquad \geq \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right] \end{aligned}$$

Over a finite horizon, global incentive compatibility is equivalent to the absence of a profitable one-shot deviation after every history (Fernandes and Phelan 2000; Kapička 2013). I follow the first-order approach of Kapička (2013), Farhi and Werning (2013), and Golosov et al. (2016), replacing these constraints with their local counterpart. Let $u_t(\theta^t)$ be the lifetime utility an agent obtains from reporting truthfully at every date. Truth-telling is locally optimal when the envelope condition holds.

$$\dot{u}_t(\theta^t) = U_\theta \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) + \beta\, w_{2,t}(\theta^t)$$

$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta_{t+1})\, \partial_{\theta_t} f_{t+1}(\theta_{t+1}|\theta_t)\, d\theta_{t+1}$ is the marginal continuation value of type θ^t, and the dot denotes a derivative with respect to the current draw θ_t. The continuation term is the future information rent of the current draw: with i.i.d. shocks, $\partial_{\theta_t} f_{t+1} = 0$ and it vanishes, since today's skill then carries no information about tomorrow's. Under Assumptions 3.1 and 3.2, this local condition is necessary for incentive compatibility (Kapička 2013). The monotonicity requirements on the allocation belong to the sufficiency direction, not to necessity. The local conditions define a relaxed problem and are necessary but not sufficient for global incentive compatibility. Indeed, no general primitive restriction guarantees sufficiency in dynamic models with multiple goods (Golosov et al. 2016). I solve the relaxed problem throughout and verify global incentive compatibility numerically in Section 6.

The aggregate resource constraint must hold in every period,

$$\int_0^\infty \mathbb{E}_0[c_t(\theta^t) + q_t(\theta^t)|\theta]\, dF_0(\theta) + K_{t+1} \leq \int_0^\infty \mathbb{E}_0[y_t(\theta^t)|\theta]\, dF_0(\theta) + RK_t$$

where K_t is the aggregate capital stock, with K_0 given and $K_{T+1} = 0$. Dividing each period's constraint by R^t, summing over t, and using these boundary conditions to telescope the capital terms consolidates the sequence into a single intertemporal constraint,

$$\int_0^\infty \mathbb{E}_0\left[\sum_{t=0}^T \frac{c_t(\theta^t) + q_t(\theta^t)}{R^t}\middle|\theta\right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0\left[\sum_{t=0}^T \frac{y_t(\theta^t)}{R^t}\middle|\theta\right] dF_0(\theta) + RK_0$$

The planner chooses allocations to maximize welfare subject to the local incentive constraint, the intertemporal resource constraint, and the pollutant law of motion.

$$\max_{\{c_t(\theta^t), q_t(\theta^t), y_t(\theta^t), S_t\}_{\theta^t \in \Theta^t,\ t=0,\dots,T}} \int_0^\infty \alpha(\theta) \left(\mathbb{E}_0\left[\sum_{t=0}^T \beta^t U\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)\middle|\theta\right]\right) dF_0(\theta)$$

Three wedges, the implicit marginal taxes on the labor, savings, and commodity margins, describe the distortions the optimal allocation imposes.

$$
\begin{aligned}
1-\tau_t^y(\theta^t) &\equiv -\frac{U_l\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\theta_t\, U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)} \\
1-\tau_t^s(\theta^t) &\equiv \frac{1}{\beta R}\, \frac{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\mathbb{E}_t\left[U_c\left(c_{t+1}(\theta^{t+1}), q_{t+1}(\theta^{t+1}), \frac{y_{t+1}(\theta^{t+1})}{\theta_{t+1}}, S_{t+1}\right)\right]} \\
1+\tau_t^q(\theta^t) &\equiv \frac{U_q\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}
\end{aligned}
$$

I define six preference statistics used throughout. Two are the standard own-curvature elasticities, namely the inverse Frisch elasticity and the inverse intertemporal elasticity of substitution.

$$
\epsilon_t \equiv \frac{U_{ll,t}\, l_t}{U_{l,t}} \qquad \sigma_t \equiv -\frac{U_{cc,t}\, c_t}{U_{c,t}} \tag{1}
$$

Four are cross-partials that measure how the marginal utilities of the two goods and of the pollutant move with labor as well as with each other. In order, they represent the labor complementarity with the clean good, the labor complementarity with the dirty good, the labor complementarity of environmental quality, and the cross-elasticity between the two goods.

$$
\gamma_{c,t} \equiv \frac{U_{cl,t}\, l_t}{U_{c,t}} \qquad \gamma_{q,t} \equiv \frac{U_{ql,t}\, l_t}{U_{q,t}} \qquad \xi_t \equiv \frac{U_{Sl,t}\, l_t}{U_{S,t}} \qquad \sigma_{cq,t} \equiv -\frac{U_{cq,t}\, q_t}{U_{c,t}} \tag{2}
$$

Note that complementarity is measured with respect to labor rather than leisure here, so a positive γ signals a good whose marginal utility rises as the agent works more. To get the leisure complementarity interpretation, it suffices to flip the sign. Most of the commodity-wedge results hinge on the gap $\gamma_{c,t} - \gamma_{q,t}$ between the clean and dirty goods' complementarities with labor. If the gap is positive, the clean good is the relative labor complement, meaning an agent who works more wants more of it. Likewise, the results pertaining to the social cost of carbon hinge on the gap $\xi_t - \gamma_{c,t}$, which measures how the marginal damage of pollution varies with labor supply. If the gap is positive, an agent who works more suffers more from pollution and a cleaner environment is a relative labor complement.

One combination of these statistics appears whenever the commodity margin is constrained:

$$
\mathcal{I}_t \equiv \frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t\,(1+\tau_t^q)} \tag{3}
$$

This is the income-effect gap evaluated at the prevailing consumer price of the dirty good. It is positive when the dirty good is normal since the marginal propensity to spend on it is proportional to $\mathcal{I}_t$ (Appendix D.3).

4 Optimal distortions

4.1 Planner's problem

First, following Kapička (2013)[2], we can write the planner's problem recursively. Appendix B.1 shows that the presence of externalities does not invalidate the approach. No individual report moves the aggregate pollution stock, so the agent's incentive problem is the one above with the pollution stock entering utility as a date-specific parameter. Aggregates are deterministic, so internalizing them appends a single new state, the emission window $\mathbf{S}_- = (Q_{t-s}, \dots, Q_{t-1})$ of lagged aggregate emissions feeding current and future pollution. The continuation value receives the updated window, with the current aggregate Q_t appended.

$$V_t(\hat{w}, \hat{w}_2, \theta_-, \mathbf{S}_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, \mathbf{S}) \right] f_t(\theta|\theta_-) d\theta$$

$$\text{s.t.}$$

$$\dot{u}(\theta) = U_\theta\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right) + \beta w_2(\theta)$$

$$\hat{w} = \int_0^\infty u(\theta)\bar{\alpha}_t(\theta) f_t(\theta|\theta_-) d\theta$$

$$\hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta$$

$$u(\theta) = U\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right) + \beta w(\theta)$$

$$S = S\left(\cdots, \int q(\theta) f(\theta) d\theta\right)$$

The planner minimizes resource cost while offering an agent expected lifetime utility $\hat{w}$ with

[2] Kapicka's consumption set is convex and compact (his Assumption 1), so consumption can be a vector rather than a scalar. This paper uses only the necessity direction, his Theorem 4, which is dimension-free and does not invoke his Assumption 5. That assumption, that U is affine in the consumption vector, carries the sufficiency direction, and he gives two routes to it: utility separable across the components and affine in the shock, $U(x, \theta) = U_1(x_1) + \theta U_2(x_2)$, which a non-separable $U(c, q, l, S)$ does not satisfy, or convexification by lotteries over consumption, which would replace the deterministic allocations solved here. Sufficiency is accordingly not assumed but verified numerically in Section 6.

marginal continuation value $\hat{w}_2$, now internalizing what the assigned dirty good does to the pollution stock. $\mathbf{S}$ is the updated emission state, $\mathbf{S}_-$ with Q appended. $\bar{\alpha}_t(\theta)$ is the continuation Pareto weight attached to delivering utility to type θ. It equals the primitive Pareto weight in the initial period, $\bar{\alpha}_0(\theta) = \alpha(\theta)$, and $\bar{\alpha}_t(\theta) = 1$ for $t > 0$. As in Golosov et al. (2016), the lifetime Pareto weights enter only at date 0. From date 1 on, the weight collapses to one.

4.2 Optimal distortions

Proposition 4.1 below describes the planner's optimality conditions in terms of labor supply, savings, and the split between the two goods.

Proposition 4.1 (optimal wedges)**.**
The labor wedge is

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R\,\frac{\tau_{t-1}^y}{1-\tau_{t-1}^y}\,D_t(\theta)$$

$$A_t(\theta) = 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta)$$
$$B_t(\theta) = \frac{1-F_t(\theta)}{\theta f_t(\theta)}$$
$$C_t(\theta) = \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right)d\tilde{x}\right]\left(1-\lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x)\right)\frac{f_t(x)dx}{1-F_t(\theta)}$$
$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}}\frac{U_{c,t}(\theta)}{U_{c,t-1}}\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]f_{2,t}(x)dx}{\theta f_t(\theta)}$$

Let $\widehat{U}_{c,t} \equiv U_{c,t}/(1-\chi_t)$, with $\chi_t \equiv \frac{\gamma_{c,t}}{1+\epsilon_t-\gamma_{c,t}}\frac{\tau_t^y}{1-\tau_t^y}$ as the complementarity-weighted labor wedge. Savings satisfy the generalized inverse Euler equation

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}}\right]$$

The carbon wedge departs from the social cost of carbon by

$$\frac{\tau_t^q - \mathrm{SCC}_t}{1+\tau_t^q} = (\gamma_{c,t}-\gamma_{q,t})\left[B_t(\theta)C_t(\theta) + \beta R\,\frac{\tau_{t-1}^y}{1-\tau_{t-1}^y}\,\frac{D_t(\theta)}{A_t(\theta)}\right] = \frac{\gamma_{c,t}-\gamma_{q,t}}{1+\epsilon_t-\gamma_{c,t}}\,\frac{\tau_t^y}{1-\tau_t^y} \tag{4}$$

with SCC_t characterized in Proposition 4.2.

Proof. See Appendices A.1, A.2, and B.2. □

The first formula is the standard ABCD labor wedge formula, as in Golosov et al. (2016) and Farhi and Werning (2013). None of the terms change, except for term C_t which represents the social value of the resources extracted from types above θ, where the exponential term captures the income effect induced on those types induced by the redistribution. This term now includes the cross-curvature between the two goods, such that the wedge rises all things equal when the goods are Edgeworth substitutes ($\sigma_{cq,t} < 0$) and falls when they are complements ($\sigma_{cq,t} > 0$). Intuitively, when the goods are substitutes, taking away the numéraire from these types renders the dirty good more pleasurable and allows the planner to extract more resources from them.

The second formula is the generalized inverse Euler condition, expressed as a reweighted inverse Euler condition similar to Hellwig (2021). When preferences are non-separable between labor and consumption, the cost of delivering utility is weighted by $1 - \chi_t$, where χ_t is the complementarity-weighted labor wedge. When the numéraire complements labor, a unit of consumption makes work less painful and shrinks the information rent that truth-telling requires, rebating χ_t from the delivery cost. Neither the presence of the second good nor of externalities changes the structure of the savings wedge.

The third formula is new and describes how the planner allocates each good across types. It is the dynamic analogue to the optimal commodity wedge derived in Jacobs and Mooij (2015). The left-hand side is the commodity distortion net of the social cost of carbon, described in the next section, and the right-hand side is the labor distortion scaled by the relative labor complementarities of the two goods. When preferences are weakly separable, $\gamma_{c,t} = \gamma_{q,t}$ and the optimal goods wedge is zero, as shown in Golosov et al. (2003). Otherwise, when one of the goods is relatively more complementary to labor than the other, the planner would like to distort the weaker complement to labor more heavily. Because the labor wedge is history dependent, so is the goods wedge, which then inherits both the shape and the time path of the labor wedge, bent by the ratio of relative labor complementarities and labor response elasticities. This is the dynamic version of the Corlett-Hague rule that states that the weaker complement of labor should be distorted more heavily since this acts as an indirect tax on leisure.

4.3 Social cost of carbon

The social cost of carbon in (4) is not the textbook Pigouvian price, but rather the discounted stream of marginal damages where each damage term is reweighted by how pollution interacts with labor distortions.

Proposition 4.2 (the social cost of carbon)**.**
The shadow price of current emissions (20) is

$$\mathrm{SCC}_t = -\sum_{j=0}^{s} \frac{1}{R^j}\, \mathbb{E}_t \int \left[1 + \left(\frac{\tau^y_{t+j}}{1-\tau^y_{t+j}}\right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}}\right] \frac{U_{S,t+j}}{U_{c,t+j}}\, \Phi_j\, f_{t+j}\, d\theta$$

with $\Phi_j \equiv \partial S_{t+j}/\partial Q_t$ the effect of current aggregate emissions on the date-$(t+j)$ pollution stock ($\Phi_j = 0$ for $j > s$ by construction) and the expectation taken over future skill histories. The Pigouvian price of the same emission is the same stream with every bracket set to one,

$$\mathrm{SCC}_t^P = -\sum_{j=0}^{s} \frac{1}{R^j}\, \mathbb{E}_t \int \frac{U_{S,t+j}}{U_{c,t+j}}\, \Phi_j\, f_{t+j}\, d\theta$$

and $\mathrm{SCC}_t = \mathrm{SCC}_t^P$ when $\xi_{t+j} = \gamma_{c,t+j}$ at every future node or when labor is undistorted.

Proof. See Appendix B.2. □

The bracket is a fiscal-adjustment factor, which is active whenever labor is distorted and pollution interacts with labor supply. Suppose that $\xi_{t,j}, \gamma_{c,t+j} > 0$. When $\xi_{t+j} > \gamma_{c,t+j}$, pollution harm discourages labor supply more so than consuming more of the numéraire encourages it. Pollution then causes direct harm to utility and indirect harm through labor discouragement, both of which the planner would like to correct. Hence, the social cost of carbon rises above the Pigouvian level. When $\xi_{t+j} < \gamma_{c,t+j}$, pollution discourages labor less than consuming more of the numéraire encourages it, so the planner would like to underdistort the dirty good, shrinking the social cost of carbon. This correction operates on every term of the discounted stream of damages and is the dynamic extension of Jacobs and Mooij (2015), under different notation.

4.4 Separation

Proposition 4.3 (three-instrument separation)**.**
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, the corrective and redistributive margins separate.

$$\tau_t^q = \mathrm{SCC}_t \qquad \frac{\tau_t^y}{1-\tau_t^y} = A_t B_t C_t + \beta R\, \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y}\, D_t \qquad \frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R}\, \mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}}\right]$$

The carbon price carries no redistributive term: the income tax and the savings wedge do all the redistribution and insurance whereas the goods wedge is purely corrective.

Proof. Immediate from (4) at $\gamma_{c,t} = \gamma_{q,t}$. □

Corollary 4.1 (dynamic Atkinson-Stiglitz)**.**
With no externality, weak separability gives $\tau_t^q = 0$ and the goods wedge vanishes.

Proof. Proposition 4.3 at $\mathrm{SCC}_t = 0$. □

Redistributing through an optimal nonlinear income tax separates externality correction from redistribution. The carbon price is set at the social cost of carbon, unlike a dynamic Ramsey planner with representative agents like that in Barrage (2019) who prices carbon below it. Douenne et al. (2026) recover an average near-Pigouvian price with linear instruments, whereas the separation here is exact at every history. However, the social cost of carbon may be below or above the Pigouvian level depending on whether a clean environment complements labor supply. Corollary 4.1 is the dynamic Atkinson-Stiglitz result of Golosov et al. (2003) and Proposition 4.3 generalizes the static Mirrlees irrelevance result of Jacobs and Mooij (2015) to a dynamic setting.

5 Constrained optimum

These results assume that the planner is free to jointly optimize the three wedges. However, governments in practice only reform the tax system to move only one wedge at a time. Hence, I consider a problem where the planner adjusts allocations to optimize one wedge, holding the other two constant to an exogenous schedule. This schedule applies node-by-node and can take an arbitrary form, whether linear or nonlinear.

5.1 Constrained-optimal commodity wedges

When setting a carbon tax, a relevant question is how to account for preexisting distortions. This can be translated into our framework by considering a planner that distorts both goods under the constraint that labor and savings distortions must stay fixed at an exogenous level. In practice, as we will see in the quantitative exercise of Section 6, this can be done by fixing the wedges to the levels implied by the existing tax system. At every type, the labor wedge is held to an exogenous schedule $\bar{\tau}_t^y(\theta) < 1$.

$$\Gamma(\theta) \equiv -\frac{U_l\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{\theta\, U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} - (1 - \bar{\tau}_t^y(\theta)) = 0, \tag{5}$$

Here, the multiplier is $\eta(\theta) f_t(\theta|\theta_-)$. Moreover, the savings wedge is held at $\bar{\tau}_t^s(\theta) < 1$.

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}_t^s(\theta_t))\, \beta R\ \mathbb{E}_t\big[U_{c,t+1}(\theta^{t+1})\big]\,. \tag{6}$$

The two constraints behave differently: (5) is static and enters the period Hamiltonian pointwise, whereas (6) links marginal utilities across dates and enters through the marginal-utility promise $\hat{M}$ as an additional state, with multiplier ϑ_t. The planner's problem is

$$V_t(\hat{w}, \hat{w}_2, \hat{M}, \theta_-, \mathbf{S}_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), M(\theta), \theta, \mathbf{S}) \right] f_t(\theta|\theta_-)\, d\theta$$

with constraints (5) and (6) added to the planner's problem of Section 4.1.

Lemma 5.1 (recursive formulation)**.**
Adding (5) and (6) to the problem of Section 4.1 requires the marginal-utility promise $\hat{M}$ as an additional state. Neither constraint involves u, w, or w_2, so the envelope condition, the costate equation, and the boundary conditions $\psi(0) = \psi(\infty) = 0$ are unchanged, and $\partial V_t / \partial \hat{M} = -\vartheta_t$ with $\vartheta_0 = 0$.

Proof. See Appendix C.1. □

Three statistics organize the results, alongside the normalized shadow price $\psi U_{c,t}/(\theta f_t)$ of the local incentive constraint. All are evaluated at the constrained allocation with $y = \theta l$.

$$\Gamma_{c,t} \equiv (1 - \bar{\tau}_t^y)\, \frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{y_t} \qquad \Lambda_t \equiv \frac{\Gamma_{c,t}}{1 - \bar{\tau}_t^y} = \frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{(1 - \bar{\tau}_t^y)\, y_t} \qquad \Delta_t \equiv (1 - \bar{\tau}_t^y)\, \frac{\sigma_t}{c_t} + \frac{\epsilon_t - 2\gamma_{c,t}}{y_t}$$

$\Gamma_{c,t} = \partial \Gamma / \partial c$ is the response of the implied net-of-tax rate to the numéraire, positive when leisure is a normal good, so that delivering consumption depresses labor supply (Appendix C.2). Λ_t is the same object measured per unit of net-of-tax earnings rather than per unit of gross earnings; it is the drag a marginal unit of consumption puts on labor supply. Δ_t is the response of the implied rate along a utility-compensated increase in consumption and labor. It is proportional to the second-order condition of the agent's hours choice at the frozen rate, so $\Delta_t > 0$ wherever the schedule decentralizes an interior labor choice, which I assume throughout.

The two constraints enter every first-order condition additively, and each is priced by its own multiplier. Write

$$\eta_t^s(\theta^t) \equiv \left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}_t^s)\, \beta R^2} \right] U_{c,t}(\theta^t) \tag{7}$$

for the resource value of relaxing the intertemporal restriction at a node, so that η_t prices the labor schedule and η_t^s the savings schedule. One combination of the two carries the whole departure from Pigouvian pricing under weak separability. The labor multiplier reaches the carbon price a second time, through the Corlett-Hague term, and that channel closes when the two goods are equally complementary with labor.

Definition 5.1 (net shadow price of the given income tax).

$$\Xi_t \equiv \eta_t \, (1 - \bar{\tau}_t^y) - \eta_t^s$$

Lemma 5.2 (the two shadow prices).
At every history,

$$\eta_t \, \Delta_t = \frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} - A_t \, \frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s \, \Lambda_t$$

with $A_t = 1 + \epsilon_t - \gamma_{c,t}$. Given η_t^s, the labor multiplier is pinned node by node: η_t is positive where the schedule overtaxes labor relative to the virtual wedge $A_t \frac{\psi U_c}{\theta f}$ corrected for the savings error, negative where it undertaxes, and zero where the two cross.

Proof. See Appendix C.3. □

The lemma is the section's engine, and its reading changes from Section 4. There, combining the first-order conditions for the numéraire and for labor delivered the labor wedge, $\tau_t^y/(1-\tau_t^y) = A_t \frac{\psi U_c}{\theta f}$. Here the same combination cannot determine the wedge, which is frozen. It determines the shadow price of freezing it. The virtual wedge is not the Section 4 wedge either: ψ solves the same costate equation with a source term modified by both constraints (Appendix C.7), so $\psi U_c/(\theta f)$ carries the hazard shape and the persistence of the ABCD decomposition evaluated in the constrained allocation. Setting $\eta_t \equiv \eta_t^s \equiv 0$, which requires both schedules to coincide with their virtual counterparts at every node, returns every formula of Section 4.

The savings side keeps the planner's inverse-Euler logic, now with both credits inside the delivery cost.

Proposition 5.1 (the intertemporal condition).
Define the delivery adjustment and the net delivery value

$$\chi_t \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t \, \Gamma_{c,t} + \eta_t^s \, \frac{\sigma_t}{c_t} \qquad \mathcal{G}_t \equiv 1 - \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \, \Gamma_{c,t}$$

so that $1 - \chi_t = \mathcal{G}_t - \eta_t^s \sigma_t / c_t$. The generalized inverse Euler equation holds in the adjusted marginal utility $U_{c,t}/(1-\chi_t)$,

$$\frac{1-\chi_t}{U_{c,t}} = \frac{1}{\beta R} \, \mathbb{E}_t \left[\frac{1-\chi_{t+1}}{U_{c,t+1}} \right]$$

Proof. See Appendix C.4. □

The delivery-cost accounting extends the one-good logic by two terms. Delivering a util through the numéraire costs $1/U_{c,t}$ in resources and earns the incentive credit $\gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t}$ when the numéraire complements labor. The labor schedule adds a second charge. The same unit of consumption moves the implied net-of-tax rate by $\Gamma_{c,t}$, and with the rate frozen, labor supply must fall to restore (5); the planner values the induced fall at η_t. The savings schedule adds a third. Raising the marginal utility of the numéraire at a node serves the inherited requirement and tightens the one passed forward, and $\eta^s_t \sigma_t / c_t$ is the net of those two.

Because the intertemporal restriction is a ratio across dates rather than a level within one, its multiplier is pinned by a law of motion rather than by a static condition.

Lemma 5.3 (the savings multiplier's law of motion)**.**
Define the virtual savings wedge by $1 - \tau^{s,*}_t \equiv \mathcal{G}_t / \mathcal{G}_{t+1}$. When the date-$(t+1)$ draw is degenerate,

$$\eta^s_t \frac{\sigma_t}{c_t} - (1 - \bar{\tau}^s_t)\, \eta^s_{t+1} \frac{\sigma_{t+1}}{c_{t+1}} = \mathcal{G}_{t+1} \left(\bar{\tau}^s_t - \tau^{s,*}_t\right)$$

with initial condition $\vartheta_0 = 0$ and terminal value $\eta^s_T = \vartheta_T\, U_{c,T}$. In general the right-hand side carries an additional covariance term, given in Appendix C.4.

Proof. See Appendix C.4. □

The law of motion differs from Lemma 5.2 in what the pricing error determines. A frozen labor wedge is a static restriction, so its gap sets η_t node by node. A frozen savings wedge restricts a ratio of marginal utilities across dates, so its gap injects drift into η^s_t and the level is tied down only at the two ends of life, $\vartheta_0 = 0$ at birth and $\eta^s_T = \vartheta_T U_{c,T}$ at the close. One consequence is immediate and it constrains the whole exercise: η^s cannot be one-signed.

Lemma 5.4 (the savings multiplier changes sign)**.**
If $\eta^s_t \geq 0$ at every node, or $\eta^s_t \leq 0$ at every node, then $\vartheta \equiv 0$ and $\eta^s \equiv 0$. A savings schedule away from its virtual counterpart therefore produces a multiplier that takes both signs over the life cycle.

Proof. See Appendix C.4. The argument uses only the chain $\vartheta_{t+1}(\theta^t) = (1 - \bar{\tau}^s_t)\, \beta R^2 \left[\vartheta_t - \eta^s_t / U_{c,t}\right]$ and its boundary conditions, neither of which involves η_t. □

Now the carbon price. It is the only margin left free, and one statistic summarizes what the two frozen margins do to it.

Proposition 5.2 (the carbon price under a given income tax)**.**
Let $\mathcal{I}_t$ be the income-effect gap (3), positive exactly when the dirty good is normal, evaluated at the endogenous price $1 + \tau^q_t$. At every history,

$$\frac{\tau^q_t - \mathrm{SCC}_t}{1 + \tau^q_t} = (\gamma_{c,t} - \gamma_{q,t}) \left(\frac{\psi U_{c,t}}{\theta f_t} + \frac{\eta_t}{y_t} \right) - \Xi_t \, \mathcal{I}_t$$

where the social cost of carbon keeps its discounted-damage form with the damage weight

$$1 + (\xi_{t+j} - \gamma_{c,t+j}) \left(\frac{\psi U_{c,t+j}}{\theta f_{t+j}} + \frac{\eta_{t+j}}{y_{t+j}} \right) + \Xi_{t+j} \, \frac{\sigma_{t+j} + \xi_{c,t+j}}{c_{t+j}}$$

at each future node, with $\xi_{c,t} \equiv U_{Sc,t} c_t / U_{S,t}$ the complementarity of the pollutant with the numéraire.

Proof. See Appendices C.5 and C.6. □

Corollary 5.1 (separation fails through one statistic)**.**
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$,

$$\frac{\tau_t^q - \mathrm{SCC}_t}{1 + \tau_t^q} = -\,\Xi_t \, \mathcal{I}_t$$

so when the dirty good is normal, $\operatorname{sign}(\tau_t^q - \mathrm{SCC}_t) = -\operatorname{sign}(\Xi_t)$. The carbon price lies below the social cost of carbon where the given income tax is jointly too tight, $\Xi_t > 0$, and above it where the tax is jointly too loose.

The two errors reach the carbon price through the same channel and can cancel. Delivering the numéraire is what both schedules distort: the labor constraint makes a unit of consumption drag labor supply down, priced at $\eta_t(1 - \bar{\tau}_t^y)$ per unit of the drag, and the savings constraint makes the same unit move the marginal-utility requirement, priced at $-\eta_t^s$. The planner tilts the assigned bundle toward the good that carries less of the net drag, and the relative price of the dirty good moves with it. Whether the tilt raises or lowers the carbon price depends on the net, not on either schedule separately.

Corollary 5.2 (accidental separation)**.**
Under weak separability, the carbon price equals the social cost of carbon at any node where $\eta_t\,(1 - \bar{\tau}_t^y) = \eta_t^s$, and at every history if the equality holds everywhere. Exact Pigouvian pricing therefore survives an income tax that is misset on both margins, provided the two errors are misset in offsetting proportions.

Proof. Immediate from Corollary 5.1 at $\Xi_t = 0$. □

That the deviations can cancel is not a curiosity, because the two multipliers have different shapes. By Lemma 5.2 the labor multiplier is a within-date object: it takes whatever sign the schedule's error takes at each node and can keep that sign for the whole working life. By Lemma 5.4 the savings multiplier must change sign across dates. The labor error therefore sets the level of the carbon-price deviation within a date, and the savings error tilts it across dates. A schedule that overtaxes labor throughout, paired with a schedule that overtaxes savings, gives a carbon price held below the social cost of carbon by the labor error and tilted around that level by the savings error, with the tilt running from above to below as the cohort ages (Appendix C.4). Which force dominates at a given age is a quantitative question, and Section 6 answers it at US parameters.

The tilt has an exact aggregate measure. When the savings schedule does not vary with the type, taking unconditional expectations in (6) and iterating forward gives

$$\mathbb{E}_0[U_{c,t}] = \prod_{j=0}^{t-1} \frac{1}{\left(1-\bar{\tau}_j^s\right)\beta R}\ \mathbb{E}_0[U_{c,0}] \tag{8}$$

so the schedule pins the growth rate of the population mean of marginal utility at every date, whatever else the planner does, and leaves its level free. This is the allocation-side counterpart of the multiplier's level being tied down only at the two ends of life. At the calibration of Section 5.1.1, $\beta R = 1$ and $\bar{\tau}^s \approx 0.009$: mean marginal utility then compounds at $\bar{\tau}^s/(1-\bar{\tau}^s) \approx 0.9\%$ per period and rises by about seventy percent over the sixty-year life of the quantitative model. A frozen savings wedge does not so much distort the intertemporal margin date by date as bend the whole path of marginal utility. The carbon-price tilt of Corollary 5.1 prices that bending date by date.

Lemma 5.4 disciplines the sign of η_t^s, not the size of ϑ_t. The chain (28) compounds the promise multiplier at $(1-\bar{\tau}_t^s)\,\beta R^2$ per period, so ϑ_t grows with the accumulated bending in (8) and can end the life cycle orders of magnitude above the per-period gap $\bar{\tau}_t^s - \tau_t^{s,*}$, even when the schedule sits close to its virtual counterpart at every date. The formulas price η_t^s, a difference of adjacent multipliers that nets out the accumulated level; by Lemma 5.3 its drift is of the order of that gap. A large promise multiplier and a small carbon-price deviation therefore coexist: ϑ_t measures how far the constraint has bent the path of marginal utility, η_t^s how much bending the current date adds.

Retirement separates the two forces cleanly.

Corollary 5.3 (retirement)**.**
Suppose agents retire at a date $T_R \le T$. From T_R on there is no labor to report, so $\psi \equiv 0$ and (5) has no content, giving $\eta_t \equiv 0$ and $\Xi_t = -\eta_t^s$. Under weak separability the retiree's carbon price satisfies

$$\frac{\tau_t^q - \mathrm{SCC}_t}{1+\tau_t^q} = \eta_t^s\,\mathcal{I}_t$$

so the deviation is the savings error alone. Retirees face pure Pigouvian pricing when the carbon price is chosen anonymously, Corollary 5.5 below; that result survives a misset labor schedule and fails against a misset savings schedule.

Proof. See Appendix D.12 for $\psi \equiv 0$; the rest is Corollary 5.1 at $\eta_t = 0$. $\square$

The asymmetry is informative. A misset labor schedule distorts the allocation of the numéraire across types within a date, so its damage dies with the labor margin. A misset savings schedule distorts the allocation of the numéraire across dates within a life, so it operates through retirement, where the virtual savings wedge is zero and any nonzero frozen wedge keeps injecting error. Retirees' carbon price deviates from marginal damages for reasons that have nothing to do with screening their labor.

Two limits close the characterization. Setting both schedules flat and choosing their levels optimally recovers the Ramsey instruments.

Proposition 5.3 (the optimal flat pair)**.**
Let both schedules be flat, $\bar{\tau}_t^y(\theta) = \bar{\tau}_t^y$ and $\bar{\tau}_t^s(\theta) = \bar{\tau}_t^s$, with their levels chosen optimally at every date. The optimal levels satisfy

$$\mathbb{E}_0[\eta_t] = 0 \qquad \text{and} \qquad \mathbb{E}_0[\vartheta_{t+1}\, U_{c,t}] = 0$$

Equivalently, for the labor rate,

$$\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y} = \frac{\mathbb{E}_0\left[s_{yy,t}\left(A_t\,\frac{\psi U_{c,t}}{\theta f_t} - \eta_t^s\,\Lambda_t\right)\right]}{\mathbb{E}_0[s_{yy,t}]} \qquad s_{yy,t} \equiv \left.\frac{\partial y}{\partial\,(1-\bar{\tau}_t^y)}\right|_u = \frac{1}{(1-\bar{\tau}_t^y)\,\Delta_t}$$

with $s_{yy,t}$ the compensated response of earnings to the net-of-tax rate, and $\mathbb{E}_0[\eta_t^s] = \mathbb{E}_0[\vartheta_t U_{c,t}]$ for the savings rate, which equals zero only at $t = 0$.

Proof. See Appendix C.8. $\square$

The flat labor rate is the compensated-earnings-response-weighted average of the virtual Mirrleesian wedges, net of the savings error at each node. It mirrors the optimal anonymous carbon price of Proposition 5.5 below, with s_{yy} in place of $|s_{qq}|$: a single rate averages the schedule the planner would set node by node, weighting each node by how elastically its earnings respond. The savings condition prices only what the date-t rate controls, namely the requirement passed forward, so the multiplier averages to zero at birth and afterward drifts with the inherited term. Even at the optimal flat pair, Ξ_t is generically nonzero node by node, so the carbon-price deviations of Corollary 5.1 do not vanish. Choosing the levels optimally disciplines their average, not their profile.

At the other limit, $\eta_t \equiv \eta_t^s \equiv 0$ returns Section 4. Between the two, the family sharpens the reading of Kaplow (2012). His argument frees Pigouvian pricing from any optimality requirement on the income tax: whatever the schedule, an environmental reform can be packaged with a type-by-type income-tax adjustment that keeps the distribution whole, so moving to marginal-damage pricing is a Pareto improvement. Freezing both income-side margins forbids the package. The compensating adjustment is exactly the instrument the constraints remove, and $-\Xi_t \mathcal{I}_t$ prices the missing degree of freedom. The rigidity reaches the social cost of carbon itself: its fiscal factor now carries the schedules' errors, and even the $U = V(h(c, q, S), l)$ benchmark, which made the second-best social cost of carbon purely Pigouvian in Section 4.3, no longer does so unless $\Xi_t = 0$.

The question of whether a controlled margin should be priced at social marginal cost when another margin carries a distortion the planner cannot remove predates the Mirrleesian literature, where Guesnerie (1979) calls it the case of irreducible distortions. Boadway and Harris (1977) ask it for proportional pricing in a control sector given fixed distortions elsewhere and reach a negative verdict. Piecemeal policy requires that the uncompensated demand derivatives of every other sector with respect to control-sector prices vanish, a condition of which weak separability is only a necessary consequence, and which holds universally across partitions only for the linear logarithmic class. Even where it holds, the proportionality factor is not unity, so control-sector prices are proportional to marginal cost rather than equal to it. Jewitt (1981) settles the single-consumer case at implicit separability, and Blackorby et al. (1991) show that with many consumers the requirement tightens to a two-sector representation of community preferences, for which implicit separability of every consumer's preferences is neither necessary nor sufficient. Their conclusion is that single-consumer results give a misleadingly optimistic view of when piecemeal policy is valid.

Corollary 5.1 carries that conclusion into a Mirrleesian environment and sharpens it. In the partition this subsection studies, the labor margin is the only distorted one, and Boadway and Harris (1977) note that a single distorted margin makes the problem trivial when the numéraire is in fixed supply and lump-sum transfers are available: proportional pricing in the control sector then attains the first best. Private information destroys the triviality. The frozen schedules bind type by type, the transfer the planner still controls is a scalar rather than a schedule, and the residue is $-\Xi_t \mathcal{I}_t$. Weak separability kills the Corlett-Hague term and leaves that residue standing. Golosov et al. (2014a) reach related restricted-instrument comparisons from the variational side, characterizing optimal systems within classes such as age-independent, linear, or separable.

5.1.1 Calibrating the given schedules to the US tax system

The schedules the exercise needs are not hypothetical. For the labor margin, take the two-parameter tax and transfer function of Bénabou (2002) and Heathcote et al. (2017),

$$T_t(y) = y - \lambda_t^T\, y^{1-p_t} \qquad \Longrightarrow \qquad 1 - \bar{\tau}_t^y(y) = \lambda_t^T\,(1 - p_t)\, y^{-p_t} \tag{9}$$

where p_t indexes progressivity and λ_t^T the level. Heathcote et al. (2017) estimate $p = 0.181$ for

US households aged 25 to 60 by regressing log post-government on log pre-government earnings in the PSID, with an R^2 of 0.91. Heathcote and Tsujiyama (2021) pair it with $\lambda^T = 0.840$. Progressivity is flat in age. Re-estimating with both parameters conditioned on age, Heathcote et al. (2020) find no significant age dependence in p, so an age-invariant progressivity is a feature of the US system rather than an approximation error. Their null covers the slope alone; the level λ_t^T is left free to vary with age.

Indexing the schedule on earnings rather than on type changes one statistic and nothing else.

Lemma 5.5 (an earnings-indexed schedule)**.**
Let the frozen labor wedge follow (9), so that $\bar{\tau}_t^y$ depends on the type only through assigned earnings $y_t(\theta^t)$. Then $\Gamma_{c,t}$, $\Gamma_{q,t}$, and $\Gamma_{S,t}$ are unchanged, while

$$\Gamma_{l,t} = (1-\bar{\tau}_t^y)\,\frac{\epsilon_t - \gamma_{c,t} + p_t}{l_t} \qquad \text{and} \qquad \Delta_t = (1-\bar{\tau}_t^y)\,\frac{\sigma_t}{c_t} + \frac{\epsilon_t + p_t - 2\gamma_{c,t}}{y_t}$$

Every result of this subsection holds verbatim with ϵ_t replaced by $\epsilon_t + p_t$ inside Δ_t alone. Since $p_t > 0$ under a progressive schedule, Δ_t rises and the interior-hours requirement $\Delta_t > 0$ is weaker than under a type-indexed schedule.

Proof. See Appendix C.9. □

Freezing a tax function rather than a type-indexed wedge schedule is the more faithful exercise and it strengthens the reading of Corollary 5.1. A tax code conditions on earnings, so a planner constrained by (9) cannot make the type-by-type adjustment Kaplow (2012) relies on even in principle, whereas a planner constrained by a type-indexed schedule is denied it only by assumption.

For the savings margin there is no counterpart to (9) with the same standing, since the fitted tax functions of Heathcote et al. (2017) and Guner et al. (2014) are functions of earnings and imply nothing about the intertemporal margin. The practical route is a linear capital income tax. With private saving at gross return R taxed at rate τ_t^k on the interest component, the household's Euler equation and the definition of the savings wedge give

$$1 - \bar{\tau}_t^s = \frac{1 + (R-1)\left(1 - \tau_t^k\right)}{R} \tag{10}$$

so a single number fixes the whole schedule. With $\beta R = 1$ as in Section 6, (10) collapses to $\bar{\tau}_t^s = (1-\beta)\,\tau_t^k$: at $\beta = 0.95$ the frozen wedge is one twentieth of the capital income tax rate, because the tax falls on the interest component alone. A wedge of under one percent of the gross return is therefore a tax of tens of percent on the margin it actually distorts, and I report the schedule in tax units throughout to keep that visible.

The rate (10) asks for is a marginal one, since it comes from the household's Euler equation on a marginal unit of saving. Burnham and Carloni (2022) put the US effective marginal rate on capital income near 0.18, and I take that as the benchmark. The aggregate effective rates computed by the method of Mendoza et al. (1994), which put τ^k near 0.36 for the US (Trabandt and Uhlig 2011), are averages: capital tax revenue divided by capital income in the national accounts. They coincide with the marginal rate in the linear-tax representative-agent settings they were built for, which is precisely the identification a Mirrleesian model drops. They do capture corporate and property taxes that the marginal measure omits, so I carry 0.36 as an upper bound in the sensitivity analysis of Section 6.4 and use 0.18 everywhere else. If anything the benchmark is generous, since a large share of US household saving sits in tax-deferred and Roth accounts whose marginal intertemporal rate is near zero.

At $\tau^k = 0.18$ the frozen wedge is $\bar{\tau}^s = 0.009$, nine tenths of a percent of the gross return per year. Section 6 computes the virtual savings wedge, the one the unconstrained mechanism would set, as falling from 0.0056 to 0.0002 over the working life; in the same tax units that is a capital income tax of about 11% at labor-market entry declining to nearly zero by retirement. The US system therefore overtaxes the intertemporal margin at every age, and by more as the cohort ages, at any τ^k above 0.11 — which every measure of the US rate clears.

The exercise inherits one sensitivity worth stating before the numbers. Whether the given schedules sit far from their virtual counterparts, and therefore whether Ξ_t is large, depends on the Pareto weights. Under a utilitarian objective Heathcote and Tsujiyama (2021) find a welfare gain of about 2% of consumption from replacing the estimated US schedule with the fully optimal one, so the schedule is far from optimal and the deviations of Corollary 5.1 are substantial. Under the Pareto weights that rationalize observed US progressivity within their parametric class, the same gain falls to about 0.05%, the schedule is nearly optimal, and the deviations nearly vanish. I report the utilitarian case, matching the planner of Section 3, and the empirically weighted case as a lower bound.

5.2 A given carbon price

Real carbon prices are legislated rather than derived, anonymous rather than personalised, and set without reference to the income tax. This subsection holds the carbon wedge to an exogenous schedule $\bar{\tau}^q_t(\theta)$ and asks how the income-side wedges should respond to it. The restriction that the price be a common rate, which is the form every carbon tax in force actually takes, is one member of the family: the schedule is flat in type. The problem in which the planner still chooses that common rate, rather than inheriting it, is a second member. Both are read off the same multiplier.

The planner solves the problem of Section 4.1 subject to

$$\frac{U_q\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} = 1 + \bar{\tau}_t^q(\theta) \equiv \bar{p}_t(\theta), \tag{11}$$

with multiplier $\eta^q(\theta) f_t(\theta|\theta_-)$. The constraint is static, so the multiplier is pinned by a static condition, and the regularity it needs is the second-order condition $\mathcal{D}_t < 0$ below, guaranteed by Assumption 3.1.

Three demand statistics organise the results, alongside the income-effect gap $\mathcal{I}_t$ of (3) and the normalised shadow price $\psi U_{c,t}/(\theta f_t)$ of the local incentive constraint. Let $\mathcal{D} \equiv U_{qq} - 2\bar{p}\,U_{cq} + \bar{p}^2 U_{cc} < 0$ be the curvature of utility along the budget line, negative at every price by Assumption 3.1. Define

$$m_q \equiv \bar{p}\,\frac{\partial q^d}{\partial x} = \bar{p}^2\,|s_{qq}|\,\mathcal{I}_t \qquad s_{qq} \equiv \left.\frac{\partial q^d}{\partial \bar{p}}\right|_u = \frac{U_c}{\mathcal{D}} < 0$$

the marginal budget share of the dirty good and its compensated own-price response (Appendix D.3). Two more objects carry the pricing error itself:

$$\varpi_t \equiv \frac{\bar{\tau}_t^q - \mathrm{SCC}_t}{\bar{p}_t} \qquad \text{and} \qquad \varpi_t^* \equiv (\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}$$

ϖ_t is the screening component of the given price, the ad-valorem wedge net of the part that pays for damages. ϖ_t^* is the wedge the planner would set at that node if the instrument were free, which is the commodity wedge of Proposition 4.1. The whole subsection turns on the gap between them.

When the schedule is flat in type, the restricted economy collapses to a one-good economy in expenditure.

Lemma 5.6 (anonymity and the one-good reduction)**.**
Suppose $\bar{p}_t(\theta) = \bar{p}_t$ for every type. An allocation satisfies (11) if and only if it assigns each type the private split of some pair (x, y) of expenditure and output at the common price. With indirect utility

$$\tilde{U}(x, l; \bar{p}) \equiv \max_{c,q \geq 0} U(c, q, l) \qquad \text{s.t.} \qquad c + \bar{p}\,q \leq x,$$

the split is unique since $\mathcal{D} < 0$, and incentive compatibility in the restricted economy is that of a one-good economy in (x, y) with period utility $\tilde{U}(x, y/\theta; \bar{p}_t)$. The envelope condition is unchanged,

$$\dot{u}(\theta) = -\frac{\tilde{U}_l\,l}{\theta} + \beta\,w_2(\theta) = -\frac{U_l\,l}{\theta} + \beta\,w_2(\theta)$$

Proof. See Appendix D.2. □

The reduction is what makes an anonymous carbon price tractable: the planner assigns expenditure and the agent splits it, so the two-good screening problem is a one-good screening problem in a reshaped utility. It fails for a schedule that varies with type, since then no single budget line rationalises the assigned bundles. Section 6 solves the restricted mechanism through exactly this reduction.

Lemma 5.7 (the price of a given carbon wedge)**.**
At every history,

$$\eta_t^q = \bar{p}_t \, |s_{qq,t}| \, (\varpi_t^* - \varpi_t)$$

The multiplier is the pricing error converted into resources by the compensated substitution response: $\eta_t^q > 0$ where the schedule underprices the dirty good relative to the node's Corlett-Hague level, $\eta_t^q < 0$ where it overprices it, and $\eta_t^q = 0$ where the two cross.

Proof. See Appendix D.4. □

Nodes at which the consumer substitutes away from the dirty good easily convert a given pricing error into a large multiplier, which is the deadweight-loss weighting of the Corlett-Hague tradition. Every result below is this multiplier read at a different schedule.

Proposition 5.4 (wedges under a given carbon price)**.**
At every history the labor wedge and the savings adjustment are

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t \frac{\psi U_{c,t}}{\theta f_t} + \eta_t^q \, \bar{p}_t \left[\mathcal{I}_t - \frac{\gamma_{c,t} - \gamma_{q,t}}{(1-\tau_t^y)\, y_t} \right] \qquad \chi_t^q \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q \, \bar{p}_t \, \mathcal{I}_t$$

with the generalised inverse Euler equation holding in $U_{c,t}/(1-\chi_t^q)$ and, to second order,

$$\tau_t^s \approx \mathrm{Var}_t(\ln U_{c,t+1}) + \left(\chi_t^q - \mathbb{E}_t \, \chi_{t+1}^q\right) + \mathrm{Cov}_t\!\left(\chi_{t+1}^q, \, \ln U_{c,t+1}\right)$$

The social cost of carbon keeps its discounted-damage form with the damage weight

$$1 + (\xi - \gamma_c) \, \frac{\psi U_c}{\theta f} + \eta^q \left[\bar{p} \, \frac{\sigma + \xi_c}{c} - \frac{\sigma_{cq} + \xi_q}{q} \right]$$

at each future node, where $\xi_{q,t} \equiv U_{Sq,t} \, q_t / U_{S,t}$ is the complementarity of the pollutant with the dirty good.

Proof. See Appendices D.5, D.6, and D.11. □

The labor wedge the proposition reports is the income-tax wedge alone, and it is not what the screening condition pins down. When an agent earns a marginal unit of output on a compensated path, the planner collects τ_t^y through the income margin and $\bar{\tau}_t^q\, dq^d/dy|_u$ through the commodity margin, where $dq^d/dy|_u \equiv (1-\tau^y)\partial_x q^d + \partial_l q^d/\theta$ is the utility-compensated response of dirty-good demand to earnings. Their sum

$$\Omega_t \equiv \frac{\tau_t^y}{1-\tau_t^y} + \frac{\bar{\tau}_t^q}{1-\tau_t^y}\left.\frac{dq_t^d}{dy}\right|_u = \tilde{A}_t\,\frac{\psi U_{c,t}}{\theta f_t}$$

is the comprehensive wedge the mechanism sets, the dynamic counterpart of the effective labor wedge of Jacobs and Boadway (2014), and $\tilde{A}_t$ is built on composite statistics that average the two goods' curvatures by their marginal expenditure shares (Appendix D.7). One property of those composites is worth stating in the text: the effective inverse Frisch elasticity satisfies $\tilde{\epsilon} \le \epsilon$, a Le Chatelier property, because the agent's freedom to re-optimise the split cushions labor shocks, so delivering expenditure is a weakly better hedge than delivering a rigid bundle. The comprehensive wedge inherits the full $ABCD$ structure of Proposition 4.1 in those composites (Appendix D.8), and Proposition 5.4 then dictates how Ω_t splits between the income-tax wedge and the commodity-revenue term.

The correction to the income-tax wedge has two parts of opposite sign. The expenditure part $(1-\tau^y)\partial_x q^d$ is positive for a normal good. The labor part $\partial_l q^d/\theta$ is the statistic of Christiansen (1984), negative when the dirty good is a labor substitute, since working more at fixed expenditure shifts spending away from it. When the labor-substitute part dominates, discouraging labor is fiscally cheaper than it looks, because the agent who works less spends more on the taxed good and part of the forgone output returns as commodity revenue.

Under weak separability the whole apparatus collapses to a single statistic.

Corollary 5.4 (the income tax collects the carbon the carbon price misses)**.**
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, we have $\varpi_t^* \equiv 0$, so $\eta_t^q \bar{p}_t = -\bar{p}_t^2\,|s_{qq,t}|\,\varpi_t$ and the corrections collapse to the marginal budget share times the uncollected carbon price:

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t\,\frac{\psi U_{c,t}}{\theta f_t} + m_{q,t}\,\frac{\mathrm{SCC}_t - \bar{\tau}_t^q}{\bar{p}_t} \qquad\qquad \chi_t^q = \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} - m_{q,t}\,\frac{\mathrm{SCC}_t - \bar{\tau}_t^q}{\bar{p}_t}$$

A carbon price given at the social cost of carbon is harmless: $\bar{\tau}_t^q = \mathrm{SCC}_t$ gives $\varpi_t = 0$, hence $\eta_t^q = 0$, and returns every wedge to its Section 4 form. A price given below it raises the labor wedge above the virtual level; one given above it lowers the labor wedge.

Proof. See Appendix D.5. □

The mechanism is embedded carbon. A marginal compensated dollar of earnings is spent, a fraction $m_{q,t}$ of it on the dirty good, and each of those units carries uncollected marginal damage $(\mathrm{SCC}_t - \bar{\tau}^q_t)/\bar{p}_t$. The income tax adds that charge, taxing the carbon content of marginal spending that the carbon price misses. The savings side applies the same charge intertemporally: delivering consumption at date $t+1$ rather than t carries the uncollected damage of the extra dirty consumption it drags along, so the planner taxes savings where underpricing worsens over the life cycle and subsidises it where pricing improves. A legislated price that stays flat while the social cost of carbon grows is the worsening case. Both corrections persist into retirement, since the pricing error is a within-period object that never expires. This is the Mirrleesian counterpart of Hänsel et al. (2022), who find that a mandated uniform carbon tax reshapes the optimal income tax; here the reshaping has a closed form, and it is the marginal budget share times the pricing error.

5.2.1 When the planner chooses the common rate

Suppose now that the price is anonymous but not inherited: the schedule is flat, $\bar{p}_t(\theta) = \bar{p}_t$, and the planner chooses its level at every date. This is the problem a government solves when it may legislate the carbon tax but not personalise it, and it is the configuration every carbon tax in force approximates.

Proposition 5.5 (the optimal anonymous price)**.**
The optimal level sets the population average of the multiplier to zero at every date, $\mathbb{E}_0[\eta^q_t] = 0$. Equivalently, the screening component is the substitution-weighted average of the virtual node wedges,

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}|\ \varpi^*_t]}{\mathbb{E}_0[|s_{qq,t}|]}$$

and the optimal price splits additively into a corrective and a screening part,

$$1 + \tau^q_t = 1 + \mathrm{SCC}_t + \bar{\tau}^{\text{screen}}_t \qquad \frac{\bar{\tau}^{\text{screen}}_t}{1+\tau^q_t} = \frac{\mathbb{E}_0\left[|s_{qq,t}|\,(\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[|s_{qq,t}|]}$$

To second order in the dispersion of ϖ^*_t, the date-t resource-equivalent loss from charging this common rate rather than the node-specific schedule of Proposition 4.1 is

$$\mathrm{Loss}_t \approx \frac{1}{2}\,\bar{p}^3_t\ \mathbb{E}_0[|s_{qq,t}|]\ \mathrm{Var}^{|s|}_t(\varpi^*_t)$$

with $\mathrm{Var}^{|s|}_t$ the $|s_{qq,t}|$-weighted cross-sectional variance over date-t histories, minimised at the level above.

Proof. See Appendices D.9, D.11, and D.10. □

A single instrument averages the schedule the planner would set node by node, weighting each history by how easily the consumer can substitute away from the dirty good there.[3] The condition is implicit: the weights and the node wedges are themselves evaluated at the restricted allocation, so the level is characterised as a fixed point rather than in closed form. The cost of not personalising is the substitution-weighted cross-sectional variance of the wedge the planner forgoes. Because that wedge tracks the labor wedge by Proposition 4.1, its dispersion is first-order in the dispersion of τ^y, and what the tracking is worth is what anonymity gives up. Every object in the formula is a demand-side or tax-side statistic, so this is a sufficient-statistics answer to the cost-of-mispricing question, and the near-sufficiency of a flat carbon price documented in Section 6 is precisely this variance being small.

Corollary 5.5 (when the screening component vanishes)**.**
The screening component of the optimal anonymous price is zero, and the price is exactly Pigouvian, whenever $\varpi_t^* \equiv 0$ across the date-t cross-section. Two cases deliver this. Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$ pointwise, so $\varpi_t^* \equiv 0$ at every history and the restricted optimum coincides with the unconstrained optimum of Corollary 4.1. From a retirement date T_R on there is no labor to report, so $\psi \equiv 0$ and again $\varpi_t^* \equiv 0$: the screening component of the carbon price dies with the labor margin and retirees face pure Pigouvian pricing. More generally the restriction is costless at date t if and only if ϖ_t^* is constant across the date-t cross-section, since only its dispersion enters the loss.

Proof. See Appendix D.12 for $\psi \equiv 0$; the rest is immediate from Proposition 5.5. □

The additive split is the section's main reading, and it is a statement about which margin the restriction binds. The corrective part of the optimal dirty-good price was uniform already at the unconstrained optimum, since a ton of emissions carries the same social cost whoever emits it and the emissions margin cannot screen. What anonymity constrains is only the Corlett-Hague screening part, forced from a history-dependent schedule down to a substitution-weighted scalar. A legislated uniform carbon price therefore sacrifices nothing on the corrective margin: the whole cost of the restriction falls on screening, which the loss formula measures. The level of the social cost of carbon is nonetheless co-determined with the constrained allocation, its damage weight carrying the multiplier node by node through Proposition 5.4; this reprices carbon in the constrained second best but is not a loss on the corrective margin, which has

[3] The construction parallels the many-person Ramsey rule of Diamond (1975) in collapsing a heterogeneous population into a single linear-tax condition: there the optimal excise rule, here $\mathbb{E}_0[\eta_t^q] = 0$. The weighting differs in kind. Diamond aggregates through the distributional characteristic, the covariance of demand with the net social marginal utility of income; here the weights are pure compensated substitution responses, with the redistributive content carried inside the node wedges being averaged rather than in the weights.

no cross-sectional dispersion to forgo. This is the separation of Section 4.4 restated for the instrument governments actually use.

Between the two readings of this subsection lies the whole family. A price given at the social cost of carbon leaves every wedge at its Section 4 value. A price given anywhere else routes the uncollected damage into the income tax at the rate Corollary 5.4 sets. And a price chosen anonymously but optimally sits at the zero-average point, $\mathbb{E}_0[\eta_t^q] = 0$, where the errors cancel in the cross-section without vanishing node by node.

6 Quantitative analysis

How large is the Mirrleesian correction to the carbon price, how much welfare does it deliver over a flat carbon tax, and how much of the answer survives an income tax the planner did not choose? I answer all three by solving the model's optimal mechanism globally, calibrated to US data, rather than evaluating the wedge formulas at borrowed sufficient statistics.

The exercise runs in two worlds that differ in one respect. In the first the labor and savings wedges are set optimally alongside the carbon price, which is Section 4. In the second they are frozen at the schedules the US tax system implies, which is Section 5.1. Everything else is held fixed: the same preferences, the same calibration, the same solver family, and the same simulated skill histories. Every result below is a comparison across those two worlds, and the contrast is sharper than the level of either.

The solver extends the dual-costate recursion of Farhi and Werning (2013) and Golosov et al. (2016) to two consumption goods, a clean numéraire and a polluting energy good, and prices the externality as a constant marginal damage e entering the planner's resource constraint, so the social producer price of energy is $1 + e$. This is the single-cohort reading of the social cost of carbon in Section 4.3: for a cohort too small to move the pollution stock, the discounted damage weights Φ_j collapse to one constant per-unit cost, no pollution-stock state is carried, and the Pigouvian benchmark is $\tau^q = \mathrm{SCC} = e$. The pollution-stock dynamics are a separate exercise.

6.1 Model, solvers, and calibration

The life-cycle block is Farhi and Werning (2013)'s, replicated and validated before the two-good extension. Agents live sixty years, work for forty and retire for twenty, so period t is age $24+t$: labor-market entry is at 25, the last working year is 64, and retirement runs from 65 to 84. I report ages throughout; log-skills follow a geometric random walk with innovation variance $\hat{\sigma}^2 = 0.0095$; $\beta = 1/R = 0.95$, so $\beta R = 1$; skills are private information and the incentive constraints are handled by the first-order approach. The planner is utilitarian. Following Farhi and Werning (2013), newborns are ex ante identical, so the period-one cross-section is the first innovation of the random walk: the initial skill dispersion is already the $\hat{\sigma}^2 = 0.0095$ above, there is no separate initial-skill variance to set, and the common initial promised value is pinned by the economy's resource constraint. Each period the agent consumes a clean numéraire and an energy good (motor fuel, electricity, gas, fuel oil), both with producer price one. Period utility is

$$U(c, q, l) = \frac{1}{\rho} \log\left[(1 - b_q)\, c^{\rho} + b_q\, e^{-\varkappa l^{\alpha}}\, (q - \bar{q})^{\rho}\right] - \frac{l^{\alpha}}{\alpha} \qquad \rho \equiv 1 - \frac{1}{\varsigma}$$

a CES aggregate over the clean good and energy net of subsistence $\bar{q}$, inside the log shell that preserves Farhi and Werning (2013)'s balanced-growth homogeneity, with isoelastic labor disutility of curvature $\alpha = 3$ (Frisch elasticity one half). ς is the elasticity of substitution between the goods, b_q the energy weight, and $\bar{q} = 0$ in the homothetic benchmark; $\bar{q} > 0$ is the Stone-Geary necessity variant. The twist $e^{-\varkappa l^{\alpha}}$ makes energy a relative leisure complement: its utility weight falls with labor effort, so $U_{ql} \neq 0$ with the sign, in the notation of Section 3, that makes the complementarity gap $\gamma_c - \gamma_q$ positive. This is the empirically relevant case (West and Williams (2007); Christiansen (1984)), and the one that pushes the optimal carbon tax above Pigouvian.

The two-good calibration is disciplined by four US moments (Table 1). The energy expenditure share is 7% (Consumer Expenditure Survey, 2023). The income elasticity of energy is 0.45, below one because energy is a necessity, from residential-energy demand studies; the homothetic benchmark raises it to one, and the Stone-Geary variant keeps it at 0.45. The own-price elasticity is 0.4, from meta-analyses of energy demand; at a 7% share it pins the substitution elasticity $\varsigma \approx 0.4$. Both are long-run elasticities, the horizon a forty-year model asks for; short-run energy elasticities are roughly half as large, and the welfare numbers scale with the long-run value. The share moment pins the small CES energy weight b_q, and under Stone-Geary the income-elasticity moment implies a subsistence level $\bar{q} \approx 0.04$, about 57% of mean energy spending, so more than half of energy consumption is price-inelastic at the mean. The leisure complementarity $\varkappa$ is set so that, with the externality off, the second-best energy wedge at mid-career is about 10% ad valorem: West and Williams (2007)'s zero-externality gasoline tax, tempered for the household-energy composite, whose labor cross-effects are weaker than motor fuel's. The marginal damage is $e = 0.494$: the EPA's central social cost of carbon of \$190 per ton (U.S. Environmental Protection Agency 2023) times the carbon intensity of household energy spending, 0.0026 tons per producer dollar. Because $p_2 = 1$, the Pigouvian wedge $\tau^q = e$ is a 49% ad valorem tax on energy.

The frozen schedules of the second world are the ones calibrated in Section 5.1.1: the Heathcote et al. (2017) tax function at $p = 0.181$ for the labor margin, and a capital income tax at the US effective marginal rate $\tau^k = 0.18$ for the savings margin, which by (10) implies $\bar{\tau}^s = 0.009$ at $\beta R = 1$. Because the carbon price turns out to be far more sensitive to the savings schedule than to the labor schedule, Section 6.4 solves the model across a grid of τ^k running up to the 0.36 of the aggregate effective-rate measures; everything else in this section is at the benchmark.

Table 1: Calibration

parameter	value	moment or source
life-cycle block (Farhi and Werning 2013)		
discount factor $\beta = 1/R$	0.95	FW13 ($\beta R = 1$)
labor-disutility curvature α	3	Frisch elasticity 0.5
log-skill innovation variance $\hat{\sigma}^2$	0.0095	US male wages, via FW13
working / retirement periods	40/20	FW13
two-good block		
energy expenditure share	0.07	Consumer Expenditure Survey 2023
income elasticity of energy	0.45	residential-energy demand (necessity)
own-price elasticity of energy	0.4	energy-demand meta-analyses
leisure complementarity $\varkappa$	0.25	mid-career zero-externality wedge $\approx$ 10% (West and Williams 2007)
marginal damage e	0.494	\$190/tCO_2 (U.S. Environmental Protection Agency 2023) $\times$ 0.0026 tCO_2/\$
frozen schedules, Section 5.1.1		
labor progressivity p	0.181	Heathcote et al. (2017)
capital income tax τ^k	0.18	effective marginal rate, Burnham and Carloni (2022)
implied savings wedge $\bar{\tau}^s$	0.9% per year	(10) at $\beta R = 1$

Three solvers deliver the results, and each is anchored on the one before it. The *second best* assigns full consumption bundles, so the commodity wedge is unrestricted; this solver requires homothetic preferences. *Linear commodity taxation* layers an anonymous rate τ^q on the optimal nonlinear income and savings mechanism through the exact one-good reduction of Lemma 5.6, which is valid with subsistence and hence usable for the necessity calibration. The *third best* adds the two frozen schedules, with the marginal-utility promise $\hat{M}$ of Lemma 5.1 carried as a third state, and runs either with the commodity margin free or with it restricted to a flat rate. Two agreements discipline the stack. At $\gamma_c = \gamma_q$ the second best sets $\tau^q = e$ at every history and every age, the dynamic Atkinson-Stiglitz result, and coincides with a flat Pigouvian tax to solver precision. With the frozen schedules set at their virtual counterparts the third-best solver returns the second best and its multipliers vanish, which is the $\eta_t \equiv \eta_t^s \equiv 0$ limit of Section 5.1. All welfare comparisons use common random numbers over 60,000 simulated histories, so the same agents face every regime, and every regime pays for its own emissions in the planner's budget at the social price $1+e$ whatever the consumer price: no rung is flattered by unpriced damages.

The incentive compatibility of the first-order-approach solution is verified ex post on the simulated second-best benchmark.

6.2 The carbon wedge over the life cycle

Figure 1 is the paper's central quantitative fact, and the two curves in it are the same object computed under the same calibration.

When the income tax is set optimally, the carbon wedge barely leaves the social cost of carbon. At labor-market entry the labor wedge is near zero, there is no incentive motive yet, and the carbon wedge sits at the Pigouvian level $e = 0.494$. It rises concavely to about 0.63 near retirement, a premium over the Pigouvian gross price that tops out around 9%, and at retirement it falls back to exactly 0.494: with no labor to report the costate vanishes, the screening term dies with it, and retirees face pure Pigouvian pricing (Corollary 5.5). The whole path lies within a tenth of the social cost of carbon.

When the income tax is instead the one the US actually levies, the wedge starts in the same place and then rotates away from it. It opens at 51%, within a point of the social cost of carbon, crosses the Pigouvian level at 28, falls to 22% by 45 and -8% at 64, and reaches a 34% energy *subsidy* at 84. That is a rotation through 85 points of ad-valorem price, against the 14 points the second-best wedge covers over the same sixty years, and its working-life mean is 23%, about half the social cost of carbon. The freeze does not move the level at which the carbon price starts; it bends the whole path.

The premium in the first world is a scaled labor wedge (Table 2). Its first-order value is $\varkappa\, l^{\alpha}\, \tau^{y}/(1-\tau^{y})$, and this identity tracks the solved premium to within 2.1% at every age. The identity sits just below the solved premium throughout, by about 1% through mid-career and by 2.1% at 64, where the labor wedge is flattest and the first-order approximation to it is worst. The non-Pigouvian part of the carbon wedge therefore inherits the labor wedge's level and its life-cycle rise, which is the proportionality result of Section 4.2 made quantitative. The income-side wedges are in turn essentially undisturbed by the second good: the labor wedge runs from 0.02 to 0.36 over the working life against 0.02 to 0.37 in the one-good economy, and the savings wedge falls from 0.56% to 0.02%, the Farhi and Werning (2013) pattern intact. The commodity margin adds a corrective and screening instrument without reshaping the other two, which is the separation showing up in the solved policies.

History dependence separates the two worlds as sharply as the path does. The second-best wedge's cross-sectional standard deviation fans out from 0.1% early in life to 5.5% near retirement, mirroring the spreading of the labor wedge as skill histories diverge. The third-best wedge's reaches 18 points in the early sixties, more than three times as wide, and the bands in Figure 1 separate from mid-career on. Against a given income tax, the price the mechanism would charge depends heavily on which history an agent has drawn.

Table 2: Cross-sectional mean wedges by age over the working life, income tax optimal. The premium is measured against the consumer price $1+\tau^q$, the normalization of (4), so that it and the identity below it are the same object. Measured instead against the Pigouvian gross price $1+e$, it reaches 0.092 at 64. The last two rows carry a fourth decimal because at three the first column reads 0.007 against 0.006, which is a rounding artifact of a 1.4% gap. The identity lies below the premium at every age, by between 1.0% and 2.1%.

age	25	35	45	55	64
labor wedge τ^y	0.021	0.180	0.280	0.341	0.363
carbon wedge τ^q	0.504	0.574	0.609	0.625	0.632
s.d. across histories	0.001	0.005	0.015	0.031	0.055
premium $(\tau^q - e)/(1+\tau^q)$	0.0065	0.0507	0.0714	0.0806	0.0844
identity $\varkappa l^\alpha \tau^y/(1-\tau^y)$	0.0065	0.0501	0.0707	0.0796	0.0826

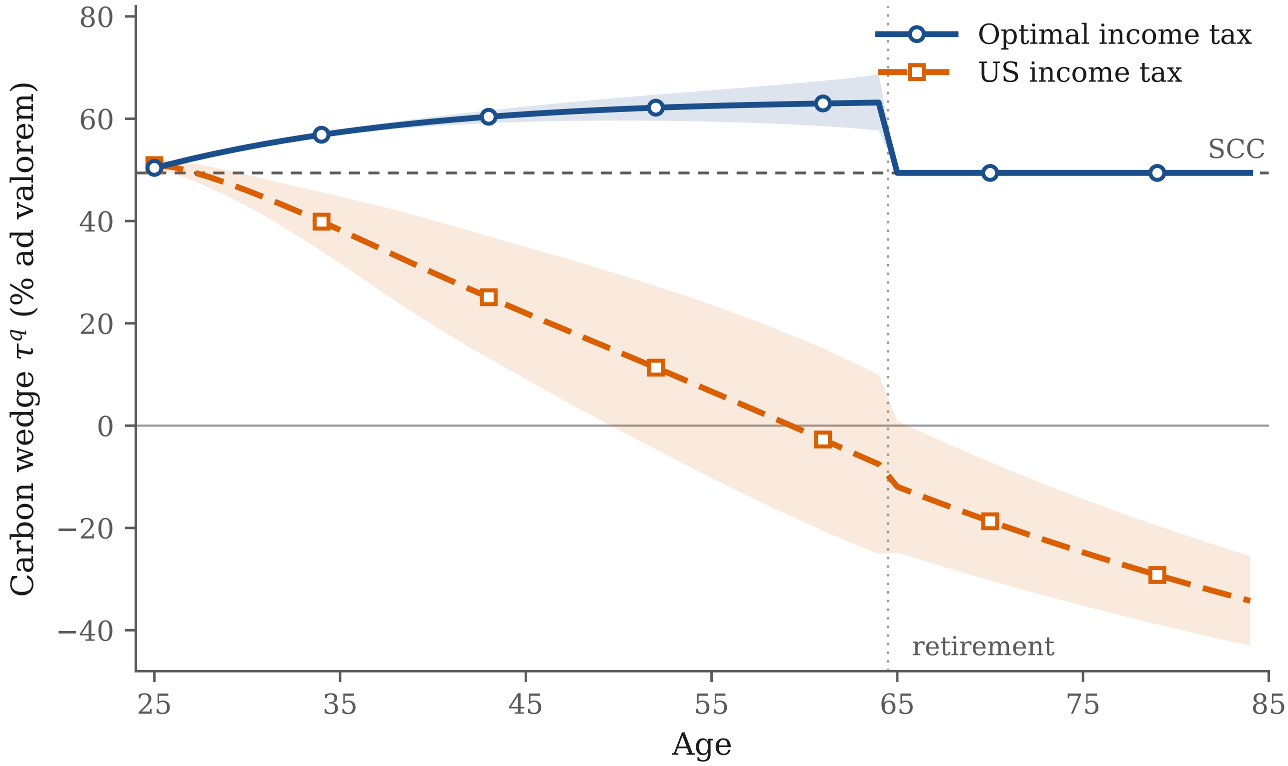

Figure 1: The optimal carbon wedge over the life cycle, benchmark calibration, when the income tax is set optimally alongside it and when it is frozen at the US schedules. Bands are ± 1 cross-sectional standard deviation across skill histories; the dashed horizontal line is the Pigouvian level $\tau^q = e = 0.494$. The optimal-income-tax path is solved over the working life; from retirement on the costate vanishes and $\tau^q = e$ exactly (Appendix D.12), so that segment is analytical rather than simulated.

6.3 Who bears the wedge

The optimal carbon wedge rises with skill in both worlds (Figure 2). Under an optimal income tax it is indistinguishable across quintiles early in life, when every history looks alike, and fans out from mid-career: by 64 it reaches about 70% for the top skill quintile against 57% for the bottom, a 14-point spread. Under the US income tax the same ranking holds and the spread is three times as wide, 44 points at 64, running from a 29% energy subsidy for the bottom quintile to a 15% tax on the top. High current productivity signals a history whose incentive constraints are expensive, and the commodity margin is the planner's lever on exactly those histories, so the screening value of the instrument concentrates on high-skill agents whether or not the income tax is optimal.

Two readings follow, and the second is the one that survives dropping the optimality assumption.

The optimal carbon wedge is not regressive: the planner taxes the energy of high-skill households more, not less, so the redistributive worry about carbon pricing is misplaced at the level of the optimal type-contingent wedge. And this is not an artifact of assuming the rest of the tax system is right. Freezing the income tax at the observed US schedules leaves the progressivity intact and sharply strengthens it. The regressivity of real carbon taxes comes from elsewhere: from the flat-rate restriction, since an anonymous point-of-sale tax cannot condition on skill, and, in the necessity calibration, from energy's falling budget share.

Figure 2 is a schedule of marginal wedges by skill, not a map of the burden each household bears. That burden is set jointly by the income tax and transfers alongside the wedge, so the schedule's progressivity speaks to the corrective instrument itself, not to the full incidence of carbon pricing. Note also that the two panels are drawn on different vertical scales, which the level gap of Section 6.2 forces; the comparison to read is the slope, and the spreads are printed on the figure for that reason.

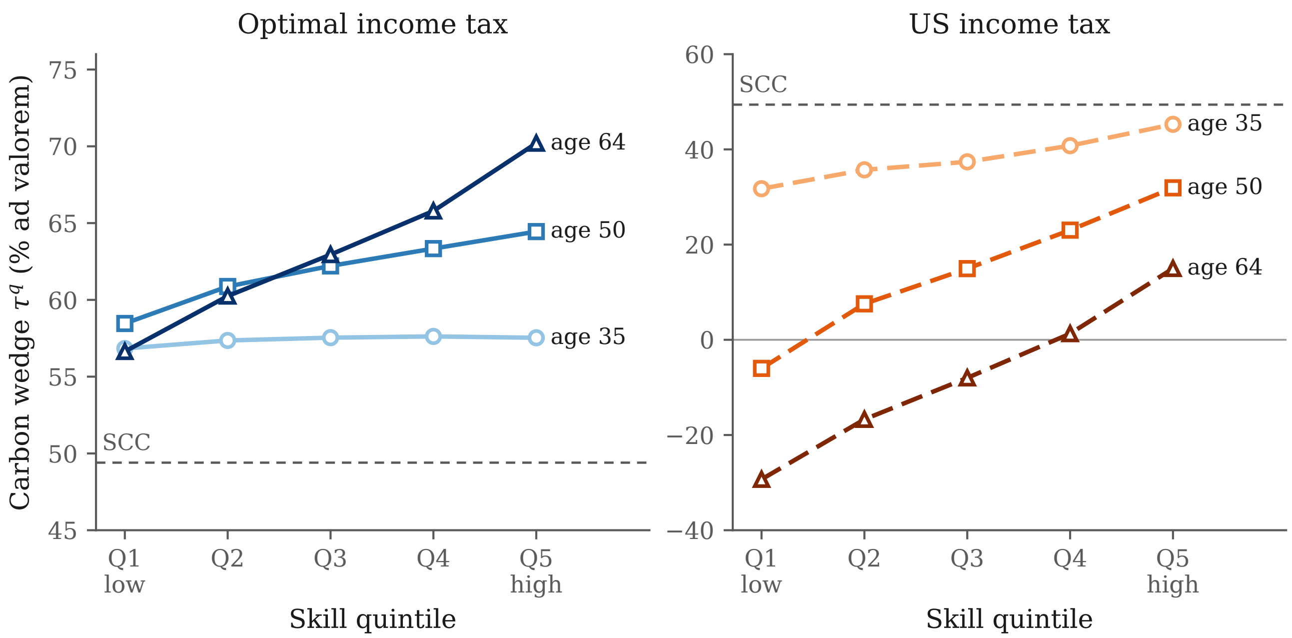

Figure 2: The optimal commodity wedge τ^q by skill quintile at ages 35, 50 and 64, when the income tax is set optimally (left) and when it is frozen at the US schedules (right). Shading runs light to dark in age. The dashed horizontal line is the Pigouvian level. Note the different vertical scales: the comparison the figure is drawn for is the slope in skill, whose magnitudes are given in the text.

6.4 Where the rotation comes from

Section 5.1 predicts that the two frozen margins reach the carbon price through one statistic, $\Xi_t = \eta_t(1-\bar{\tau}_t^y) - \eta_t^s$, and that they enter it differently: the labor error is a within-date object that can hold one sign for the whole working life and therefore sets the *level* of the deviation, while the savings error must change sign across dates (Lemma 5.4) and therefore *tilts* the profile. Varying τ^k while holding the labor schedule fixed separates the two forces exactly, because at $\tau^k = 0$ only the labor freeze binds.

The separation is visible in Table 3 and the left panel of Figure 3. With the labor schedule frozen and the savings margin free, the deviation is negative at every age and nearly flat: $\delta_1 = -0.27$ against $\delta_{40} = -0.02$. The US labor schedule overtaxes labor relative to the virtual Mirrleesian wedge, so $\eta_t > 0$, $\Xi_t > 0$, and by Corollary 5.1 the carbon price sits below the social cost of carbon throughout. Turning on the capital tax rotates the profile around that level, and the rotation grows monotonically in τ^k: at the benchmark $\tau^k = 0.18$ the deviation runs from $+0.01$ at 25 to -1.32 at 84, and at the $\tau^k = 0.36$ upper bound from $+0.29$ to -3.23. The sign pattern is the one the two-period example of Appendix C.4 derives analytically, with the price above the social cost of carbon early and below it late.

Two features of the rotation are worth separating from its size. The age at which $\hat{\eta}^s$ changes sign sits at 45 at every capital-tax rate on the grid: the multiplier's shape is a property of the boundary conditions $\vartheta_0 = 0$ and $\eta_T^s = \vartheta_T U_{c,T}$, not of the size of the pricing error, exactly as Lemma 5.4 implies. And the deviation survives into retirement while the labor error does not, which is Corollary 5.3: from T_R on the labor constraint has no content and $\Xi_t = -\eta_t^s$, so the entire retirement deviation is the savings error. A misset labor schedule distorts the allocation across types within a date and its damage dies with the labor margin; a misset savings schedule distorts the allocation across dates within a life and keeps injecting error through twenty years in which the virtual savings wedge is zero.

The deviation at labor-market entry is monotone in τ^k and crosses zero at $\tau^k \approx 0.17$, just below the benchmark (Figure 3, right panel). At the US effective marginal rate on capital income the labor error and the savings error therefore very nearly cancel at entry: the young face a carbon price within a point of the social cost of carbon, and the whole third-best deviation is a rotation with age rather than a shift in level. This is Corollary 5.2 realized at an observed capital-tax rate rather than at a knife-edge. Exact Pigouvian pricing survives an income tax that is misset on both margins, provided the two errors are misset in offsetting proportions, and at the rate the US actually levies they nearly are.

Nothing in the section rests on that coincidence. The grid is reported in full precisely so that the reading does not depend on where the benchmark falls, and every qualitative feature survives at every rate on it: the labor error holds the price below the social cost of carbon, the savings error rotates the profile from above to below as the cohort ages, the flip age is invariant, and the deviation outlives the labor margin into retirement. What the benchmark fixes is the size of the rotation and the age at which the price crosses the social cost of carbon, both of which rise with τ^k.

Table 3: The carbon price against a given income tax, across the capital-tax grid. $\delta_t \equiv (\tau_t^q - \mathrm{SCC})/(1+\tau_t^q)$ is the deviation of (4), reported as a visited mean over simulated histories; "crossing" is the age at which it changes sign. The labor schedule is frozen at the Heathcote et al. (2017) function in every row, so the $\tau^k = 0$ row isolates the labor error. $\mathrm{SCC} = e = 0.494$.

τ^k	$\bar{\tau}^s$ (%/yr)	δ at 25	crossing age	δ at 64	δ at 84	working-life mean τ^q
0	0	−0.27	—	−0.02	−0.10	0.36
0.025	0.13	−0.26	—	−0.13	−0.27	0.29
0.05	0.25	−0.22	—	−0.22	−0.42	0.27
0.18	0.90	+0.01	28	−0.68	−1.32	0.23
0.36	1.80	+0.29	36	−1.57	−3.23	0.23

Note: $\tau^k = 0.18$ is the benchmark of Table 1 and $\tau^k = 0.36$ the upper bound of Section 5.1.1. A dash in the crossing column means the deviation keeps one sign over the whole life cycle.

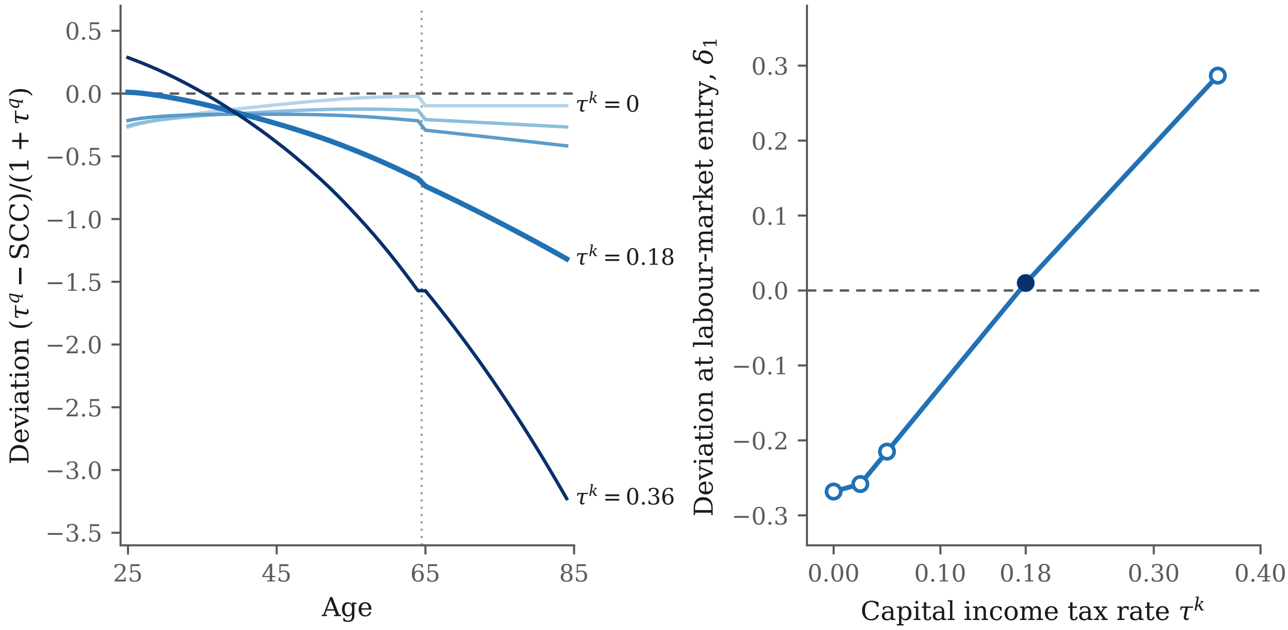

Figure 3: Left: the deviation $\delta_t = (\tau_t^q - \text{SCC})/(1+\tau_t^q)$ over the life cycle across the capital-tax grid, with the labor schedule frozen at the US function throughout. Shading runs light to dark in τ^k over the grid $\{0, 0.025, 0.05, 0.18, 0.36\}$; the benchmark $\tau^k = 0.18$ is drawn heavier. The dashed horizontal line is Pigouvian pricing and the dotted vertical line is the retirement date. Right: the same deviation at labor-market entry, plotted against τ^k, with the benchmark rate marked by a filled circle.

6.5 What carbon pricing is worth

The welfare accounting inverts between the two worlds, and the inversion is the section's main result (Figure 4).

6.5.1 When the income tax is optimal

Table 4 reports consumption-equivalent gains over having no carbon price, under the optimal nonlinear income and savings mechanism throughout. Four findings, in order of size.

The value of a carbon price is overwhelmingly Pigouvian. At the EPA social cost of carbon, having any carbon price is worth 0.31% of consumption, and a flat rate at the social cost of carbon alone captures 98% of the full second-best value while cutting emissions by 13.5%. For scale, the entire insurance value of Farhi and Werning (2013)'s optimal income-tax apparatus over laissez-faire is 1.56% of consumption: the carbon instrument is worth about a fifth of that at this social cost of carbon.

The optimal flat rate exceeds the social cost of carbon by about 14%, $\tau^{q*} = 0.56$ against $e = 0.494$, the flat reflection of the Corlett-Hague premium. But the welfare gain from that uplift over pricing at the social cost of carbon is 0.003%: the objective is very flat in the rate, so "price at the social cost of carbon" is a near-optimal rule of thumb even with nonseparable preferences.

Sophistication buys almost nothing beyond the level. Making the rate age-dependent adds about 0.001 percentage points over the optimal flat rate, and full history dependence about 0.001 more, both at the numerical resolution. The premium varies over life and across histories by a few points of a 6% budget share with a substitution elasticity of 0.4, so the deadweight loss from mistracking it is second-order squared.

The pure Corlett-Hague motive is tiny, and necessity halves the stakes. With the externality off, the entire commodity-wedge apparatus of the second best is worth 0.0036% of consumption, and its optimal flat rate is 5%, materially positive as in West and Williams (2007) but negligible in welfare. Under the Stone-Geary calibration, in which energy's budget share falls with income, the value of the flat carbon tax falls to 0.15% and its abatement to 6%, because the subsistence part of energy does not respond to price, while the optimal flat rate is essentially unchanged. This is the calibration in which a flat carbon tax is regressive, yet the mechanism absorbs the incidence through the income side, which is why the flat instrument loses so little welfare despite its incidence. As a check on the whole apparatus, switching the leisure complementarity off ($\gamma_c = \gamma_q$) makes the second best and the flat Pigouvian tax coincide to five decimal places in welfare and in emissions: dynamic Atkinson-Stiglitz, and the exact knife-edge of Proposition 4.3, reproduced through two structurally different solvers.

Table 4: Welfare value of carbon pricing under progressively simpler rules, income tax optimal (consumption-equivalent gain over $\tau^q = 0$, % of consumption)

	benchmark ($e = 0.494$)	Stone-Geary (necessity)	$\gamma_c = \gamma_q$ (placebo)	no externality ($e = 0$)
second best (nonlinear τ^q)	0.315	—	0.252	0.0036
age-dependent linear $\tau^q(t)$	0.314	—	—	0.0038
flat τ^{q*} (optimized)	0.313	0.153	—	0.0026
optimal flat rate τ^{q*}	0.56	0.56	—	0.051
flat Pigouvian $\tau^q = e$	0.309	0.151	0.252	—
emissions vs $\tau^q = 0$: second best	−15.0%	—	−13.5%	−1.8%
flat Pigouvian	−13.5%	−6.3%	−13.5%	−1.8%

Note: welfare entries are in % of consumption; the optimal-flat-rate row is an ad-valorem rate, and the final two rows report emissions changes relative to $\tau^q = 0$. Differences below the 0.002-point numerical resolution, including the apparent ranking of the age-dependent rate above the second best in the no-externality column, are simulation noise. Dashes mark three cases. *Not computed*: the second best requires homothetic preferences, so it and the age-dependent rung are blank under the non-homothetic Stone-Geary calibration. *Degenerate*: under the $\gamma_c = \gamma_q$ placebo the second best already coincides with the flat Pigouvian tax, so the intermediate rungs add nothing. *Not applicable*: with no externality ($e = 0$) there is no Pigouvian benchmark.

6.5.2 When the income tax is the one the US levies

Every one of those conclusions reverses. Table 5 runs the same rate grid with the two income-side wedges frozen, and Figure 4 plots the two rate curves side by side. Both panels come from the same solver family at the same grid with the same simulated histories, so the only difference between them is the freeze.

A flat carbon tax at the social cost of carbon is welfare-negative. It costs 0.115% of consumption relative to having no carbon price at all, and 0.155% at the $\tau^k = 0.36$ upper bound. This is not an artifact of ignoring the damages the tax avoids: the planner's budget prices energy at the social price $1 + e$ in every regime, including the one with no carbon tax, so the comparison is damage-inclusive by construction. With the income tax stuck at US levels, legislating the social cost of carbon as a flat rate overshoots the constrained optimum badly enough that the economy would rather have no carbon price.

The optimal flat rate is a fifth of the social cost of carbon, and it is worth almost nothing. It is about 0.10, against 0.56 when the income tax is optimal, and it buys 0.02% of consumption over charging nothing. A single rate must average a profile that runs from above the social cost of carbon at entry to a deep subsidy in retirement, weighted by substitution responses, and the twenty retirement years drag that average far down. The below-damages prescription of the Ramsey tradition reappears here, and more starkly than in the linear-instrument literature, because a frozen income tax cannot absorb any of the burden.

The near-sufficiency of flat carbon taxation is overturned. When the income tax is optimal, the entire nonlinear fine structure of the wedge is worth 0.003 percentage points over the best flat rate. When it is frozen at the US schedules, the nonlinear wedge is worth 1.38 percentage points — more than five hundred times as much, and four times the entire second-best value of carbon pricing. Under a misset income tax the commodity margin's history dependence performs income-tax repair, and no anonymous rate can imitate it.

That last number is best read as a statement about commodity taxation rather than about carbon. Energy is the only margin the constrained planner still controls freely, so it absorbs the whole repair job; with more consumption goods the same logic would spread across every commodity wedge and each would move less. What the exercise establishes is that when the income tax is given, a history-dependent commodity instrument is worth a great deal, and that the instrument governments actually use cannot deliver it.

Table 5: Welfare of a flat carbon tax when the income-side wedges are frozen at the US schedules (consumption-equivalent gain over $\tau^q = 0$ of the same regime, % of consumption). The "income tax optimal" column is the same rate grid solved with the income and savings wedges free, at the same grid resolution and on the same simulated histories, and is reported for comparison.

flat rate τ^q	income tax optimal	income tax given
0 (no carbon price)	0	0
0.10	+0.119	+0.021
0.20	+0.203	+0.014
0.35	+0.282	−0.035
Pigouvian, $e = 0.494$	+0.314	−0.115
0.65	+0.314	−0.226
best flat rate τ^{q*}	0.56	0.10
its welfare gain	+0.318	+0.021
nonlinear τ^q	see Table 4	+1.401

Note: entries are consumption-equivalent gains in % of consumption; the best-flat-rate row is an ad-valorem rate. The right-hand column is the benchmark $\tau^k = 0.18$; at the $\tau^k = 0.36$ upper bound the same rows read +0.010 at the best flat rate 0.09, −0.155 at the Pigouvian rate, and +1.649 for the nonlinear wedge, so nothing in the reading turns on the capital-tax measure. The best flat rate under a given income tax is a fitted optimum confirmed by a solve at the fitted rate; the welfare surface is flat between 0.10 and 0.13. The nonlinear entry is left to Table 4 in the left-hand column because the second best is solved on a finer grid than this rate sweep and the two levels are not directly comparable; the within-column comparison there gives 0.315 against 0.313 for the best flat rate.

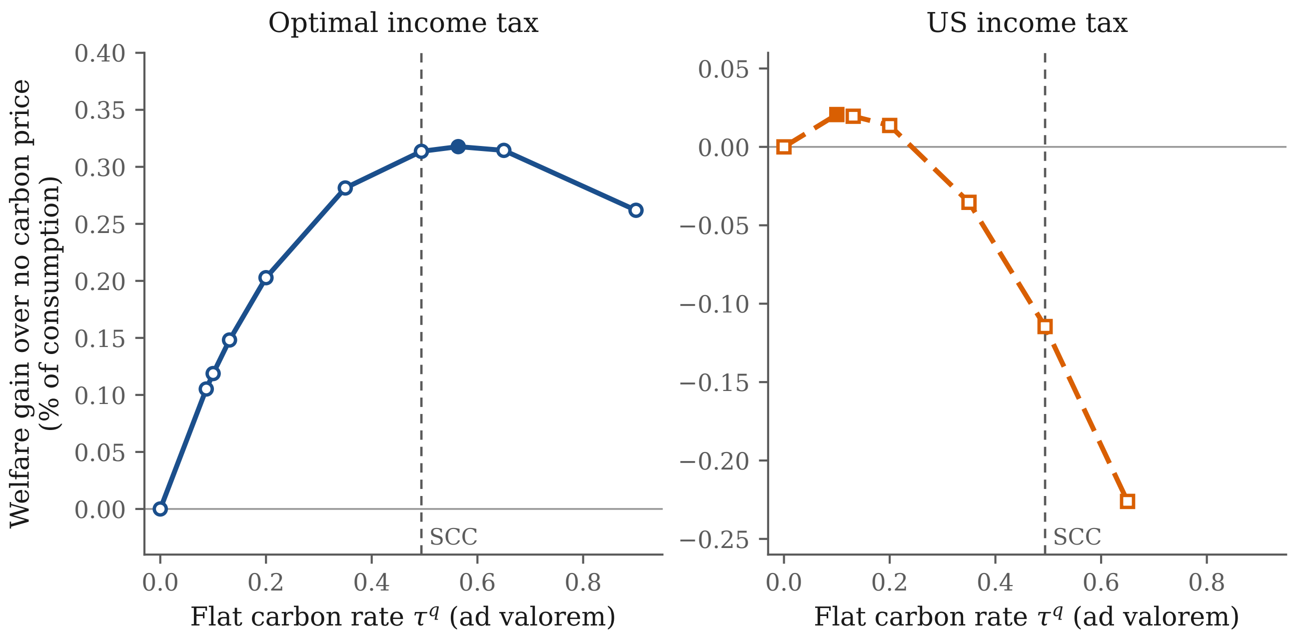

Figure 4: Consumption-equivalent welfare gain from a flat carbon tax, against the rate, when the income tax is set optimally alongside it (left) and when it is frozen at the US schedules (right). Both panels use the same solver family, the same grid, and the same simulated histories; only the freeze differs. The filled marker is the welfare-maximising rate on each grid and the dashed vertical line is the social cost of carbon. Note the different vertical scales.

6.6 Reading and limitations

The practical hierarchy for carbon policy depends on a condition that is easy to leave implicit. If the income tax is set optimally alongside the carbon price, the hierarchy is clean: get the carbon price to the social cost of carbon, worth 0.31% of consumption; nudge it up by the premium for leisure complementarity, worth a further 0.003%; and spend the sophistication budget on the income tax, where Farhi and Werning (2013)'s own accounting shows the returns to age- and history-dependence are two orders of magnitude larger. The redistributive value embedded in the carbon tax is small, which is the separation principle in welfare terms, and it agrees with Douenne et al. (2026)'s near-Pigouvian conclusion, reached here from the Mirrleesian rather than the Ramsey side.

If the income tax is the one actually in force, that hierarchy does not hold, and the direction of the failure is informative. The below-damages prescription is not a property of distortionary taxation as such — with the income tax optimal, the carbon price is exactly the social cost of carbon at every history under weak separability. It is a property of income-tax misspecification. Set the income tax right and the correction vanishes; set it wrong and the correction is large, larger than the Ramsey literature's 8–24%, and a flat tax at the social cost of carbon becomes worse than no carbon tax. The two results are one claim: the case for pricing carbon at the social cost of carbon is a case for fixing the income tax, and the two reforms are not separable in practice even though the wedges separate in theory.

This does not rehabilitate weakening the carbon price on its own. A government that lowers the carbon price while leaving the income tax misset moves along the right-hand panel of Figure 4 and recovers at most 0.02% of consumption; one that fixes the income tax moves to the left-hand panel and gains 0.31% from the carbon price alone, on top of the income tax's own insurance value. The ordering of the two reforms matters, and it runs from the income tax to the carbon price rather than the other way around.

Five limitations bound the numbers. The externality is a constant marginal damage, so the pollution-stock dynamics of Section 4.3, which move the *level* of the tax rather than the wedge structure measured here, are a separate exercise. The second best requires homothetic preferences, so the necessity calibration is read off the exact linear-tax reduction rather than the full nonlinear mechanism. The log consumption index is required by the balanced-growth homogeneity the solver relies on. The energy good carries a single carbon intensity, so within-energy heterogeneity is outside the model. And the third-best exercise freezes the savings wedge through retirement as well as through the working life, which is the right reading of a capital income tax that does not stop at 65 but is nonetheless a modelling choice doing real work: it is what keeps the deviation growing after T_R and what pulls the optimal flat rate down.

None of these bears on the qualitative results, though the results rest on different footings. The premium's proportionality to the labor wedge is structural, the identity $\varkappa\, l^{\alpha}\, \tau^{y}/(1-\tau^{y})$, as is the near-sufficiency of a flat Pigouvian price under an optimal income tax, which follows from the objective's flatness in the rate. The sign of the third-best rotation is structural too: it is Lemma 5.4 and the two-period example of Appendix C.4, and the solved profiles reproduce both. The premium's size and the rotation's magnitude are calibration-contingent. They rest on energy being a leisure complement ($\varkappa > 0$), the empirical input from West and Williams (2007), and on the capital income tax rate. The premium scales roughly linearly in $\varkappa$, so the gasoline-only upper bound $\varkappa = 0.45$ about doubles the mid-career premium and $\varkappa = 0$ collapses it to the Atkinson-Stiglitz benchmark; the welfare stakes in the second-best world stay small across this range.

7 Conclusion

A pollution externality placed inside a dynamic Mirrlees economy leaves the corrective and redistributive tasks separate. When preferences are weakly separable, the carbon wedge is exactly the social cost of carbon and the labor and savings wedges are unchanged, so the below-Pigouvian result that the Ramsey tradition blames on fiscal distortion does not survive an optimal nonlinear income tax. Only nonseparability bends the carbon wedge away from the social cost of carbon, and it does so in the direction the good's complementarity with labor dictates.

Real carbon taxes are anonymous linear prices, not the history-dependent wedge the theory prescribes. Under that restriction the two-good problem collapses to a one-good problem in expenditure; the optimal uniform rate is a substitution-weighted average of the wedges it replaces, and its welfare cost is their dispersion, which vanishes under weak separability. With the externality the uniform price splits additively into the social cost of carbon and a screening term, so the restriction binds only the redistributive part and never the corrective one.

Carbon pricing is resisted as a burden on the poor, and governments answer by capping it, tilting it, or outright repealing it. This paper suggests a different response: once the income tax is set optimally, the carbon price has no real redistributive role and should simply be set at the social cost of carbon.

This is a claim about optimal design, not about political durability. Canada paired its consumer carbon tax with a per-household dividend and repealed it anyway, so whether households perceive the compensation and trust it to last can matter as much as whether it exists. The model takes the income tax and the dividend as credible and delivered; making them so is a question of salience and commitment that it does not address.

This reading carries one caveat. The separation is a property of the jointly optimal tax system, and presumes the labor and savings wedges are set optimally alongside the carbon price. Section 5 holds both to the schedules an actual income tax implies and finds that a single statistic carries the deviation, the labor error net of the savings error. Where the schedule overtaxes labor, part of the redistribution it fails to carry falls back on the carbon price, which drops below the social cost of carbon; where the schedule undertaxes, the price rises above it. A misset savings wedge works differently: its error cannot share one sign across the life cycle, so it tilts the carbon-price profile rather than moving its level, and the two errors generically cancel at some age. And the flow runs both ways: a carbon price legislated below the social cost of carbon raises the optimal labor wedge by the marginal budget share of the dirty good times the uncollected carbon price, the income tax collecting the carbon the carbon price misses. The optimal linear instruments of the Ramsey tradition emerge inside the same family, as the levels at which the constraint multipliers average to zero.

Three extensions stand out. First, the paper characterizes wedges, not their implementation; how to replicate the optimal commodity wedge with real instruments is open. Second, the model follows a single generation, but climate change is intergenerational. The natural next step is an overlapping-generations extension, in which the single-generation model becomes the component problem of an infinite-horizon planner. The fixed point between the carbon-price path and the per-cohort mechanism-design problems would then endogenize the savings-wedge paths that Section 6 takes from calibrated optima. Third, outside the environmental context, the optimal commodity wedge is a time-varying object, which raises the question of age-dependent commodity taxation. Age-dependent commodity taxes may sound fanciful, yet age-based pricing of some goods and services, and discounts for low-income consumers, already implements a time-varying, income-based commodity wedge.

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A Optimal wedge derivation without externalities

Because $U(c, q, l)$ can be written as $U(c, q, y/\theta)$, I can write U_θ as $-U_l l/\theta$. The Hamiltonian of the recursive problem is

$$
\begin{aligned}
\mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta) \right] f \\
& + \psi \left[-U_l(c, q, y/\theta) \frac{l}{\theta} + \beta w_2 \right] \\
& - \lambda_1 u(\theta) \bar{\alpha}_t f_t \\
& + \lambda_2 u(\theta) f_2 \\
& + \phi \left[u - U(c, q, y/\theta) - \beta w \right]
\end{aligned}
$$

Differentiating yields the following first-order conditions

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql} \frac{l}{\theta} = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l = \frac{1}{\theta} \psi \left[U_{ll} l + U_l \right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta
\end{aligned}
$$

The three wedges can be derived from here. The labor wedge is derived in Section A.1, the savings wedge in Section A.2, and the commodity wedge in Section A.3.

A.1 Labor wedge

Dividing the first-order condition for c by U_c and using $\gamma_c \equiv U_{cl} l / U_c$,

$$
\phi = \frac{f}{U_c} - \frac{\psi}{\theta} \gamma_c. \tag{12}
$$

From the first-order condition for l and since $U_{ll}l + U_l = U_l(1+\epsilon)$ with $\epsilon \equiv U_{ll}l/U_l$,

$$-\theta f - \phi U_l = \frac{\psi}{\theta}U_l(1+\epsilon) \implies \phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon)$$

Equating the two expressions for ϕ and collecting the terms in ψ,

$$\frac{f}{U_c} + \frac{\theta f}{U_l} = -\frac{\psi}{\theta}(1+\epsilon-\gamma_c)$$

The labor wedge is $1-\tau^y = -U_l/(\theta U_c)$, so $\theta U_c/U_l = -1/(1-\tau^y)$, making the left-hand side $\frac{f}{U_c}\left(1-\frac{1}{1-\tau^y}\right) = -\frac{f}{U_c}\frac{\tau^y}{1-\tau^y}$. Substituting and rearranging,

$$\frac{\tau^y}{1-\tau^y} = (1+\epsilon-\gamma_c)\frac{\psi U_c}{\theta f} = A_t(\theta)\,\frac{\psi U_c}{\theta f} \tag{13}$$

Next, substitute ϕ from (12) into $\dot{\psi} = \lambda_1\bar{\alpha}_t f - \lambda_2 f_2 - \phi$,

$$\dot{\psi} - \frac{\gamma_c}{\theta}\psi = \lambda_1\bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}$$

This is a linear first-order ordinary differential equation. Multiplying by the integrating factor $\exp\left[-\int^{\theta}\gamma_c(\tilde{x})\,d\tilde{x}/\tilde{x}\right]$ and integrating from θ to ∞ with the boundary condition $\psi(\infty)=0$,

$$\psi(\theta) = \int_{\theta}^{\infty}\exp\left[-\int_{\theta}^{x}\gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{f(x)}{U_c(x)} - \lambda_1\bar{\alpha}_t(x)f(x) + \lambda_2 f_2(x)\right)dx. \tag{14}$$

The remaining boundary condition $\psi(0)=0$ pins down λ_1,

$$\lambda_{1,t} = \frac{\int_0^{\infty}\exp\left[-\int_0^{x}\gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{1}{U_{c,t}(x)}f_t(x) + \lambda_{2,t}f_{2,t}(x)\right)dx}{\int_0^{\infty}\exp\left[-\int_0^{x}\gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\bar{\alpha}_t(x)f_t(x)dx}. \tag{15}$$

Substituting (14) into (13),

$$\frac{\tau^y}{1-\tau^y} = (1+\epsilon-\gamma_c)\frac{U_c(\theta)}{\theta f(\theta)}\int_{\theta}^{\infty}\exp\left[-\int_{\theta}^{x}\gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{f(x)}{U_c(x)} - \lambda_1\bar{\alpha}_t(x)f(x) + \lambda_2 f_2(x)\right)dx$$

I split the integral into the $[f/U_c - \lambda_1 \bar{\alpha}_t f]$ part, which yields the intratemporal term $A_t B_t C_t$, and the $\lambda_2 f_2$ part, which yields the intertemporal term. For the first part, carry $U_c(\theta)/U_c(x)$ into the integrand. Along the optimal allocation $U_{c,t}$ depends on the type through $c(\theta)$, $q(\theta)$, and $l(\theta) = y(\theta)/\theta$. Hence,

$$\frac{d}{d\theta} \ln U_{c,t} = \frac{U_{cc}\dot{c} + U_{cq}\dot{q} + U_{cl}\dot{l}}{U_c} = -\sigma_t \frac{\dot{c}}{c} - \sigma_{cq,t}\frac{\dot{q}}{q} + \gamma_c \frac{\dot{l}}{l}$$

Since $l = y/\theta$ gives $\dot{l}/l = \dot{y}/y - 1/\theta$, integrating from θ to x and multiplying by the integrating factor gives the following:

$$\frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp\left[-\int_\theta^x \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] = \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_c(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right) d\tilde{x}\right], \quad (16)$$

Here, the $\gamma_c/\tilde{x}$ term from $\dot{l}/l$ cancels the integrating factor. The cross-curvature $\sigma_{cq,t}$ enters because, under nonseparability, extracting resources from a higher type shifts the marginal utility of the numéraire through the adjustment of both goods. The first part therefore equals $A_t B_t C_t$. For the second part, two preliminaries pin down $\lambda_{2,t}$. The first-order condition for w_2 rearranges to $\partial V_{t+1}/\partial w_2 = -\beta R\,\psi/f$, and the envelope theorem applied to the period-$(t+1)$ problem gives $\partial V_{t+1}/\partial \hat{w}_2 = -\lambda_{2,t+1}$, the multiplier with which threat keeping enters the Lagrangian. Equating the two expressions and shifting the date back one period,

$$\lambda_{2,t}(\theta^{t-1}) = \beta R\,\frac{\psi_{t-1}}{f_{t-1}}$$

Next, evaluating (13) at $t-1$ and solving for the costate,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\,\frac{\theta_{t-1}}{U_{c,t-1}\,A_{t-1}}$$

The $\lambda_2 f_2$ block of the wedge is $\lambda_{2,t}$, a constant in the current type, times the same outer structure as the first part,

$$A_t(\theta)\,\frac{U_{c,t}(\theta)}{\theta f_t(\theta)}\,\lambda_{2,t}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)\,dx$$

Substituting the two preliminaries and collecting the ratios,

$$\beta R\,\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\,\frac{A_t(\theta)}{A_{t-1}}\,\frac{U_{c,t}(\theta)}{U_{c,t-1}}\,\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)\,dx}{\theta f_t(\theta)} = \beta R\,\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\,D_t(\theta)$$

Collecting both parts,

$$\frac{\tau^y}{1-\tau^y} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}D_t(\theta)$$

$$\begin{aligned}
A_t(\theta) &= 1+\epsilon_t(\theta)-\gamma_{c,t}(\theta)\\
B_t(\theta) &= \frac{1-F_t(\theta)}{\theta f_t(\theta)}\\
C_t(\theta) &= \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})}+\sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})}-\gamma_{c,t}(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right)d\tilde{x}\right]\left(1-\lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x)\right)\frac{f_t(x)dx}{1-F_t(\theta)}\\
D_t(\theta) &= \frac{A_t(\theta)}{A_{t-1}}\frac{U_{c,t}(\theta)}{U_{c,t-1}}\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)dx}{\theta f_t(\theta)}
\end{aligned}$$

This is Golosov et al. (2016)'s labor-wedge decomposition, with the cross-curvature $\sigma_{cq,t}$ in C_t.

A.2 Savings wedge

The savings wedge follows from the same first-order conditions. I define the incentive credit χ_t, the share of the cost of delivering utility through the numéraire that the planner recoups because consumption complements labor, relaxing the incentive constraint:

$$\chi_t(\theta^t) \equiv \frac{\gamma_{c,t}}{1+\epsilon_t-\gamma_{c,t}}\frac{\tau^y_t}{1-\tau^y_t}$$

Using $\frac{\tau^y_t}{1-\tau^y_t} = \frac{\psi U_c}{\theta f}(1+\epsilon_t-\gamma_{c,t})$, $\phi = f/U_c - \psi\gamma_c/\theta$ from (12) can be rewritten as follows:

$$\frac{\phi}{f} = \frac{1}{U_c}\left(1-\gamma_c\psi U_c/(\theta f)\right) = \frac{1-\chi_t}{U_{c,t}}. \tag{17}$$

$\chi_t > 0$ when the numéraire is complementary with labor and labor is taxed and zero when preferences are separable in the numéraire against labor or labor is undistorted. The first-order condition for w and the envelope condition $\partial V_t/\partial\hat{w} = \lambda_{1,t}$ one period ahead give

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1-\chi_t}{U_{c,t}}. \tag{18}$$

A second expression for $\lambda_{1,t}$ comes from integrating the costate equation over the whole type space. With $\psi(0) = \psi(\infty) = 0$,

$$0 = \int_0^\infty \dot{\psi}\, d\theta = \lambda_{1,t} \int_0^\infty \bar{\alpha}_t f \, d\theta - \lambda_{2,t} \int_0^\infty f_2 \, d\theta - \int_0^\infty \phi \, d\theta$$

The Pareto weights are normalized so that $\int \bar{\alpha}_t f \, d\theta = 1$, and $\int f_t(\theta|\theta_-)\, d\theta = 1$ for every θ_- implies $\int f_{2,t}\, d\theta = 0$. Hence $\lambda_{1,t} = \int_0^\infty \phi\, d\theta$, which by (17) is

$$\lambda_{1,t} = \mathbb{E}_{t-1}\left[\frac{1-\chi_t}{U_{c,t}}\right], \tag{19}$$

where $\mathbb{E}_{t-1}$ integrates over θ_t conditional on θ_{t-1}. The multiplier on promise keeping is set before the current draw is known. Evaluating (19) at $t+1$ and equating with (18),

$$\frac{1-\chi_t}{U_{c,t}} = \frac{1}{\beta R}\, \mathbb{E}_t \left[\frac{1-\chi_{t+1}}{U_{c,t+1}}\right].$$

Defining the incentive-adjusted marginal utility $\widehat{U}_{c,t} \equiv U_{c,t}/(1-\chi_t)$,

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R}\, \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}}\right]$$

This is the inverse Euler equation in the adjusted marginal utility $\widehat{U}_c$. Delivering a util through the numéraire costs $1/U_{c,t}$ in resources and, when $U_{cl} > 0$, the unit lowers the information rent $-U_l\, l/\theta$, an incentive gain worth $\chi_t/U_{c,t}$.

A.3 Commodity wedge

The commodity wedge uses the first-order conditions for c and q of the preamble. There are different equivalent ways to express it, but I will seek the familiar labor wedge $ABCD$ form. First, because the first-order conditions for each good share the same ϕ,

$$\frac{f_t}{U_q} - \psi \frac{U_{ql}}{U_q} \frac{l}{\theta} = \frac{f_t}{U_c} - \psi \frac{U_{cl}}{U_c} \frac{l}{\theta}$$

Cross-multiplying by U_q and U_c and using $\gamma_c \equiv \frac{U_{cl}\, l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql}\, l}{U_q}$,

$$U_c \left(f - \frac{\psi}{\theta} \gamma_q U_q \right) = U_q \left(f - \frac{\psi}{\theta} \gamma_c U_c \right)$$

Dividing by $U_q f$ and rearranging,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f} (\gamma_q - \gamma_c)$$

Note that, by definition, $1 + \tau^q = \frac{U_q}{U_c}$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1+\tau^q}$. Multiplying through by -1,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q)$$

The prefactor $\psi U_c / (\theta f)$ is the same object that drives the labor wedge. Solving (13) for it and substituting the $ABCD$ decomposition of Appendix A.1,

$$\frac{\psi U_c}{\theta f} = \frac{1}{A_t} \frac{\tau^y}{1 - \tau^y} = B_t C_t + \beta R \frac{\tau^y_{t-1}}{1 - \tau^y_{t-1}} \frac{D_t}{A_t}$$

Multiplying by the complementarity gap $\gamma_{c,t} - \gamma_{q,t}$,

$$\frac{\tau^q}{1 + \tau^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t C_t + \beta R \frac{\tau^y_{t-1}}{1 - \tau^y_{t-1}} \frac{D_t}{A_t} \right]$$

with A_t, B_t, C_t, D_t as in Appendix A.1. Alternatively, multiplying $\frac{\psi U_c}{\theta f} = \frac{1}{A_t} \frac{\tau^y}{1 - \tau^y}$ by the gap,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau^y_t}{1 - \tau^y_t}$$

The commodity wedge inherits every dynamic feature of the labor wedge. Its shape is that of the labor wedge, scaled by the elasticity term.

B Optimal wedge derivation with externalities

B.1 Validity of the recursive formulation

Section 4.1 takes its recursive form from Kapička (2013), whose consumption set is an arbitrary convex and compact set, so his reduction covers the consumption vector (c, q). The externality adds two ingredients his environment lacks: the pollution stock enters preferences and the planner internalizes its accumulation from aggregate emissions. This subsection shows that neither disturbs the validity of the recursive approach.

An allocation is a collection of measurable functions $c_t(\theta^t)$, $q_t(\theta^t)$, and $y_t(\theta^t)$ for $t = 0, \dots, T$. Aggregate emissions at an allocation are

$$Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] \, dF_0(\theta) \qquad t = 0, \dots, T$$

and the implied pollution stocks are $S_t = S(Q_{t-s}, \dots, Q_t)$, with pre-period emissions given. As in the main text, $\Phi_j \equiv \partial S_{t+j}/\partial Q_t$, with $\Phi_j = 0$ for $j > s$ by the truncation of the carbon cycle. Multipliers dated beyond T are zero.

Lemma B.1 (incentive constraints with an aggregate pollution stock)**.**
Fix a deterministic emission sequence $(Q_0, \dots, Q_T)$ and let $S_t = S(Q_{t-s}, \dots, Q_t)$. Define the dated utility

$$\tilde{U}_t(c, q, l) \equiv U(c, q, l, S_t)$$

Given this sequence, (i) an allocation is incentive compatible in the economy with pollution if and only if it is incentive compatible in the no-externality economy, the $\nu_t = 0$ case of Section 4.1 in which the dirty good is dirty in name only, with period utility $\tilde{U}_t$, and (ii) the local incentive constraint is

$$\dot{u}_t(\theta^t) = U_\theta\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right) + \beta\, w_{2,t}(\theta^t)$$

with no term in the pollution stock.

Proof. Step 1: deviations leave the pollution stock unchanged. Incentive compatibility compares one agent's payoffs under truthful reporting and under a reporting strategy σ, holding the allocation and all other reports fixed. Aggregate emissions integrate the allocation over the population and a single agent has measure zero in that integral, so Q_t and with it S_t take the same values under truth-telling and under the deviation. Every payoff entering the incentive constraint is therefore evaluated at the same pollution-stock sequence.

Step 2: the incentive constraints are those of a dated no-externality economy. By Step 1, the incentive constraint of Section 3 reads for every initial type θ and every reporting strategy σ,

$$\begin{aligned}\mathbb{E}_0 &\left[\sum_{t=0}^{T} \beta^t U\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right) \middle| \theta\right] \\ &\geq \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U\left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t\right) \middle| \theta\right]\end{aligned}$$

with the same fixed S_t on both sides. Substituting the definition of $\tilde{U}_t$ on both sides,

$$\begin{aligned}\mathbb{E}_0 &\left[\sum_{t=0}^{T} \beta^t \tilde{U}_t\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right) \middle| \theta\right] \\ &\geq \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t \tilde{U}_t\left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}\right) \middle| \theta\right]\end{aligned}$$

The two displays are the same statement, so an allocation satisfies one if and only if it satisfies the other. The second is the incentive constraint of the no-externality economy with period utility $\tilde{U}_t$, which proves (i).

Step 3: the local constraint. Part (i) places the reporting problem in the environment of Kapička (2013) with period utility $\tilde{U}_t$. His requirements bear on the period utility date by date and $\tilde{U}_t$ inherits them from Assumption 3.1 because S_t enters U as a fixed parameter: $\tilde{U}_t$ is twice continuously differentiable, increasing in both goods, jointly concave in (c, q, l), and differentiable in the type. The dating of $\tilde{U}_t$ adds nothing beyond the date dependence Section 4.1 already carries in f_t and $\bar{\alpha}_t$ and the horizon is finite, so the one-shot-deviation reduction of Fernandes and Phelan (2000) and the envelope theorem of Kapička (2013) apply date by date:

$$\dot{u}_t(\theta^t) = \frac{\partial}{\partial \theta_t} \tilde{U}_t\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right) + \beta\, w_{2,t}(\theta^t)$$

The derivative passes through the pollution stock, which does not vary with the agent's type:

$$\frac{\partial}{\partial \theta_t} \tilde{U}_t\left(c_t, q_t, \frac{y_t}{\theta_t}\right) = U_\theta\left(c_t, q_t, \frac{y_t}{\theta_t}, S_t\right) + U_S\left(c_t, q_t, \frac{y_t}{\theta_t}, S_t\right)\frac{\partial S_t}{\partial \theta_t} = U_\theta\left(c_t, q_t, \frac{y_t}{\theta_t}, S_t\right)$$

since $\partial S_t/\partial \theta_t = 0$. This proves (ii). □

The lemma is the formal version of the observation in Section 3 that the pollution stock enters both sides of the incentive constraint identically. To the agent, the pollution stock is a date-specific preference parameter, not a state she controls. The planner is different because it moves the pollution stock. The next proposition shows what internalizing it costs.

Proposition B.1 (recursive formulation with an externality)**.**
Let the Kapička (2013) reduction be valid for the two-good economy without an externality. The relaxed planner problem with pollution then admits the recursive formulation of Section 4.1 with the emission window $\mathbf{S}_{-} = (Q_{t-s}, \ldots, Q_{t-1})$ as the only new state, the accumulation constraint as the only new restriction, and the first-order system of the recursive problem as that of the sequence problem. In that system, (i) given the emission price ν_t, every date-t history solves the no-externality node problem with period utility $\tilde{U}_t$ and the resource cost of the dirty good raised from 1 to $1 + \nu_t$; (ii) the pollution-stock multiplier μ_t is common to all date-t histories and equals the damage integral over the date-t cross-section; and (iii) the emission price collects the discounted pollution-stock multipliers, $\nu_t = \sum_{j=0}^{s} R^{-j}\,\mu_{t+j}\,\Phi_j$.

Proof. Step 1: the aggregates are deterministic and internalizing them is a finite-dimensional addition. Skills are the only shocks and they are idiosyncratic, allocations are measurable functions of skill histories, and agents form a continuum, so the cross-sectional distribution of histories at each date coincides with the ex-ante law of a single agent's history. Aggregate emissions Q_t are therefore numbers known at date 0 and so are the pollution stocks S_t. The relaxed planner problem is then equivalent to a joint choice of the allocation and two vectors $(Q_0, \ldots, Q_T)$ and $(S_0, \ldots, S_T)$, subject to the consistency constraint $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta]\,dF_0(\theta)$ and the accumulation constraint $S_t = S(Q_{t-s}, \ldots, Q_t)$ at each date. Substituting the two constraints back into the objective recovers the original problem, so the two problems select the same allocations.

Step 2: Lagrangian. Attach the multiplier ν_t in present value ν_t/R^t to the date-t consistency constraint and μ_t in present value μ_t/R^t to the date-t accumulation constraint, with the sign convention of the Hamiltonian in Appendix B.2. The Lagrangian of the relaxed problem is

$$\mathcal{L} = \mathcal{L}^{alloc}\left(\{c_t, q_t, y_t\}; \{S_t\}\right) + \sum_{t=0}^{T} \frac{\nu_t}{R^t}\left(\int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta]\, dF_0(\theta) - Q_t\right)$$
$$- \sum_{t=0}^{T} \frac{\mu_t}{R^t}\left[S_t - S(Q_{t-s}, \dots, Q_t)\right]$$

where $\mathcal{L}^{alloc}$ collects every term of the Section 4.1 Lagrangian: the resource cost, the local incentive constraints, and the promise- and threat-keeping constraints, all with period utility $\tilde{U}_t$, as Lemma B.1 requires. The pollution stocks enter $\mathcal{L}^{alloc}$ only as the dates of $\tilde{U}_t$. The aggregates enter only the two constraint blocks.

Step 3: given the aggregates, the allocation problem is the Section 4.1 problem, re-dated and re-priced. Fix $(Q_t, S_t, \nu_t)_{t=0}^{T}$. Merging the consistency block with the resource cost turns the date-t flow cost of a history into

$$c_t(\theta^t) + (1 + \nu_t)\, q_t(\theta^t) - y_t(\theta^t)$$

plus the term $-\nu_t Q_t$, a constant with respect to the allocation. The allocation problem is therefore the no-externality problem with two changes: period utility $\tilde{U}_t$ and dirty-good resource cost $1+\nu_t$. Both stay inside the class the Section 4.1 reduction covers. Dating the utility adds nothing beyond the date dependence already present in f_t and $\bar{\alpha}_t$, and over the finite horizon the recursion is a backward induction from $V_{T+1} \equiv 0$, so no stationarity is used. Re-pricing keeps the period cost linear, hence continuous and convex, and increasing in q for $\nu_t > -1$. Since ν_t will turn out to be the social cost of carbon, this asks only that the carbon price not subsidize the dirty good by more than its unit resource cost. The factorization of the incentive-constraint set into current controls and continuation promises never involves the objective, so the re-pricing leaves it untouched. Given the aggregates, the allocation problem is therefore recursive in the Section 4.1 states $(\hat{w}, \hat{w}_2, \theta_-)$ and its value functions solve the Bellman equation of that section with $\tilde{U}_t$ and the re-priced dirty good. This is (i).

Step 4: first-order conditions for the aggregates. The pollution stock S_t appears in $\mathcal{L}$ in two places: the accumulation block, with derivative $-\mu_t/R^t$, and the date-t utility terms of $\mathcal{L}^{alloc}$, at every date-t history. Within a history, utility enters two constraints: the local incentive constraint with costate ψ and the definition of u with multiplier ϕ, exactly as in the Hamiltonian of Appendix B.2. Differentiating each with respect to S,

$$\frac{\partial}{\partial S}\psi\left[-U_l\left(c, q, \frac{y}{\theta}, S\right)\frac{l}{\theta} + \beta w_2\right] = -\psi\, U_{Sl}\frac{l}{\theta} \qquad \frac{\partial}{\partial S}\phi\left[u - U\left(c, q, \frac{y}{\theta}, S\right) - \beta w\right] = -\phi\, U_S$$

The pollution stock is common to all date-t histories, so stationarity of $\mathcal{L}$ in S_t sums these contributions across the date-t cross-section,

$$\mu_t = \int_0^\infty \left(-\psi\, U_{Sl}\frac{l}{\theta} - \phi\, U_S\right) d\theta$$

where the integral runs over the date-t population of skill histories. This is the first-order condition for S in Appendix B.2 whose type integral is to be read over that cross-section: at $t = 0$ the cross-section is F_0 and the reading is literal and at later dates the social-cost-of-carbon formula applies it by writing the date-$(t+j)$ damage integrals as population averages, the $\mathbb{E}_t \int (\cdot)\, f_{t+j}\, d\theta$ terms of Proposition 4.2. Because there is no aggregate risk, these averages are deterministic: $\mathbb{E}_t$ denotes a cross-section, the same for every date-t history, not a forecast specific to the current node. In particular, μ_t is one number per date, which is (ii).

Current aggregate emissions Q_t appear in the consistency block with derivative $-\nu_t/R^t$ and, in the accumulation blocks of dates t, through $t+s$ since $S_\tau = S(Q_{\tau-s}, \ldots, Q_\tau)$ carries Q_t as an argument exactly when $t \le \tau \le t+s$. Stationarity of $\mathcal{L}$ in Q_t gives

$$-\frac{\nu_t}{R^t} + \sum_{\tau=t}^{t+s} \frac{\mu_\tau}{R^\tau} \frac{\partial S_\tau}{\partial Q_t} = 0 \quad \Longleftrightarrow \quad \nu_t = \sum_{j=0}^{s} \frac{1}{R^j}\, \mu_{t+j}\, \Phi_j$$

which is (iii).

Step 5: recursive problem of Section 4.1 encodes the same system. Two checks remain: the state and the first-order conditions. From date t on, the aggregates matter only through the dated utilities $\tilde{U}_\tau$ for $\tau \ge t$, hence through $S_\tau = S(Q_{\tau-s}, \ldots, Q_\tau)$ for $\tau \ge t$, and each of these depends on past aggregates only through the window $\mathbf{S}_- = (Q_{t-s}, \ldots, Q_{t-1})$, the rest being current and future choices. Appending $\mathbf{S}_-$ to the Section 4.1 state is therefore sufficient and the window updates recursively, appending Q_t and dropping Q_{t-s}. The conditions: differentiate the Hamiltonian of Appendix B.2. The condition for q carries the extra term $\left[\mu_t S_q + \frac{1}{R}\frac{\partial V_{t+1}}{\partial Q}\right] f$, so the dirty good's effective resource cost is $1 + \nu_t$ with $\nu_t = \mu_t S_q + \frac{1}{R}\frac{\partial V_{t+1}}{\partial Q_t}$, and the envelope chain of that appendix resolves $\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^{s} R^{-(j-1)} \mu_{t+j} \Phi_j$, so that

$$\nu_t = \mu_t\, \Phi_0 + \frac{1}{R}\sum_{j=1}^{s} \frac{1}{R^{j-1}}\, \mu_{t+j}\, \Phi_j = \sum_{j=0}^{s} \frac{1}{R^j}\, \mu_{t+j}\, \Phi_j$$

The condition for S is the population damage integral of Step 4. The conditions for (c, l, u, w, w_2) contain no aggregate term and are those of the Section 4.1 problem with $\tilde{U}_t$, matching Step 3.

The two first-order systems coincide condition by condition. □

The argument concerns the relaxed problem, as everywhere in the paper: the envelope condition stands in for global incentive compatibility, and the sufficiency caveat of Section 3 is unchanged since Lemma B.1 maps the externality economy into the class for which that caveat was stated. Section 6 verifies global incentive compatibility numerically at the computed optima.

B.2 Commodity wedge

The Hamiltonian of this problem is as follows:

$$
\begin{aligned}
\mathcal{H} = &\left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, \mathbf{S})\right] f \\
&+ \psi \left[-U_l(c, q, y/\theta, S)\frac{l}{\theta} + \beta w_2\right] \\
&- \lambda_1 u(\theta) \bar{\alpha}_t f_t \\
&+ \lambda_2 u(\theta) f_2 \\
&+ \phi \left[u - U(c, q, y/\theta, S) - \beta w\right] \\
&- \mu \left[S - S(\dots, Q)\right]
\end{aligned}
$$

This yields the following first-order conditions.

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql}\frac{l}{\theta} + \left[\mu S_q + \frac{1}{R}\frac{\partial V_{t+1}}{\partial Q}\right] f = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: - \theta f - \phi U_l = \frac{1}{\theta}\psi \left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S\right) d\theta = \mu
\end{aligned}
$$

Here $S_q \equiv \partial S/\partial Q$ is the marginal effect of current aggregate emissions on the current pollution stock and the density f in the first-order condition for q comes from differentiating the aggregate $Q = \int q(\theta)f(\theta)\,d\theta$ which enters both the current accumulation constraint and, as the newest entry of the emission state $\mathbf{S}$, the continuation value through $\partial V_{t+1}/\partial Q \equiv \partial V_{t+1}/\partial Q_t$. The scalar current stock S is not itself an argument of $V_{t+1}(\cdot, \mathbf{S})$; the continuation value sees current emissions only through $\mathbf{S}$, via $\partial V_{t+1}/\partial Q$, which is why no V term appears in the first-order condition for S. Define the full shadow price of current emissions,

$$\nu_t \equiv \mu_t S_q + \frac{1}{R}\frac{\partial V_{t+1}}{\partial Q}. \tag{20}$$

As in the previous proof,

$$\phi = \frac{f}{U_c} - \frac{\psi U_{cl}\frac{l}{\theta}}{U_c}$$

Because c and q share the same ϕ,

$$\phi = \frac{f}{U_q} - \frac{\psi U_{ql}\frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q}$$

Hence,

$$\frac{f}{U_q} - \frac{\psi U_{ql}\frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q} = \frac{f}{U_c} - \frac{\psi U_{cl}\frac{l}{\theta}}{U_c}$$

Multiplying by U_q and U_c,

$$U_c\left(f - \psi U_{ql}\frac{l}{\theta} + \nu_t f\right) = U_q\left(f - \psi U_{cl}\frac{l}{\theta}\right)$$

Using $\gamma_c \equiv \frac{U_{cl}\,l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql}\,l}{U_q}$, or, equivalently, $\gamma_c U_c = U_{cl}\,l$ and $\gamma_q U_q = U_{ql}\,l$,

$$U_c\left(f - \frac{\psi}{\theta}\gamma_q U_q + \nu_t f\right) = U_q\left(f - \frac{\psi}{\theta}\gamma_c U_c\right)$$

Then, putting the γ terms (as well as the externality term) on one side,

$$U_c f - U_q f = \frac{\psi}{\theta} U_c U_q (\gamma_q - \gamma_c) - \nu_t U_c f$$

Finally, dividing by $U_q f$,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f}(\gamma_q - \gamma_c) - \nu_t \frac{U_c}{U_q}$$

Note that, by definition, $1 + \tau^q = U_q / U_c$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1+\tau^q}$. Multiplying through by -1,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) + \nu_t \frac{U_c}{U_q}$$

Since $\frac{U_c}{U_q} = \frac{1}{1+\tau^q}$,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) + \frac{\nu_t}{1+\tau^q}$$

Rearranging,

$$\frac{\tau^q - \nu_t}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$$

The right-hand side is exactly that of the no-pollution primitive condition $\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$ of Appendix A.3, with τ^q replaced by the deviation $\tau^q - \nu_t$. The first-order conditions for c and l and the costate equation for ψ are untouched by the externality, so the integrating-factor steps there carry over verbatim and give

$$\frac{\tau_t^q - \nu_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right]$$

with A_t, B_t, C_t, D_t the labor-wedge terms of Appendix A.1. The externality leaves this block untouched and enters only by replacing τ^q with the deviation $\tau^q - \nu_t$ on the left.

Now consider ν_t. Its first piece is the date-t pollution-stock multiplier μ_t, the marginal damage of a unit of the current pollution stock. Substituting ϕ from the first-order condition for c into the first-order condition for S, then dividing and multiplying by f,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \left[\frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c} \right] U_S \right) d\theta = \int_0^\infty \left(-\frac{\psi}{\theta f} U_{Sl} l - \left[1 - \frac{\psi}{\theta f} U_{cl} l \right] \frac{U_S}{U_c} \right) f\, d\theta$$

Rewrite $\psi/\theta f$ by noting that

$$\frac{\psi}{\theta f} = \left(\frac{\tau^y}{1-\tau^y} \right) \frac{1}{U_c} \frac{1}{1+\epsilon-\gamma_c}$$

Hence, also defining $\xi \equiv \frac{U_{Sl} l}{U_S}$,

$$\begin{aligned}
\mu_t &= \int \left(-\left(\frac{\tau^y}{1-\tau^y} \right) \frac{1}{1+\epsilon-\gamma_c} \frac{U_{Sl} l}{U_c} - \left[1 - \left(\frac{\tau^y}{1-\tau^y} \right) \frac{1}{1+\epsilon-\gamma_c} \frac{U_{cl} l}{U_c} \right] \frac{U_S}{U_c} \right) f\, d\theta \\
&= \int \left(-\left(\frac{\tau^y}{1-\tau^y} \right) \frac{1}{1+\epsilon-\gamma_c} \frac{U_{Sl} l}{U_c} \frac{U_S}{U_S} - \left[1 - \left(\frac{\tau^y}{1-\tau^y} \right) \frac{\gamma_c}{1+\epsilon-\gamma_c} \right] \frac{U_S}{U_c} \right) f\, d\theta \\
&= \int \left(-\left(\frac{\tau^y}{1-\tau^y} \right) \frac{\xi}{1+\epsilon-\gamma_c} \frac{U_S}{U_c} - \left[1 - \left(\frac{\tau^y}{1-\tau^y} \right) \frac{\gamma_c}{1+\epsilon-\gamma_c} \right] \frac{U_S}{U_c} \right) f\, d\theta \\
&= -\int \left[1 + \left(\frac{\tau^y}{1-\tau^y} \right) \frac{\xi-\gamma_c}{1+\epsilon-\gamma_c} \right] \frac{U_S}{U_c} f\, d\theta
\end{aligned}$$

The second piece of ν_t prices the echo of current emissions in future pollution stocks. By the envelope theorem applied to the period-$(t+1)$ problem, Q_t enters continuation costs only through the accumulation constraints of the next s periods: $S_{t+j} = S(Q_{t+j-s}, ..., Q_{t+j})$ carries Q_t as an argument for $j \le s$ and drops it afterward. Each envelope step peels off one date,

$$\frac{\partial V_{t+1}}{\partial Q_t} = \mu_{t+1} \Phi_1 + \frac{1}{R} \frac{\partial V_{t+2}}{\partial Q_t} \qquad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t}$$

and iterating until Q_t drops out ($\Phi_j = 0$ for $j > s$ by construction),

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^{s} \frac{1}{R^{j-1}} \mu_{t+j} \Phi_j$$

Substituting into (20), with $\Phi_0 = S_q$,

$$\nu_t = \mu_t \, \Phi_0 + \frac{1}{R} \sum_{j=1}^{s} \frac{1}{R^{j-1}} \, \mu_{t+j} \, \Phi_j = \sum_{j=0}^{s} \frac{1}{R^j} \, \mu_{t+j} \, \Phi_j$$

Substituting the damage block above for each μ_{t+j} and writing the date-$(t+j)$ population integrals as expectations over future skill histories,

$$\mathrm{SCC}_t \equiv \nu_t = -\sum_{j=0}^{s} \frac{1}{R^j} \, \mathbb{E}_t \int \left[1 + \left(\frac{\tau^y_{t+j}}{1-\tau^y_{t+j}} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \, \Phi_j \, f_{t+j} \, d\theta$$

Here, s is the number of periods emissions affect the pollution stock.

C Constrained-optimal commodity wedge

C.1 Validity of the recursive formulation

The labor constraint (5) restricts the date-t allocation (c, q, l, S) at the current node only. It involves neither the promised value u nor the continuation objects w and w_2, so it enters the period Hamiltonian pointwise with multiplier $\eta(\theta)f$ and adds no state.

The savings constraint (6) involves next period's controls through $\mathbb{E}_t[U_{c,t+1}]$. Rewriting it as a promise, the date-t problem passes each successor the requirement

$$M(\theta) = \frac{U_c\Big(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\Big)}{(1 - \bar{\tau}_t^s(\theta))\, \beta R}$$

which the date-$(t+1)$ problem must meet in expectation,

$$\hat{M} = \int_0^\infty U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right) f_t(\theta|\theta_-)\, d\theta \tag{21}$$

$M(\theta)$ is pinned by the current allocation and is not a free control.

Proposition C.1 (recursive formulation with frozen wedges)**.**
Let the reduction of Proposition B.1 be valid for the two-good economy with pollution. The relaxed planner problem with the added constraints (5) and (6) then admits the recursive formulation of Section 5.1 with the marginal-utility promise $\hat{M}$ as the only new state and the delivery constraint (21) as the only new restriction, and the first-order system of the recursive problem is that of the sequence problem. In that system, (i) an allocation satisfies the incentive constraints under the added restrictions if and only if it satisfies them without, so the local envelope condition is that of Section 4.1; (ii) the labor constraint enters the date-t component problem pointwise with multiplier $\eta(\theta)f_t$ and adds no state; and (iii) the savings constraint of each history is imposed exactly once, as the delivery requirement of the component problem its successors solve, with $\partial V_t/\partial \hat{M} = -\vartheta_t$, $\vartheta_0 = 0$, and no requirement passed at $t = T$.

Proof. Step 1: the constraints restrict allocations, not reports. Incentive compatibility compares one agent's payoffs under truthful reporting and under a reporting strategy σ, holding the allocation fixed. Constraints (5) and (6) restrict the functions (c_t, q_t, y_t) the planner chooses: no reporting strategy enters them, and they appear on neither side of the incentive inequality. The constrained problem therefore selects from a smaller set of allocations; whether any given allocation is incentive compatible is decided exactly as before. The one-shot-deviation reduction of Fernandes and Phelan (2000) and the envelope theorem of Kapička (2013) apply to the unchanged reporting problem, with the pollution stock entering as a date-specific parameter by Lemma B.1. This is (i).

Step 2: the labor constraint is pointwise. At every date-t node, (5) involves the current bundle $(c(\theta), q(\theta), y(\theta))$ and the current pollution stock only: no promised value, no continuation object, and no other date's controls. It enters the date-t component problem the way the resource cost does, pointwise with multiplier $\eta(\theta) f_t$, and adds no state. This is (ii).

Step 3: the savings constraint factorizes into one scalar per history. Fix a history θ^t with $t < T$. Constraint (6) reads

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}_t^s(\theta_t))\,\beta R \int_0^\infty U_{c,t+1}(\theta^t, \theta')\, f_{t+1}(\theta'|\theta_t)\, d\theta'$$

The left side is pinned by the date-t allocation at θ^t. The right side integrates over exactly the cross-section that one component problem chooses: the date-$(t+1)$ problem that continues θ^t, indexed by $(w(\theta^t), w_2(\theta^t), \theta_t, \mathbf{S})$ in Section 4.1 and now also by $M(\theta^t)$. Dividing both sides by $(1 - \bar{\tau}_t^s(\theta_t))\beta R$ turns the constraint into the statement that this component problem delivers mean marginal utility $M(\theta^t)$, which is (21) at the inherited value $\hat{M} = M(\theta^t)$. The family of sequence constraints, one per history θ^t with $t < T$, is therefore in bijection with the family of delivery constraints, one per component problem at dates 1 through T. Conversely, an allocation assembled from recursion-feasible components satisfies (6) at every history, because each component delivers the requirement its parent computed from its own allocation. The two problems minimize the same cost over the same constraint set, and matching problems give matching first-order systems: differentiating the component Lagrangian gives (22), condition by condition the system of the sequence problem.

Step 4: multipliers and boundaries. With the date-t Lagrangian carrying $\vartheta_t \left[\int U_c f\, d\theta - \hat{M}\right]$, the envelope theorem gives $\partial V_t / \partial \hat{M} = -\vartheta_t$. The date-0 problem has no parent, so it carries no delivery constraint and $\vartheta_0 = 0$. The date-T problem has no successor, so it passes no requirement, the continuation term of Appendix C.2 is absent, and (7) at $t = T$ reads $\eta_T^s = \vartheta_T U_{c,T}$, the terminal value of Lemma 5.3. This is (iii). The two constraints do not interact in the reduction because (5) adds no state and (21) adds no restriction on reporting; and because neither involves u, w, or w_2, the costate equation and the boundary conditions $\psi(0) = \psi(\infty) = 0$ are unchanged, as Appendix C.2 verifies on the explicit first-order system. □

C.2 The problem and first-order conditions

The Hamiltonian is that of Appendix B.2 with two added terms:

$$
\begin{aligned}
\mathcal{H} = &\left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, M(c,q,l,S), \theta, \mathbf{S})\right] f \\
&+ \psi \left[-U_l(c,q,y/\theta,S)\frac{l}{\theta} + \beta w_2\right] \\
&- \lambda_1 u(\theta)\bar{\alpha}_t f_t \\
&+ \lambda_2 u(\theta) f_2 \\
&+ \phi \left[u - U(c,q,y/\theta,S) - \beta w\right] \\
&- \mu \left[S - S(\ldots, Q)\right] \\
&+ \eta \left[-\frac{U_l(c,q,y/\theta,S)}{\theta\, U_c(c,q,y/\theta,S)} - (1 - \bar{\tau}^y_t(\theta))\right] f \\
&+ \vartheta_t\, U_c(c,q,y/\theta,S)\, f
\end{aligned}
$$

Differentiating the continuation term with respect to a control $z \in \{c, q, l, S\}$, by the chain rule through M and the envelope condition at $t+1$,

$$
\frac{1}{R}\,\frac{\partial V_{t+1}}{\partial \hat{M}}\,\frac{\partial M}{\partial z}\, f = -\frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}^s_t)\,\beta R^2}\, U_{cz}\, f
$$

Adding the $\vartheta_t U_{cz} f$ term from the inherited requirement, each first-order condition gains $(\eta^s_t / U_{c,t})\, U_{cz} f$ with η^s_t as in (7), alongside the labor-constraint term $\eta f\,\Gamma_z$. The first-order conditions are

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: \; f - \psi U_{cl}\frac{l}{\theta} + \eta f\,\Gamma_c + \frac{\eta^s_t}{U_c} U_{cc}\, f = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: \; f - \psi U_{ql}\frac{l}{\theta} + \nu_t f + \eta f\,\Gamma_q + \frac{\eta^s_t}{U_c} U_{cq}\, f = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: \; -\theta f - \phi U_l + \eta f\,\Gamma_l + \frac{\eta^s_t}{U_c} U_{cl}\, f = \frac{1}{\theta}\psi\left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \; \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \; \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \; \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \; \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \eta f\,\Gamma_S + \frac{\eta^s_t}{U_c} U_{cS}\, f\right) d\theta = \mu
\end{aligned}
\tag{22}
$$

with $\nu_t \equiv \mu_t S_q + \frac{1}{R}\partial V_{t+1}/\partial Q$ the shadow price of current emissions. The derivatives of the labor constraint are

$$\begin{aligned}
\Gamma_c &= (1-\bar{\tau}_t^y)\,\frac{\sigma}{c} - \frac{\gamma_c}{y} \\
\Gamma_q &= (1-\bar{\tau}_t^y)\,\frac{\sigma_{cq}}{q} - \frac{\gamma_q\,(1+\tau^q)}{y} \\
\Gamma_l &= (1-\bar{\tau}_t^y)\,\frac{\epsilon-\gamma_c}{l} \\
\Gamma_S &= -\frac{U_S}{U_c}\left[\frac{\xi}{y} + (1-\bar{\tau}_t^y)\,\frac{\xi_c}{c}\right]
\end{aligned} \tag{23}$$

and in the statistics of (1) and (2), $U_{cc}/U_c = -\sigma/c$, $U_{cq}/U_c = -\sigma_{cq}/q$, $U_{cl}/U_c = \gamma_c/l$, and $U_{cS}/U_c = \xi_c\,(U_S/U_c)\,/c$, so the savings term in each condition is η_t^s times the log-derivative of U_c in the corresponding direction. Neither constraint involves u, w, or w_2, so the costate equation, the boundary conditions $\psi(0) = \psi(\infty) = 0$, and the envelope identities for $\hat{w}$ and $\hat{w}_2$ are unchanged. This proves Lemma 5.1.

That $\Gamma_c > 0$ signals normality of leisure is shown in Appendix **??** and the argument is unaffected by the savings constraint, which does not enter the agent's hours choice at the frozen rate.

C.3 The two shadow prices

Divide the first-order condition for c in (22) by f and use $U_{cl}l = \gamma_c U_c$ and $U_{cc}/U_c = -\sigma/c$:

$$\frac{U_c\,\phi}{f} = 1 - \gamma_c\frac{\psi U_c}{\theta f} + \eta\,\Gamma_c - \eta_t^s\frac{\sigma}{c} \;\equiv\; 1-\chi_t \tag{24}$$

which is the χ_t of Proposition 5.1. Solving (22) for ϕ twice, once from the condition for c and once from the condition for l using $U_{ll}l + U_l = U_l(1+\epsilon)$,

$$\begin{aligned}
\phi &= \frac{f}{U_c} - \frac{\psi}{\theta}\gamma_c + \eta\,\frac{\Gamma_c}{U_c}f + \eta_t^s\frac{U_{cc}}{U_c}\frac{f}{U_c} \\
\phi &= -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon) + \eta\frac{\Gamma_l}{U_l}f + \eta_t^s\frac{U_{cl}}{U_c}\frac{f}{U_l}
\end{aligned}$$

Equating the two and multiplying through by U_l,

$$\left(\frac{U_l}{\theta U_c}+1\right)\theta f=-\frac{\psi U_l}{\theta}\left(1+\epsilon-\gamma_c\right)+\eta U_l\left[\frac{\Gamma_l}{U_l}-\frac{\Gamma_c}{U_c}\right]f+\eta_t^s\left[\frac{U_{cl}}{U_c}-\frac{U_{cc}}{U_c}\frac{U_l}{U_c}\right]f$$

Because $\bar{\tau}_t^y=1+U_l/(\theta U_c)$, dividing by $\theta f\,(1-\bar{\tau}_t^y)$ and using $U_l=-\theta U_c\,(1-\bar{\tau}_t^y)$ term by term gives the following. The first term becomes $A_t\,\psi U_c/(\theta f)$ with $A_t=1+\epsilon-\gamma_c$. The second becomes $-\eta\,[\Gamma_l U_c/U_l-\Gamma_c]$, which by (23) equals

$$-\eta\left[-\frac{\epsilon-\gamma_c}{y}-(1-\bar{\tau}_t^y)\,\frac{\sigma}{c}+\frac{\gamma_c}{y}\right]=\eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma}{c}+\frac{\epsilon-2\gamma_c}{y}\right]=\eta\,\Delta_t$$

The third term, using $U_{cl}/U_c=\gamma_c/l$, $U_{cc}/U_c=-\sigma/c$, and $U_l/U_c=-\theta\,(1-\bar{\tau}_t^y)$, becomes

$$\frac{\eta_t^s}{\theta\,(1-\bar{\tau}_t^y)}\left[\frac{\gamma_c}{l}-\frac{\sigma}{c}\,\theta\,(1-\bar{\tau}_t^y)\right]=-\eta_t^s\left[\frac{\sigma}{c}-\frac{\gamma_c}{(1-\bar{\tau}_t^y)\,y}\right]=-\eta_t^s\,\Lambda_t$$

Collecting,

$$\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y}=A_t\,\frac{\psi U_{c,t}}{\theta f_t}+\eta_t\,\Delta_t-\eta_t^s\,\Lambda_t \tag{25}$$

which rearranges to Lemma 5.2. Two checks. At $\eta_t^s=0$ this is the frozen-labor condition of Lemma **??**. At $\eta_t=0$, reading (25) as a wedge formula rather than a multiplier condition recovers the frozen-savings labor wedge of Proposition **??**, since $\Lambda_t=\sigma_t/c_t-\gamma_{c,t}/((1-\tau_t^y)\,y_t)$ is that proposition's bracket.

C.4 The intertemporal condition and the multiplier's law of motion

The chain of Appendix A.2 is unchanged, because neither constraint enters any of its steps. From (22), the first-order condition for w and the envelope condition give

$$\lambda_{1,t+1}=\beta R\,\frac{\phi}{f}=\beta R\,\frac{1-\chi_t}{U_{c,t}}$$

and integrating the costate equation over $(0,\infty)$ with $\psi(0)=\psi(\infty)=0$,

$$\lambda_{1,t}=\int_0^\infty\phi\,d\theta=\mathbb{E}_{t-1}\left[\frac{1-\chi_t}{U_{c,t}}\right]$$

Evaluating the second at $t+1$ and equating with the first,

$$\frac{1-\chi_t}{U_{c,t}} = \frac{1}{\beta R}\, \mathbb{E}_t\left[\frac{1-\chi_{t+1}}{U_{c,t+1}}\right] \tag{26}$$

which is Proposition 5.1. Multiplying (26) by $U_{c,t}$ and substituting the frozen Euler equation (6),

$$1-\chi_t = (1-\bar{\tau}^s_t)\, \mathbb{E}_t[U_{c,t+1}]\, \mathbb{E}_t\left[\frac{1-\chi_{t+1}}{U_{c,t+1}}\right]$$

Because $\mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[XY] - \mathrm{Cov}(X,Y)$ with $X = U_{c,t+1}$ and $Y = (1-\chi_{t+1})/U_{c,t+1}$,

$$1-\chi_t = (1-\bar{\tau}^s_t)\left\{\mathbb{E}_t[1-\chi_{t+1}] - \mathrm{Cov}_t\left(U_{c,t+1},\, \frac{1-\chi_{t+1}}{U_{c,t+1}}\right)\right\}$$

This is the general form referred to in Lemma 5.3. When the date-$(t+1)$ draw is degenerate the covariance vanishes and, writing $1-\chi_t = \mathcal{G}_t - \eta^s_t \sigma_t/c_t$ on both sides,

$$\mathcal{G}_t - \eta^s_t \frac{\sigma_t}{c_t} = (1-\bar{\tau}^s_t)\left[\mathcal{G}_{t+1} - \eta^s_{t+1}\frac{\sigma_{t+1}}{c_{t+1}}\right]$$

Collecting the η^s terms on the left and factoring $\mathcal{G}_{t+1}$ out of the right,

$$\eta^s_t\, \frac{\sigma_t}{c_t} - (1-\bar{\tau}^s_t)\, \eta^s_{t+1}\, \frac{\sigma_{t+1}}{c_{t+1}} = \mathcal{G}_{t+1}\left[\frac{\mathcal{G}_t}{\mathcal{G}_{t+1}} - (1-\bar{\tau}^s_t)\right] = \mathcal{G}_{t+1}\,(\bar{\tau}^s_t - \tau^{s,*}_t) \tag{27}$$

using $1-\tau^{s,*}_t \equiv \mathcal{G}_t/\mathcal{G}_{t+1}$. This is Lemma 5.3. Relative to Lemma **??**, the only change is that the net delivery value $\mathcal{G}_t$ carries the labor-constraint term $+\eta_t \Gamma_{c,t}$, both in the prefactor and inside the definition of the virtual savings wedge. Setting $\bar{\tau}^s_t = \tau^{s,*}_t$ at every date solves (27) with $\eta^s \equiv 0$.

The boundary conditions come from (7): at $t=0$, $\vartheta_0 = 0$ gives $\eta^s_0 = -\vartheta_1(\theta^0)U_{c,0}/((1-\bar{\tau}^s_0)\,\beta R^2)$; at $t=T$ no requirement is passed, so $\eta^s_T = \vartheta_T U_{c,T}$. The chain

$$\vartheta_{t+1}(\theta^t) = (1-\bar{\tau}^s_t)\,\beta R^2\left[\vartheta_t - \frac{\eta^s_t}{U_{c,t}}\right] \tag{28}$$

recovers the raw multipliers from the η^s sequence.

Proof of Lemma 5.4. Suppose $\eta^s \geq 0$ at every node; the case $\eta^s \leq 0$ is symmetric. Write $\kappa_t \equiv (1-\bar{\tau}^s_t)\,\beta R^2 > 0$. At $t=0$, $\vartheta_0 = 0$ and $\eta^s_0 \geq 0$ give $\vartheta_1 \leq 0$ by (28). At interior dates, $\eta^s_t \geq 0$ gives $\vartheta_{t+1} \leq \kappa_t \vartheta_t$, so $\vartheta_t \leq 0$ propagates forward for all $t \geq 1$. At the final date, $\eta^s_T = \vartheta_T U_{c,T} \geq 0$ with $U_{c,T} > 0$ forces $\vartheta_T \geq 0$, hence $\vartheta_T = 0$. Walking back, $0 = \vartheta_T \leq \kappa_{T-1}\vartheta_{T-1}$ gives $\vartheta_{T-1} \geq 0$, hence $\vartheta_{T-1} = 0$, and by induction $\vartheta_t = 0$ at every node, so $\eta^s \equiv 0$. Nothing in the argument involves η_t, which is why the conclusion is the same as in Corollary ??. □

The two-period tilt. Let $T=1$, preferences separable ($\gamma_c = \sigma_{cq} = 0$), the labor schedule at its virtual level so that $\eta_0 = \eta_1 = 0$ and $\mathcal{G}_t = 1$, and $\bar{\tau}^s_0 > 0$. Then $\tau^{s,*}_0 = 0$, and substituting the two boundary expressions into (27) at $t=0$,

$$-\vartheta_1 \left[\frac{U_{c,0}\,\sigma_0}{\kappa_0\, c_0} + (1-\bar{\tau}^s_0)\,\frac{U_{c,1}\,\sigma_1}{c_1} \right] = \bar{\tau}^s_0 \qquad \Longrightarrow \qquad \vartheta_1 < 0$$

so $\eta^s_0 > 0 > \eta^s_1$, giving $\Xi_0 < 0 < \Xi_1$ and, by Corollary 5.1, a carbon price above the social cost of carbon at date 0 and below it at date 1. Adding a labor schedule that overtaxes labor at both dates adds $\eta_t\,(1-\bar{\tau}^y_t) > 0$ to Ξ_t at both dates, shifting the profile down and the crossing later.

C.5 Commodity wedge

Divide the first-order condition for q in (22) by f, using $U_{ql}l = \gamma_q U_q$ and $U_{cq}/U_c = -\sigma_{cq}/q$:

$$1 - \gamma_q \frac{\psi U_q}{\theta f} + \nu_t + \eta\,\Gamma_q - \eta^s_t \frac{\sigma_{cq}}{q} = \frac{\phi U_q}{f}$$

Dividing by U_q, substituting $\phi/f = (1-\chi_t)\,/U_c$ from (24), and multiplying by U_c,

$$\frac{U_c}{U_q}\left(1 + \nu_t + \eta\,\Gamma_q - \eta^s_t \frac{\sigma_{cq}}{q}\right) - \gamma_q \frac{\psi U_c}{\theta f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta\,\Gamma_c - \eta^s_t \frac{\sigma}{c}$$

With $1+\tau^q = U_q/U_c$, multiply through by $1+\tau^q$ and collect:

$$\frac{\tau^q_t - \nu_t}{1+\tau^q_t} = (\gamma_c - \gamma_q)\,\frac{\psi U_c}{\theta f} - \eta\,\Gamma_c + \eta^s_t \frac{\sigma}{c} + \frac{1}{1+\tau^q}\left[\eta\,\Gamma_q - \eta^s_t \frac{\sigma_{cq}}{q}\right]$$

Substituting Γ_c and Γ_q from (23),

$$
\begin{aligned}
\frac{\tau_t^q-\nu_t}{1+\tau_t^q} &= (\gamma_c-\gamma_q)\frac{\psi U_c}{\theta f} - \eta\left(1-\bar{\tau}_t^y\right)\frac{\sigma}{c} + \eta\frac{\gamma_c}{y} + \eta_t^s\frac{\sigma}{c} \\
&\quad + \eta\left(1-\bar{\tau}_t^y\right)\frac{\sigma_{cq}}{q\left(1+\tau^q\right)} - \eta\frac{\gamma_q}{y} - \eta_t^s\frac{\sigma_{cq}}{q\left(1+\tau^q\right)} \\
&= (\gamma_c-\gamma_q)\left(\frac{\psi U_c}{\theta f}+\frac{\eta}{y}\right) - \left[\eta\left(1-\bar{\tau}_t^y\right)-\eta_t^s\right]\left[\frac{\sigma}{c}-\frac{\sigma_{cq}}{q\left(1+\tau^q\right)}\right]
\end{aligned}
$$

The second bracket is the income-effect gap $\mathcal{I}_t$ of (3), and the first is Ξ_t by Definition 5.1. That $\mathcal{I}_t>0$ if and only if the dirty good is normal follows as in Appendix D.3: with $\mathcal{D}<0$,

$$
\frac{\partial q^d}{\partial x} = \frac{\left(1+\tau^q\right)U_{cc}-U_{cq}}{\mathcal{D}} = \frac{U_c}{\mathcal{D}}\left[-\left(1+\tau^q\right)\frac{\sigma}{c}+\frac{\sigma_{cq}}{q}\right] = \frac{U_c\left(1+\tau^q\right)}{|\mathcal{D}|}\mathcal{I}_t
$$

Appendix C.6 identifies $\nu_t=\mathrm{SCC}_t$, completing Proposition 5.2.

C.6 The social cost of carbon

Take the first-order condition for S in (22), divide the integrand by f, and substitute $\phi/f=(1-\chi_t)/U_c$ together with $U_{Sl}l=\xi U_S$, Γ_S from (23), and $U_{cS}/U_c=\xi_c\left(U_S/U_c\right)/c$. The integrand becomes

$$
-\frac{U_S}{U_c}\left\{\xi\frac{\psi U_c}{\theta f}+(1-\chi_t)+\eta\left[\frac{\xi}{y}+\left(1-\bar{\tau}_t^y\right)\frac{\xi_c}{c}\right]-\eta_t^s\frac{\xi_c}{c}\right\}
$$

Expanding $1-\chi_t=1-\gamma_c\frac{\psi U_c}{\theta f}+\eta\left[\left(1-\bar{\tau}_t^y\right)\frac{\sigma}{c}-\frac{\gamma_c}{y}\right]-\eta_t^s\frac{\sigma}{c}$ and collecting,

$$
\begin{aligned}
\{\cdot\} &= 1+(\xi-\gamma_c)\frac{\psi U_c}{\theta f}+\eta\frac{\xi-\gamma_c}{y}+\left[\eta\left(1-\bar{\tau}_t^y\right)-\eta_t^s\right]\frac{\sigma+\xi_c}{c} \\
&= 1+(\xi-\gamma_c)\left(\frac{\psi U_c}{\theta f}+\frac{\eta}{y}\right)+\Xi_t\frac{\sigma+\xi_c}{c}
\end{aligned}
$$

Iterating the emission-price recursion $\nu_t=\mu_t S_q+\frac{1}{R}\partial V_{t+1}/\partial Q$ forward exactly as in Appendix B.2, with $\Phi_j\equiv\partial S_{t+j}/\partial Q_t$ and $\Phi_j=0$ for $j>s$,

$$
\nu_t=\mathrm{SCC}_t=-\sum_{j=0}^{s}\frac{1}{R^j}\,\mathbb{E}_t\int\left[1+\left(\xi_{t+j}-\gamma_{c,t+j}\right)\left(\frac{\psi U_{c,t+j}}{\theta f_{t+j}}+\frac{\eta_{t+j}}{y_{t+j}}\right)+\Xi_{t+j}\frac{\sigma_{t+j}+\xi_{c,t+j}}{c_{t+j}}\right]\frac{U_{S,t+j}}{U_{c,t+j}}\,\Phi_j\,f_{t+j}\,d\theta
$$

Setting $\eta = 0$ recovers the frozen-savings weight of Proposition **??**; setting $\eta^s = 0$ recovers the frozen-labor weight of Proposition **??**; setting both to zero recovers Proposition 4.2. Under $U = V(h(c, q, S), l)$, where $\xi = \gamma_c$, the weight is $1 + \Xi_{t+j}\,(\sigma_{t+j} + \xi_{c,t+j})\,/c_{t+j}$, so the social cost of carbon is Pigouvian only where the net shadow price vanishes.

C.7 The costate

The costate equation in (22) is $\dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi$, unchanged in form, with ϕ now given by (24). Substituting,

$$\dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}\left(1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta\,\Gamma_c - \eta_t^s \frac{\sigma}{c}\right)$$

so that

$$\dot{\psi} - \frac{\gamma_c}{\theta}\,\psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}\left(1 + \eta\,\Gamma_c - \eta_t^s \frac{\sigma}{c}\right)$$

This is the linear equation of Appendix **??** with the source term carrying both multipliers. Integrating with the factor $\exp\left[-\int \gamma_c\, d\tilde{x}/\tilde{x}\right]$ and imposing $\psi(0) = \psi(\infty) = 0$ delivers $\psi U_c/(\theta f)$ in the $ABCD$ form of Proposition 4.1, with C_t carrying the redistributive weight $1 + \eta\Gamma_c - \eta^s\sigma/c - \lambda_{1,t}\bar{\alpha}_t U_{c,t}$ in place of $1 - \lambda_{1,t}\bar{\alpha}_t U_{c,t}$, and every statistic evaluated at the constrained allocation. The virtual wedge $A_t \psi U_c/(\theta f)$ of Lemma 5.2 therefore inherits the hazard shape and the persistence of the unconstrained decomposition but not its level.

C.8 The optimal flat pair

Let the two schedules be flat at levels $\bar{\tau}_t^y$ and $\bar{\tau}_t^s$ and treat each as a choice variable. The derivative of the Lagrangian with respect to $\bar{\tau}_t^y$ collects the labor-constraint terms at every date-t node, since $\bar{\tau}_t^y$ enters only (5):

$$\frac{\partial \mathcal{L}}{\partial \bar{\tau}_t^y} = \int_0^\infty \eta_t(\theta)\, f_t(\theta)\, d\theta = \mathbb{E}_0[\eta_t] = 0$$

The rate $\bar{\tau}_t^s$ enters only through the requirement $M(\theta)$ passed to the successor, so

$$\frac{\partial \mathcal{L}}{\partial \bar{\tau}_t^s} = \int_0^\infty \frac{1}{R} \frac{\partial V_{t+1}}{\partial \hat{M}} \frac{\partial M(\theta)}{\partial \bar{\tau}_t^s}\, f_t(\theta)\, d\theta = \int_0^\infty \frac{\vartheta_{t+1}(\theta)\, U_{c,t}(\theta)}{(1 - \bar{\tau}_t^s)^2\, \beta R^2}\, f_t(\theta)\, d\theta = 0$$

which, with $(1-\bar{\tau}^s_t)^2\,\beta R^2 > 0$ constant across the date-t cross-section, gives $\mathbb{E}_0\left[\vartheta_{t+1}U_{c,t}\right] = 0$. Taking the cross-sectional expectation of (7) then gives $\mathbb{E}_0[\eta^s_t] = \mathbb{E}_0\left[\vartheta_t U_{c,t}\right]$, which equals zero at $t=0$ because $\vartheta_0 = 0$.

For the closed form, solve (25) for η_t and impose $\mathbb{E}_0[\eta_t] = 0$:

$$\mathbb{E}_0\left[\frac{1}{\Delta_t}\left(\frac{\bar{\tau}^y_t}{1-\bar{\tau}^y_t} - A_t\frac{\psi U_{c,t}}{\theta f_t} + \eta^s_t\Lambda_t\right)\right] = 0$$

With $s_{yy,t} = 1/((1-\bar{\tau}^y_t)\,\Delta_t)$ the compensated response of earnings to the net-of-tax rate, derived in Appendix **??**, and $\bar{\tau}^y_t$ constant across the cross-section, this rearranges to

$$\frac{\bar{\tau}^y_t}{1-\bar{\tau}^y_t} = \frac{\mathbb{E}_0\left[s_{yy,t}\left(A_t\,\frac{\psi U_{c,t}}{\theta f_t} - \eta^s_t\,\Lambda_t\right)\right]}{\mathbb{E}_0[s_{yy,t}]}$$

which is Proposition 5.3. At $\eta^s_t = 0$ it is Proposition **??**.

C.9 An earnings-indexed schedule

Let the frozen labor wedge follow (9), so $1-\bar{\tau}^y_t(y) = \lambda^T_t\,(1-p_t)\,y^{-p_t}$ with $y = \theta l$. Differentiating,

$$\frac{\partial\bar{\tau}^y_t}{\partial y} = \lambda^T_t\,(1-p_t)\,p_t\,y^{-p_t-1} = \frac{p_t\,(1-\bar{\tau}^y_t)}{y}$$

The constraint is $\Gamma = -U_l/(\theta U_c) - (1-\bar{\tau}^y_t(y))$. Its derivatives with respect to c, q, and S are unchanged, because $\bar{\tau}^y_t$ depends on none of them. Its derivative with respect to l gains one term through $y = \theta l$:

$$\Gamma_l = (1-\bar{\tau}^y_t)\,\frac{\epsilon-\gamma_c}{l} + \frac{\partial\bar{\tau}^y_t}{\partial y}\,\theta = (1-\bar{\tau}^y_t)\,\frac{\epsilon-\gamma_c}{l} + \frac{p_t\,(1-\bar{\tau}^y_t)}{l} = (1-\bar{\tau}^y_t)\,\frac{\epsilon-\gamma_c+p_t}{l}$$

using $\theta/y = 1/l$. Only Δ_t changes, since Γ_l enters the derivation of Appendix C.3 solely through $-\eta\,[\Gamma_l U_c/U_l - \Gamma_c]$:

$$-\eta\left[-\frac{\epsilon-\gamma_c+p_t}{y} - (1-\bar{\tau}^y_t)\,\frac{\sigma}{c} + \frac{\gamma_c}{y}\right] = \eta\left[(1-\bar{\tau}^y_t)\,\frac{\sigma}{c} + \frac{\epsilon+p_t-2\gamma_c}{y}\right]$$

so $\Delta_t \mapsto \Delta_t + p_t/y_t$. Every other object in Appendices C.3 through C.8 is untouched, which proves Lemma 5.5. Because $p_t > 0$ for a progressive schedule, Δ_t rises: the schedule's convexity in earnings makes the agent's objective more concave in hours, so an interior labor choice is easier to sustain than under a schedule flat in earnings.

One consequence for the quantitative exercise. The multiplier η_t is now determined at the fixed point between the assigned earnings $y_t(\theta^t)$ and the marginal rate the schedule attaches to them, since $\bar{\tau}_t^y$ appears on both sides of (25) through y. Under (9) the map is monotone in y whenever $p_t \in (0, 1)$, so the fixed point is unique, but the solver must iterate rather than read the schedule off the type.

D Derivations: a given carbon price

This appendix derives the results of Sections ?? and ?? in one pass. The commodity wedge is held to a schedule $\bar{p}_t(\theta) \equiv 1+\bar{\tau}^q_t(\theta)$: Section ?? reads every condition at a flat schedule, $\bar{p}_t(\theta) = 1+\bar{\tau}_t$, whose level is chosen optimally, and Section ?? at an arbitrary exogenous schedule. The shadow emission price ν_t of (20) is carried throughout; the no-externality economy of Section ??'s first read sets $\nu_t = 0$. Multiplier conventions follow Appendix A.1, and $\eta^q(\theta) f_t(\theta|\theta_-)$ is the multiplier on the wedge constraint. Two shorthands recur: the ad-valorem deviation and its virtual counterpart,

$$\varpi_t(\theta) \equiv \frac{\bar{\tau}^q_t(\theta) - \nu_t}{1+\bar{\tau}^q_t(\theta)} \qquad\qquad \varpi^*_t(\theta^t) \equiv (\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}$$

which at $\nu_t = 0$ and a flat schedule are the ϖ_t and ϖ^*_t of Section ??.

D.1 The problem and first-order conditions

The Hamiltonian is that of Appendix B.2 plus the constraint term $\eta^q\,[U_q/U_c - \bar{p}]\,f$. Write $\Gamma^q \equiv U_q/U_c - \bar{p}$ and Γ^q_z for its partial derivatives; each follows from the quotient rule,

$$\Gamma^q_z = \frac{U_{qz}}{U_c} - \frac{U_q}{U_c}\frac{U_{cz}}{U_c} = \frac{U_{qz}}{U_c} - \bar{p}\,\frac{U_{cz}}{U_c}$$

evaluated at the constrained allocation, where $U_q/U_c = \bar{p}$. For the numéraire, with $U_{qc}/U_c = -\sigma_{cq}/q$ and $U_{cc}/U_c = -\sigma/c$,

$$\Gamma^q_c = -\frac{\sigma_{cq}}{q} + \bar{p}\,\frac{\sigma}{c} = \bar{p}\left[\frac{\sigma}{c} - \frac{\sigma_{cq}}{\bar{p}\,q}\right] = \bar{p}\,\mathcal{I}_t, \tag{29}$$

the income-effect gap (3) evaluated at $1+\tau^q = \bar{p}$. For the dirty good, define $\sigma_{q,t} \equiv -U_{qq}\,q_t/U_q$, the own-curvature of the dirty good's marginal utility, so that $U_{qq}/U_c = (U_{qq}q/U_q)(U_q/U_c)/q = -\bar{p}\,\sigma_q/q$ and

$$\Gamma^q_q = -\bar{p}\,\frac{\sigma_q}{q} + \bar{p}\,\frac{\sigma_{cq}}{q} = -\bar{p}\,\frac{\sigma_q - \sigma_{cq}}{q} \equiv -\bar{p}\,\mathcal{J}_t, \tag{30}$$

with $\mathcal{J}_t \equiv (\sigma_{q,t} - \sigma_{cq,t})/q_t$ the mirror image of $\mathcal{I}_t$. For labor, with $U_{ql}/U_c = (U_{ql}l/U_q)(U_q/U_c)/l = \bar{p}\,\gamma_q/l$ and $U_{cl}/U_c = \gamma_c/l$,

$$\Gamma^q_l = \bar{p}\,\frac{\gamma_q}{l} - \bar{p}\,\frac{\gamma_c}{l} = -\bar{p}\,\frac{\gamma_c - \gamma_q}{l}. \tag{31}$$

For the pollution stock, with $\xi_c \equiv U_{Sc}\,c/U_S$ and $\xi_q \equiv U_{Sq}\,q/U_S$ the complementarities of the pollutant with each good, $U_{qS}/U_c = \xi_q(U_S/U_c)/q$ and $U_{cS}/U_c = \xi_c(U_S/U_c)/c$, so

$$\Gamma^q_S = \frac{U_S}{U_c}\left[\frac{\xi_q}{q} - \bar{p}\,\frac{\xi_c}{c}\right]. \tag{32}$$

The first-order conditions are those of Appendix B.2, each augmented by $\eta^q f\,\Gamma^q_z$:

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: \; f - \psi U_{cl}\frac{l}{\theta} + \eta^q f\,\Gamma^q_c = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: \; f - \psi U_{ql}\frac{l}{\theta} + \nu_t f + \eta^q f\,\Gamma^q_q = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: \; -\theta f - \phi U_l + \eta^q f\,\Gamma^q_l = \frac{1}{\theta}\psi\left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial S} &: \; \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \eta^q f\,\Gamma^q_S\right) d\theta = \mu
\end{aligned}$$

with the conditions for u, w, and w_2 unchanged, so the costate equation, the boundary conditions $\psi(0) = \psi(\infty) = 0$, and the promise-keeping chain carry over. In the no-externality economy $\nu_t = 0$ and the pollution-stock condition is absent. When the level of a flat schedule is itself chosen, one more derivative is needed; it is taken in Appendix D.9.

D.2 Implementation and first-order approach

Fix a common price $\bar{p} = 1 + \bar{\tau}$ and consider the agent's split of expenditure x,

$$\tilde{U}(x, l; \bar{\tau}) = \max_{c,q \geq 0} U(c, q, l) \quad \text{s.t.} \quad c + \bar{p}\,q = x$$

The first-order condition of the split is $U_q = \bar{p}\,U_c$. If we substitute c using the budget constraint, $U(x - \bar{p}\,q, q, l)$ and therefore $\tilde{U}_q = -U_c\bar{p} + U_q$ and $\tilde{U}_{qq} = U_{cc}\bar{p}^2 - 2U_{cq}\bar{p} + U_{qq}$. Assumption 3.1 implies that $\mathcal{D} < 0$ and that an interior split exists and is unique. A bundle (c, q) therefore satisfies the wedge constraint $U_q/U_c = \bar{p}$ if and only if it is the split of $x = c + \bar{p}q$.

For the envelope identities, differentiate $\tilde{U} = U(c^d, q^d, l)$. Differentiating the budget over x gives $\frac{\partial c^d}{\partial x} + \bar{p}\,\frac{\partial q^d}{\partial x} = 1$. By the first-order condition,

$$\tilde{U}_x = U_c \frac{\partial c^d}{\partial x} + U_q \frac{\partial q^d}{\partial x} = U_c \left(\frac{\partial c^d}{\partial x} + \bar{p}\, \frac{\partial q^d}{\partial x} \right) = U_c$$

Differentiating the budget in l at fixed x gives $\frac{\partial c^d}{\partial l} + \bar{p}\, \frac{\partial q^d}{\partial l} = 0$.

$$\tilde{U}_l = U_l + U_c \left(\frac{\partial c^d}{\partial l} + \bar{p}\, \frac{\partial q^d}{\partial l} \right) = U_l$$

A deviator who reports $\hat{\theta}$ receives the bundle assigned to $\hat{\theta}$, meaning the pair $(x(\hat{\theta}), y(\hat{\theta}))$. They split $x(\hat{\theta})$ at the same common price, so the deviation space is that of a one-good economy in (x, y) with period utility $\tilde{U}(x, y/\theta; \bar{\tau})$. The type enters $\tilde{U}$ only through $l = y/\theta$, so the envelope condition is $\dot{u} = -\tilde{U}_l\, l/\theta + \beta w_2 = -U_l\, l/\theta + \beta w_2$ and the first-order approach applies as in the one-good economy of Appendix A.1.

D.3 Demand identities

Differentiating the tangency $U_q = \bar{p} U_c$ together with the budget gives the demand derivatives. In x at fixed $l, \bar{p}$, the budget gives $\partial_x c^d = 1 - \bar{p}\, \partial_x q^d$. Substituting into $U_{qc} \partial_x c^d + U_{qq} \partial_x q^d = \bar{p}(U_{cc} \partial_x c^d + U_{cq} \partial_x q^d)$ and collecting,

$$\mathcal{D}\, \partial_x q^d = -\left(U_{cq} - \bar{p} U_{cc}\right) \qquad \Longrightarrow \qquad \partial_x q^d = \frac{U_{cq} - \bar{p} U_{cc}}{|\mathcal{D}|} = \frac{\bar{p}\, U_c\, \mathcal{I}_t}{|\mathcal{D}|}, \tag{33}$$

where the last step uses $U_{cq} - \bar{p} U_{cc} = U_c\,(\sigma \bar{p}/c - \sigma_{cq}/q) = \bar{p}\, U_c\, \mathcal{I}_t$ from the definitions of the statistics. Since $\partial_x q^d$ is the income expansion of dirty-good demand, $\mathcal{I}_t > 0$ exactly when the dirty good is normal. The budget then gives $\partial_x c^d = 1 - m_q$ with $m_q \equiv \bar{p}\, \partial_x q^d$, which satisfies $U_q \mathcal{J}_t = \bar{p} U_{cq} - U_{qq}$ and hence $\partial_x c^d = U_q \mathcal{J}_t / |\mathcal{D}|$ is positive when c is normal. In l at fixed $x, \bar{p}$, the budget gives $\partial_l c^d = -\bar{p}\, \partial_l q^d$ and the differentiated tangency yields $\mathcal{D}\, \partial_l q^d = \bar{p} U_{cl} - U_{ql}$, with $U_{cl} = \gamma_c U_c / l$ and $U_{ql} = \gamma_q \bar{p} U_c / l$,

$$\partial_l q^d = \frac{\bar{p}\, U_c\, (\gamma_c - \gamma_q)}{l\, \mathcal{D}} = \frac{\bar{p}\, U_c\, (\gamma_q - \gamma_c)}{l\, |\mathcal{D}|}. \tag{34}$$

Compensated own-price response: along an indifference curve $dc = -\bar{p}\, dq$ and differentiating the tangency in $\bar{p}$ at constant utility gives $\mathcal{D}\, dq = U_c\, d\bar{p}$. Hence,

$$s_{qq} \equiv \left. \frac{\partial q^d}{\partial \bar{p}} \right|_u = \frac{U_c}{\mathcal{D}} = -\frac{U_c}{|\mathcal{D}|} < 0 \qquad \frac{\partial q^d}{\partial \bar{p}} = s_{qq} - q^d\, \partial_x q^d \quad \text{(Slutsky)}$$

Three identities tie the constraint statistics to the demand objects. Combining (34) with s_{qq},

$$l\,\partial_l q^d = -\bar{p}\,|s_{qq}|(\gamma_c - \gamma_q)\,. \tag{35}$$

Adding $\bar{p}\,\mathcal{I}_t = (U_{cq} - \bar{p}U_{cc})/U_c$ and $\mathcal{J}_t = (\bar{p}U_{cq} - U_{qq})/U_q = (\bar{p}U_{cq} - U_{qq})/(\bar{p}U_c)$,

$$\Delta_t^q \equiv \bar{p}\,\mathcal{I}_t + \mathcal{J}_t = \frac{\bar{p}U_{cq} - \bar{p}^2 U_{cc} + \bar{p}U_{cq} - U_{qq}}{\bar{p}\,U_c} = \frac{-\mathcal{D}}{\bar{p}\,U_c} = \frac{1}{\bar{p}\,|s_{qq}|} > 0. \tag{36}$$

Combining (33) and (36), the marginal budget share is the numéraire-side curvature share,

$$m_q = \bar{p}\,\partial_x q^d = \frac{\bar{p}^2\,U_c\,\mathcal{I}_t}{|\mathcal{D}|} = \frac{\bar{p}\,\mathcal{I}_t}{\Delta_t^q}. \tag{37}$$

D.4 The multiplier

Dividing the first-order condition for c by $U_c f$, using $U_{cl} l = \gamma_c U_c$ and (29),

$$\frac{U_c\,\phi}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta^q\,\bar{p}\,\mathcal{I}_t. \tag{38}$$

Dividing the first-order condition for q by $U_q f$, multiplying by U_c, and substituting $U_c/U_q = 1/\bar{p}$ and (30),

$$\frac{U_c\,\phi}{f} = \frac{1+\nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} + \eta^q\,\frac{\Gamma_q^q}{\bar{p}} = \frac{1+\nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} - \eta^q\,\mathcal{J}_t. \tag{39}$$

Equating (38) and (39) and collecting,

$$1 - \frac{1+\nu_t}{\bar{p}} = (\gamma_c - \gamma_q)\frac{\psi U_c}{\theta f} - \eta^q\,(\bar{p}\,\mathcal{I}_t + \mathcal{J}_t)$$

Since $1 - (1+\nu_t)/\bar{p} = (\bar{\tau}_t^q - \nu_t)/(1+\bar{\tau}_t^q) = \varpi_t$ with Δ_t^q from (36),

$$\eta_t^q\,\Delta_t^q = \varpi_t^* - \varpi_t \qquad \Longleftrightarrow \qquad \eta_t^q = \bar{p}_t\,|s_{qq,t}|(\varpi_t^* - \varpi_t)\,. \tag{40}$$

At $\nu_t = 0$ and a flat schedule, this is Lemma **??**. With the externality, it is Lemma **??** with $\mathrm{SCC}_t \equiv \nu_t$. At $\eta_t^q = 0$, (40) is the primitive corrective-wedge condition of Appendix B.2, so the multiplier vanishes where the frozen deviation equals the virtual one.

D.5 Labor wedge

From the first-order condition for l, using $U_{ll}l + U_l = U_l(1+\epsilon)$,

$$\phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon) + \frac{\eta^q f\, \Gamma_l^q}{U_l}$$

With $U_l = -(1-\tau_t^y)\theta U_c$, where τ_t^y is the endogenous labor wedge, and (31),

$$\frac{\Gamma_l^q}{U_l} = \frac{-\bar{p}(\gamma_c - \gamma_q)/l}{-(1-\tau_t^y)\theta U_c} = \frac{\bar{p}\,(\gamma_c - \gamma_q)}{(1-\tau_t^y)\, y\, U_c}$$

Multiplying by U_c/f,

$$\frac{U_c\,\phi}{f} = \frac{1}{1-\tau_t^y} - (1+\epsilon)\frac{\psi U_c}{\theta f} + \eta^q\, \frac{\bar{p}\,(\gamma_c - \gamma_q)}{(1-\tau_t^y)\, y} \tag{41}$$

Equating (38) and (41) and collecting, with $1/(1-\tau_t^y) - 1 = \tau_t^y/(1-\tau_t^y)$ and $A_t = 1+\epsilon-\gamma_c$,

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t \frac{\psi U_c}{\theta f} + \eta_t^q\, \bar{p}_t \left[\mathcal{I}_t - \frac{\gamma_{c,t} - \gamma_{q,t}}{(1-\tau_t^y)\, y_t}\right] \tag{42}$$

This is the primitive labor wedge of Propositions **??** and **??**. The bracket is the response of the implied commodity wedge $\ln(U_q/U_c)$ to a utility-compensated unit of earnings. Along $(dc, dl) = (1-\tau_t^y,\, 1/\theta)$, we have $d\ln(U_q/U_c) = (1-\tau_t^y)\,\mathcal{I}_t + (\gamma_q - \gamma_c)/y$. Dividing by $1-\tau_t^y$ gives the bracket. Under weak separability the bracket is $\mathcal{I}_t$. Using (40) with $\varpi^* = 0$ and (37),

$$\eta_t^q\, \bar{p}_t\, \mathcal{I}_t = -\frac{\bar{p}_t\, \mathcal{I}_t}{\Delta_t^q}\, \varpi_t = -m_{q,t}\, \varpi_t = m_{q,t}\, \frac{\nu_t - \bar{\tau}_t^q}{1+\bar{\tau}_t^q}$$

which is Corollary 5.4.

D.6 Savings wedge

Define $\chi_t^q \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q \bar{p}_t \mathcal{I}_t$, so that (38) reads

$$\frac{\phi}{f} = \frac{1-\chi_t^q}{U_{c,t}}.$$

The promise-keeping chain of Appendix A.2 is untouched, since the wedge constraint enters none of its steps. The first-order condition for w and the envelope condition $\partial V_{t+1}/\partial \hat{w} = \lambda_{1,t+1}$ give

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1-\chi_t^q}{U_{c,t}}.$$

Integrating the costate equation over the type space with $\psi(0) = \psi(\infty) = 0$, $\int \bar{\alpha}_t f \, d\theta = 1$, and $\int f_2 \, d\theta = 0$ gives the second expression

$$\lambda_{1,t} = \int_0^\infty \phi \, d\theta = \mathbb{E}_{t-1}\left[\frac{1-\chi_t^q}{U_{c,t}}\right].$$

Evaluating this at $t+1$ and equating with the previous display one period apart,

$$\frac{1-\chi_t^q}{U_{c,t}} = \frac{1}{\beta R} \, \mathbb{E}_t\left[\frac{1-\chi_{t+1}^q}{U_{c,t+1}}\right]$$

Substituting the multiplier (40) and the curvature-share identity (37) assembles the adjustment into composite form,

$$\chi_t^q = \gamma_c \frac{\psi U_c}{\theta f} - m_q\left(\varpi_t^* - \varpi_t\right) = \left[\gamma_c - m_q(\gamma_c - \gamma_q)\right] \frac{\psi U_c}{\theta f} + m_q \, \varpi_t = \tilde{\gamma}_t \, \frac{\psi U_{c,t}}{\theta f_t} + m_{q,t} \, \varpi_t$$

with $\tilde{\gamma} \equiv (1-m_q)\gamma_c + m_q \gamma_q$ as in Proposition D.1. At $\nu_t = 0$, this is the $\chi^q = \tilde{\gamma} \, \psi U_c/(\theta f) + \varpi m_q$ of Proposition **??** and $m_q \varpi = (\bar{\tau}^q - \nu) \, \partial_x q^d$ is the commodity revenue net of marginal damages, returned by a marginal dollar of expenditure. It remains to pass from the inverse Euler equation to the savings wedge. The savings wedge τ_t^s is defined by the agent's intertemporal condition $U_{c,t} = \beta R \, (1-\tau_t^s) \, \mathbb{E}_t[U_{c,t+1}]$, so

$$1 - \tau_t^s = \frac{U_{c,t}}{\beta R \, \mathbb{E}_t[U_{c,t+1}]}.$$

The generalized inverse Euler equation gives $\beta R = U_{c,t} \, \mathbb{E}_t\left[(1-\chi_{t+1}^q)/U_{c,t+1}\right] / (1-\chi_t^q)$. Substituting,

$$1 - \tau_t^s = \frac{1-\chi_t^q}{\mathbb{E}_t\left[(1-\chi_{t+1}^q)/U_{c,t+1}\right] \, \mathbb{E}_t[U_{c,t+1}]}, \tag{43}$$

the exact wedge. To approximate it, write $g_{t+1} \equiv \ln U_{c,t+1} = \bar{g} + \hat{g}$ with $\mathbb{E}_t \hat{g} = 0$, and treat the fluctuation $\hat{g}$ and the adjustment χ^q_{t+1} as small. Expanding $e^{\pm g_{t+1}}$ to second order in $\hat{g}$ and keeping first order in χ^q,

$$\mathbb{E}_t[U_{c,t+1}] = e^{\bar{g}} \left(1 + \frac{1}{2} \operatorname{Var}_t \hat{g}\right),$$
$$\mathbb{E}_t\left[\frac{1-\chi^q_{t+1}}{U_{c,t+1}}\right] = e^{-\bar{g}} \left(1 - \mathbb{E}_t \chi^q_{t+1} + \operatorname{Cov}_t(\chi^q_{t+1}, \hat{g}) + \frac{1}{2} \operatorname{Var}_t \hat{g}\right),$$

where $\mathbb{E}_t \hat{g} = 0$ turns the linear term of the second expansion into the covariance. Their product is $1 - \mathbb{E}_t \chi^q_{t+1} + \operatorname{Cov}_t(\chi^q_{t+1}, \hat{g}) + \operatorname{Var}_t \hat{g}$ to this order, so inverting (43),

$$1-\tau^s_t = (1-\chi^q_t)\left[1 + \mathbb{E}_t \chi^q_{t+1} - \operatorname{Cov}_t(\chi^q_{t+1}, \hat{g}) - \operatorname{Var}_t \hat{g}\right] \approx 1-\chi^q_t+\mathbb{E}_t \chi^q_{t+1}-\operatorname{Cov}_t(\chi^q_{t+1}, \hat{g})-\operatorname{Var}_t \hat{g}.$$

Since $\operatorname{Cov}_t(\chi^q_{t+1}, \hat{g}) = \operatorname{Cov}_t(\chi^q_{t+1}, \ln U_{c,t+1})$,

$$\tau^s_t \approx \operatorname{Var}_t(\ln U_{c,t+1}) + \left(\chi^q_t - \mathbb{E}_t \chi^q_{t+1}\right) + \operatorname{Cov}_t(\chi^q_{t+1}, \ln U_{c,t+1}),$$

the second-order form of Proposition **??**. The baseline expansion of Appendix A.2 is the special case $\chi^q \mapsto \chi$, and the frozen-commodity expansion of Proposition **??** the same at $\nu_t = \mathrm{SCC}_t$.

D.7 Composite statistics and the comprehensive wedge

Proposition D.1 (composite statistics)**.**
Define

$$\tilde{\gamma} \equiv (1-m_q)\,\gamma_c + m_q\,\gamma_q \qquad \tilde{\epsilon} \equiv \epsilon - \frac{\bar{p}^2\, U_c^2\,(\gamma_c-\gamma_q)^2}{l\,|U_l|\,|\mathcal{D}|} \;\leq\; \epsilon \qquad \tilde{A} \equiv 1 + \tilde{\epsilon} - \tilde{\gamma}$$

Then $\tilde{\gamma} = \tilde{U}_{xl}\, l/\tilde{U}_x$ and $\tilde{\epsilon} = \tilde{U}_{ll}\, l/\tilde{U}_l$ for the indirect utility of Lemma 5.6. Under weak separability $\gamma_c = \gamma_q$, so $\tilde{\gamma} = \gamma_c$, $\tilde{\epsilon} = \epsilon$, and the restricted problem is that of Section 4.

Substituting the multiplier (40) into the labor condition (42) yields the comprehensive form of Proposition **??**. Using $\bar{p}/\Delta^q = \bar{p}^2|s_{qq}|$ from (36), the correction term expands into four pieces,

$$\eta^q \bar{p}\left[\mathcal{I} - \frac{\gamma_c-\gamma_q}{(1-\tau^y)y}\right] = \left[m_q - \frac{\bar{p}^2|s_{qq}|(\gamma_c-\gamma_q)}{(1-\tau^y)y}\right](\varpi^* - \varpi)$$

and, expanding with $\varpi^* = (\gamma_c - \gamma_q)\,\psi U_c/(\theta f)$,

$$
\begin{aligned}
m_q\,\varpi^* &= m_q(\gamma_c-\gamma_q)\frac{\psi U_c}{\theta f} = (\gamma_c-\tilde{\gamma})\frac{\psi U_c}{\theta f}\\
-\frac{\bar{p}^2|s_{qq}|(\gamma_c-\gamma_q)}{(1-\tau^y)y}\,\varpi^* &= -\frac{\bar{p}^2|s_{qq}|(\gamma_c-\gamma_q)^2}{(1-\tau^y)y}\,\frac{\psi U_c}{\theta f} = -\left(\epsilon-\tilde{\epsilon}\right)\frac{\psi U_c}{\theta f}\\
-m_q\,\varpi &= -\left(\bar{\tau}^q-\nu\right)\partial_x q^d\\
+\frac{\bar{p}^2|s_{qq}|(\gamma_c-\gamma_q)}{(1-\tau^y)y}\,\varpi &= -\frac{\bar{\tau}^q-\nu}{1-\tau^y}\cdot\frac{\partial_l q^d}{\theta}
\end{aligned}
$$

where the second line uses $l|U_l|=(1-\tau^y)\theta l\,U_c=(1-\tau^y)y\,U_c$ to match Proposition D.1's Le Chatelier correction, $\epsilon-\tilde{\epsilon}=\bar{p}^2U_c^2(\gamma_c-\gamma_q)^2/(l|U_l||\mathcal{D}|)=\bar{p}^2|s_{qq}|(\gamma_c-\gamma_q)^2/((1-\tau^y)y)$, and the fourth line uses $\partial_l q^d/\theta=-\bar{p}|s_{qq}|(\gamma_c-\gamma_q)/y$ from (35) and $\varpi\bar{p}=\bar{\tau}^q-\nu$. Moving the two fiscal pieces to the left-hand side of (42) and collecting the $\psi U_c/(\theta f)$ terms on the right,

$$
\frac{\tau_t^y}{1-\tau_t^y}+\frac{\bar{\tau}_t^q-\nu_t}{1-\tau_t^y}\left[(1-\tau_t^y)\,\partial_x q^d+\frac{\partial_l q^d}{\theta}\right]=\left[A_t+(\gamma_c-\tilde{\gamma})-(\epsilon-\tilde{\epsilon})\right]\frac{\psi U_c}{\theta f}=\tilde{A}_t\,\frac{\psi U_c}{\theta f}
$$

which is the comprehensive form, with $\bar{\tau}^q-\nu$ the screening tax and the bracket the compensated demand response $dq^d/dy|_u$. Along a perturbation dy with $dx=(1-\tau^y)dy$, utility is unchanged and $dq^d=[(1-\tau^y)\partial_x q^d+\partial_l q^d/\theta]dy$.

The composite statistics are also the curvatures of the indirect utility of Lemma 5.6, which proves the second claim of Proposition D.1. Since $\tilde{U}_x=U_c(c^d,q^d,l)$ and $\partial_l c^d=-\bar{p}\,\partial_l q^d$,

$$
\tilde{U}_{xl}=U_{cl}+(U_{cq}-\bar{p}\,U_{cc})\,\partial_l q^d
$$

Dividing by $\tilde{U}_x=U_c$, multiplying by l, and substituting (33) and (34),

$$
\frac{\tilde{U}_{xl}\,l}{\tilde{U}_x}=\gamma_c+\frac{l\,(U_{cq}-\bar{p}U_{cc})}{U_c}\cdot\frac{\bar{p}U_c(\gamma_q-\gamma_c)}{l\,|\mathcal{D}|}=\gamma_c+m_q\,(\gamma_q-\gamma_c)=\tilde{\gamma}
$$

Similarly $\tilde{U}_l=U_l(c^d,q^d,l)$ gives, using $U_{lq}-\bar{p}U_{lc}=(\bar{p}U_c/l)(\gamma_q-\gamma_c)$,

$$
\tilde{U}_{ll}=U_{ll}+(U_{lq}-\bar{p}\,U_{lc})\,\partial_l q^d=U_{ll}+\frac{\bar{p}^2U_c^2(\gamma_c-\gamma_q)^2}{l^2\,|\mathcal{D}|}
$$

and dividing by $\tilde{U}_l=U_l<0$ flips the sign of the positive correction,

$$\frac{\tilde{U}_{ll}\, l}{\tilde{U}_l} = \epsilon - \frac{\bar{p}^2\, U_c^2\, (\gamma_c - \gamma_q)^2}{l\, |U_l|\, |\mathcal{D}|} = \tilde{\epsilon} \leq \epsilon$$

Under weak separability, the corrections vanish and the problem is that of Appendix A.1.

D.8 The costate and the comprehensive-wedge decomposition

Proposition D.2 (comprehensive-wedge decomposition)**.**
At every history,

$$\Omega_t = \tilde{A}_t(\theta)\, \tilde{B}_t(\theta)\, \tilde{C}_t(\theta) + \beta R\, \Omega_{t-1}\, \tilde{D}_t(\theta)$$

where $\tilde{B}_t = B_t$ is the hazard ratio of Proposition 4.1 and $\tilde{C}_t, \tilde{D}_t$ are its C_t and D_t with the integrating kernel built on the composite $\tilde{\gamma}$ and, in $\tilde{C}_t$, the redistributive weight carrying the revenue term $1 - \bar{\tau}_t^q\, \partial_x q^d - \lambda_{1,t}\bar{\alpha}_t U_{c,t}$.

Substituting $\phi = f/U_c - \psi\gamma_c/\theta + \eta^q \bar{p}\, \mathcal{I}_t\, f/U_c$ from (38) into the costate equation and eliminating the multiplier with $\eta^q \bar{p}\mathcal{I}_t = m_q(\varpi^* - \varpi)$ and $m_q \varpi^*\, f/U_c = m_q(\gamma_c - \gamma_q)\, \psi/\theta$,

$$\dot{\psi} - \frac{\tilde{\gamma}}{\theta}\, \psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - [1 - m_q\, \varpi]\, \frac{f}{U_c}$$

Solving this ODE with the integrating factor $\exp[-\int \tilde{\gamma}\, d\tilde{x}/\tilde{x}]$ and $\psi(\infty) = 0$,

$$\psi(\theta) = \int_\theta^\infty \exp\left[-\int_\theta^x \tilde{\gamma}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] \left([1 - m_q(x)\, \varpi(x)]\, \frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx$$

Here, $\psi(0) = 0$ pins $\lambda_{1,t}$ as in (15). To carry $U_c(\theta)$ into the integrand, multiply the identity (16) by $\exp[\int_\theta^x (\gamma_c - \tilde{\gamma})\, d\tilde{x}/\tilde{x}]$, with $\gamma_c - \tilde{\gamma} = m_q(\gamma_c - \gamma_q)$:

$$\frac{U_c(\theta)}{U_c(x)} \exp\left[-\int_\theta^x \tilde{\gamma}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] = \exp\left[\int_\theta^x \left(\sigma \frac{\dot{c}}{c} + \sigma_{cq} \frac{\dot{q}}{q} - \gamma_c \frac{\dot{y}}{y} + m_q\, (\gamma_c - \gamma_q)\, \frac{1}{\tilde{x}}\right) d\tilde{x}\right] \equiv \tilde{K}(\theta, x)$$

By Appendix D.7, $\Omega_t = \tilde{A}_t\, \psi U_c/(\theta f)$, so multiplying the costate solution by $\tilde{A}_t U_c/(\theta f)$ and splitting the integral as in Appendix A.1,

$$\Omega_t = \tilde{A}_t(\theta)\, \tilde{B}_t(\theta)\, \tilde{C}_t(\theta) + \beta R\, \Omega_{t-1}\, \tilde{D}_t(\theta)$$

$$\tilde{B}_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$\tilde{C}_t(\theta) = \int_\theta^\infty \tilde{K}(\theta, x)\left(1 - m_q(x)\,\varpi(x) - \lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x)\right)\frac{f_t(x)dx}{1 - F_t(\theta)}$$

$$\tilde{D}_t(\theta) = \frac{\tilde{A}_t(\theta)}{\tilde{A}_{t-1}}\frac{U_{c,t}(\theta)}{U_{c,t-1}}\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \tilde{\gamma}_t(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)dx}{\theta f_t(\theta)}$$

The persistence block uses $\lambda_{2,t} = \beta R\,\psi_{t-1}/f_{t-1}$ and, from $\Omega_{t-1} = \tilde{A}_{t-1}\,\psi_{t-1}U_{c,t-1}/(\theta_{t-1}f_{t-1})$,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\theta_{t-1}\,\Omega_{t-1}}{\tilde{A}_{t-1}\,U_{c,t-1}}$$

Hence, the recursing object is the comprehensive wedge, which is Proposition D.2. The revenue term $-m_q\varpi$ inside the $\tilde{C}_t$ weight is the only trace of the second good in the redistributive valuation.

D.9 The optimal uniform price

Let the schedule be flat, $\bar{\tau}_t^q(\theta) = \bar{\tau}_t$, with the level chosen by the planner. The level enters the date-0 Lagrangian only through the date-t wedge constraints, each contributing $\eta^q f\,\partial[U_q/U_c - (1+\bar{\tau}_t)]/\partial\bar{\tau}_t = -\eta^q f$. The R^{-t} discounting and the transition densities convert the sum over date-t nodes into the unconditional population expectation, as in Step 4 of Proposition B.1. By the envelope theorem,

$$\frac{dW_0}{d\bar{\tau}_t} = -\frac{1}{R^t}\,\mathbb{E}_0[\eta_t^q]$$

Hence, the optimal level sets $\mathbb{E}_0[\eta_t^q] = 0$. In the reduction bookkeeping, the same condition emerges from Roy's identity offsetting the incentive value of compensation. The constraint formulation nets the two automatically and only the substitution distortion against the screening benefit survives. Substituting (40) with $\bar{p}_t$ and ϖ_t common across the date-t cross-section,

$$\mathbb{E}_0[\eta_t^q] = \bar{p}_t\left\{\mathbb{E}_0[|s_{qq,t}|\,\varpi_t^*] - \varpi_t\,\mathbb{E}_0[|s_{qq,t}|]\right\} = 0 \qquad \Longrightarrow \qquad \varpi_t = \frac{\mathbb{E}_0\left[|s_{qq,t}|\,(\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[|s_{qq,t}|]}$$

Multiplying this by $\bar{p}_t\,\mathbb{E}_0[|s_{qq}|]$ and replacing $\bar{p}|s_{qq}|(\gamma_c - \gamma_q)\,\psi U_c/(\theta f) = -\left(\psi U_c/(\theta f)\right) l\,\partial_l q^d$ and $\bar{p}\varpi_t = \bar{\tau}_t - \nu_t$,

$$(\bar{\tau}_t - \nu_t)\, \mathbb{E}_0[|s_{qq,t}|] = -\, \mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t}\, l_t\, \partial_l q_t^d\right] \qquad \Longleftrightarrow \qquad (\bar{\tau}_t - \nu_t)\, \mathbb{E}_0[s_{qq,t}] = \mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t}\, l_t\, \partial_l q_t^d\right]$$

At node-specific prices the same condition without the integral gives $\varpi(\theta^t) = \varpi^*(\theta^t)$, the unconstrained wedge of Appendix A.3. The uniform tax is its $|s_{qq}|$-weighted average. Under weak separability, every node wedge is zero, so $\bar{\tau}_t = \nu_t$ (Corollaries **??** and **??**). In retirement, $\psi \equiv 0$, so $\psi U_c/(\theta f) \equiv 0$ and $\bar{\tau}_t = \nu_t$ (Corollary 5.5).

D.10 The cost of uniformity

By the envelope theorem applied node by node, the date-t resource value of moving a single node's price is $-\eta^q(\theta) f(\theta)$ per unit of $\bar{\tau}^q(\theta)$. By (40),

$$-\eta^q\, f = \bar{p}\, |s_{qq}|(\varpi - \varpi^*)\, f$$

This value zeroes out at $\varpi = \varpi^*$, reproducing the node-specific optimum. The marginal resource cost of moving the node's ad-valorem wedge is $\bar{p}^2$ times this since $\varpi = 1 - (1+\nu)/\bar{p}$ gives $d\bar{\tau}^q = \bar{p}^2\, d\varpi/(1+\nu)$, which at $\nu = 0$ is $d\bar{\tau} = \bar{p}^2 d\varpi$. The loss from setting ϖ rather than ϖ^* at a node is then, to second order

$$\int_{\varpi^*}^{\varpi} \bar{p}^3\, |s_{qq}|\, (v - \varpi^*)\, dv = \frac{1}{2}\, \bar{p}^3\, |s_{qq}|\, (\varpi - \varpi^*)^2$$

Aggregating over the date-t cross-section, with $\bar{p}$ common because the price is uniform, and minimizing the resulting quadratic over the common ϖ yields the weighted mean $\varpi = \mathbb{E}_t^{|s|}[\varpi_t^*]$ of Appendix D.9 and, at that minimizer,

$$\text{Loss}_t \approx \frac{1}{2}\, (1 + \bar{\tau}_t)^3\, \mathbb{E}_0[|s_{qq,t}|]\, \operatorname{Var}_t^{|s|}(\varpi_t^*)$$

where $\mathbb{E}_t^{|s|}$ and $\operatorname{Var}_t^{|s|}$ are the $|s_{qq,t}|$-weighted population mean and variance over date-t histories, which is Proposition 5.5.

D.11 The externality and the social cost of carbon

The conditions above already carry the shadow emission price: every deviation is measured net of ν_t, so all of Section **??**'s results hold with $\bar{\tau}_t = \tau_t^q - \nu_t$ the screening tax and every statistic evaluated at the externality allocation, and the optimal flat level of Appendix D.9 delivers the additive split of Proposition 5.5. What remains is the pollution-stock multiplier. Substitute the three pieces of the first-order condition for S,

$$-\psi U_{Sl}\frac{l}{\theta} = -\frac{\psi U_c}{\theta f}\,\xi\,\frac{U_S}{U_c}\,f \qquad -\phi U_S = -\,(1-\chi_t^q)\,\frac{U_S}{U_c}\,f \qquad \eta^q f\,\Gamma_S^q = \eta^q\,\frac{U_S}{U_c}\left[\frac{\xi_q}{q} - \bar{p}\,\frac{\xi_c}{c}\right] f$$

the second from (38) and the third from (32), to get

$$\mu_t = -\int_0^\infty \left[\frac{\psi U_c}{\theta f}\,\xi + 1 - \chi_t^q - \eta^q\left(\frac{\xi_q}{q} - \bar{p}\,\frac{\xi_c}{c}\right)\right]\frac{U_S}{U_c}\,f\,d\theta$$

Expanding $1-\chi_t^q = 1 - \gamma_c\frac{\psi U_c}{\theta f} + \eta^q \bar{p}\mathcal{I}_t$ with $\bar{p}\mathcal{I}_t = \bar{p}\,\sigma/c - \sigma_{cq}/q$ and collecting,

$$\mu_t = -\int_0^\infty \left[1 + (\xi - \gamma_c)\,\frac{\psi U_c}{\theta f} + \eta^q\left(\bar{p}\,\frac{\sigma + \xi_c}{c} - \frac{\sigma_{cq} + \xi_q}{q}\right)\right]\frac{U_S}{U_c}\,f\,d\theta$$

The shadow emission price aggregates these date-$(t+j)$ damages exactly as in Appendix B.2. Current emissions enter the continuation value only through the accumulation constraints of the next s periods, so each envelope step peels off one date,

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^{s}\frac{1}{R^{j-1}}\,\mu_{t+j}\,\Phi_j, \qquad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t},$$

and, with $\Phi_0 = S_q$, (20) gives $\nu_t = \sum_{j=0}^{s} R^{-j}\mu_{t+j}\Phi_j$. Substituting this expression for each μ_{t+j} and writing the date-$(t+j)$ population integrals as expectations over future skill histories,

$$\mathrm{SCC}_t \equiv \nu_t = -\sum_{j=0}^{s}\frac{1}{R^j}\,\mathbb{E}_t\!\int\left[1 + (\xi_{t+j} - \gamma_{c,t+j})\,\frac{\psi U_{c,t+j}}{\theta f_{t+j}} + \eta_{t+j}^q\left(\bar{p}_{t+j}\,\frac{\sigma_{t+j} + \xi_{c,t+j}}{c_{t+j}} - \frac{\sigma_{cq,t+j} + \xi_{q,t+j}}{q_{t+j}}\right)\right]\frac{U_{S,t+j}}{U_{c,t+j}}\,\Phi_j\,f_{t+j}\,d\theta$$

whose bracket is the damage weight of Proposition 5.4 at each future node. The new piece is the response of the damage valuation to a budget-neutral clean-for-dirty swap: along $(dc, dq) = (\bar{p}, -1)$, which holds expenditure at consumer prices fixed, $d\ln(U_S/U_c) = \bar{p}(\sigma + \xi_c)/c - (\sigma_{cq} + \xi_q)/q$, using $\partial_c \ln(U_S/U_c) = (\sigma + \xi_c)/c$ and $\partial_q \ln(U_S/U_c) = (\sigma_{cq} + \xi_q)/q$. At $\eta^q \equiv 0$ the weight collapses to the fiscal factor of Appendix B.2; at the optimal flat level the multiplier averages to zero but is not zero node by node, so the weight carries the term even in Section **??**, which is the sense in which Proposition **??** says the fiscal factor gains a term proportional to η^q. The corrective component of the price is nonetheless uniform across types and histories, because damages depend only on aggregate emissions, so the emissions margin carries no information about types and cannot screen; the uniformity restriction binds only the Corlett-Hague screening component and never the corrective one.

D.12 Retirement

This subsection proves Corollary 5.5 and the retirement claims of Section 5. Retirement dates are $t \geq T_R$. Output is zero and the period utility is $U(c, q, 0, S_t)$, so the current type enters neither preferences nor technology.

Lemma D.1 (retirement kills the costate)**.**
At every retirement date an optimal allocation is measurable with respect to the retirement-entry history θ^{T_R-1}, the marginal promises vanish, $w_{2,t} \equiv 0$ for $t \geq T_R - 1$, the retirement component problem is deterministic node by node, and the costate on the local incentive constraint is $\psi_t \equiv 0$ for $t \geq T_R$. At the junction, the envelope condition of the last working date loses its future-rent term, $\dot{u}_{T_R-1} = U_\theta$.

Proof. Step 1: payoff-irrelevant reports cannot be separated. At a retirement date the report moves the assignment but neither preferences nor an output requirement, so the utility an agent obtains from reporting $\hat{\theta}$,

$$U\Big(c(\hat{\theta}), q(\hat{\theta}), 0, S_t\Big) + \beta\, w(\hat{\theta})$$

does not depend on the true type. Incentive compatibility then forces it to be constant in $\hat{\theta}$ on the support. Were it higher at some report, every type would prefer that report and truth-telling would fail for a positive mass of types. Hence, u_t is constant across the date-t draws given θ^{t-1} at every retirement date. The assignment can then be taken flat as well: every draw faces the same delivery problem, minimizing $c + q + \frac{1}{R}V_{t+1}$ over the bundles and continuation promises that deliver the common utility, so an optimal assignment common to all draws exists, and without loss the allocation from T_R on is measurable with respect to θ^{T_R-1}.

Step 2: the marginal promise dies at the junction. For $t \geq T_R - 1$, the marginal continuation value integrates a constant against the derivative of a density. By Step 1, $u_{t+1}(\theta^t, \theta')$ is constant in θ', equal to some $\bar{u}_{t+1}(\theta^t)$, so

$$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta')\, \partial_{\theta_t} f_{t+1}(\theta'|\theta_t)\, d\theta' = \bar{u}_{t+1}(\theta^t)\, \frac{\partial}{\partial \theta_t} \int_0^\infty f_{t+1}(\theta'|\theta_t)\, d\theta' = 0$$

since the density integrates to one at every θ_t. At the junction, $w_2 = 0$ enters the date-$(T_R - 1)$ problem as a feasibility restriction rather than a first-order condition: the retirement block can deliver no other value, so the choice of w_2 disappears, the last working date's costate is left free, and its envelope condition reads $\dot{u} = U_\theta$.

Step 3: the retirement block is deterministic and carries no costate. With the allocation measurable in θ^{T_R-1} and $w_2 \equiv 0$, the envelope condition at retirement dates reads $\dot{u} =$

$U_\theta + \beta w_2 = 0$, which the flat profile satisfies identically. The type integral drops from the component problem, which becomes deterministic node by node,

$$V_t(\hat{w}, \mathbf{S}_-) = \min_{c,q,w,S} \; c + q + \frac{1}{R} V_{t+1}(w, \mathbf{S}) \qquad \text{s.t.} \qquad \hat{w} = U(c, q, 0, S) + \beta w$$

with the emission state and its aggregate constraint carried as in Section 4.1. There is no envelope constraint and no threat-keeping constraint, hence no costate and no threat multiplier: $\psi_t \equiv 0$ and $\lambda_{2,t} = 0$ for $t \geq T_R$. The working-age first-order system confirms the consistency of this reading: at $\lambda_2 = 0$, $\bar{\alpha}_t = 1$, and a flat allocation, the condition for the numéraire gives $\phi = f/U_c$, promise keeping gives $\lambda_{1,t} = \int \phi \, d\theta = 1/U_{c,t}$, and the costate equation integrates to

$$\dot{\psi} = \lambda_1 f - \phi = \frac{f}{U_{c,t}} - \frac{f}{U_{c,t}} = 0$$

which, with $\psi(0) = 0$, gives $\psi \equiv 0$.

Step 4: the relaxed problem alone would not deliver this. With persistent skills, retirement draws remain informative about working-life types and the relaxed problem, which keeps only the envelope condition, would tilt retirement utility against those draws to relax the last working date's incentive constraint through w_2. Step 1 rules the tilt out: nothing disciplines a payoff-irrelevant report, so no such tilt is globally incentive compatible. The measurability restriction of Step 1 is therefore a restriction to the implementable class, not an approximation, and it is how the life-cycle block of Section 6, following Farhi and Werning (2013), treats retirement.

Step 5: the formulas do not need the costate. In retirement, the formulas of interest are the commodity wedge, the savings wedge, and the social cost of carbon. In each of them, ψ enters multiplied by one of γ_c, γ_q, or ξ. Each of these statistics is proportional to l through its defining cross-partial, so each vanishes at $l = 0$. The retirement conclusions, a zero screening tax, the standard inverse Euler equation and Pigouvian carbon pricing therefore hold whatever the costate. □

Corollary 5.5 follows: $\psi \equiv 0$ gives $\psi U_c/(\theta f) \equiv 0$ at every retirement node, so the optimal uniform level of Appendix D.9 is $\bar{\tau}_t = \nu_t$, the pure Pigouvian price. The retirement statements of Sections **??** and **??** follow by the same steps. With no hours choice to decentralize, the frozen labor-wedge constraint is not imposed in retirement, so $\eta \equiv 0$ there while the frozen savings wedge keeps operating across retirement dates, which is why its distortions survive where the labor-frozen ones die.